COMMENT LIRE ET DÉCRYPTER UNE CARTE D'APPROCHE IFR?

Qu'est-ce qu'une carte d'approche ?

Tout d'abord, il existe plusieurs types de cartes d'approche:

- **La première, c'est la carte d'approche visuelle (appelée carte VAC pour *Visual Approach Chart)* que vous pouvez retrouvez ici.**
- **La seconde, celle qui nous intéresse aujourd'hui, c'est la carte d'approche aux instruments appelée IAC pour *Instrumental Approach Chart*). Il en existe aussi plusieurs types: RNAV, NDB,LOC,RADAR,VOR et ILS**

Un peu d’histoire pour comprendre d’où viennent ces cartes d’approche.

C’est en 1941 grâce aux avancées technologique comme la radiocommunication que les premières cartes d’approche ont été élaborées permettant ainsi aux pilotes d’atterrir par visibilité réduite.

Durant ces années là, l’aviation étaient devenue une préoccupation mondiale et chaque pays par l’intermédiaire d’organisations diverses et variées créaient leurs propres normes de cartes d’approche.

Un peu d’histoire pour comprendre d’où viennent ces cartes d’approche.

À la fin de la seconde guerre mondiale, il est apparu indispensable d’harmoniser ces cartes dans le but de soutenir les voyages aériens internationaux c’est pourquoi l’OACI (*l’Organisation de l’Aviation Civil Internationale*) à vue le jour.

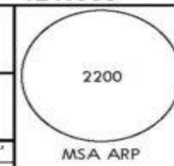
ESSB/BMA BROMMA

JEPPESEN 15 MAY 20 Eff 21 May (21-2) CAT A, B & C

STOCKHOLM, SWEDEN ILS or LOC Rwy 30

*D-ATIS	STOCKHOLM Control (APP)	*BROMMA Tower	*Ground
122.455	120.155	118.105	121.605

LOC SBA	Final Apch Crs	D6.6 SBA	ILS DA(H)	Apt Elev 47' Rwy 43'
109.7	299°	2500'(2457')	243'(200')	

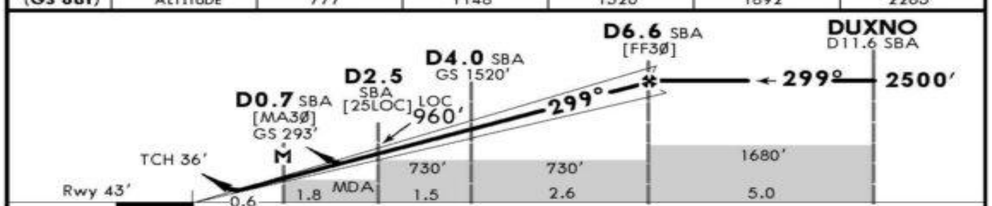


MISSED APCH: Climb STRAIGHT AHEAD to D5.2 SBA, turn LEFT and proceed to COR NDB climbing to 2500'. Join COR NDB holding.

Alt Set: hPa | Rwy Elev: 2 hPa | Trans level: By ATC | Trans alt: 5000'

1. DME required. 2 Initial Approach by RNAV STAR or RADAR.

LOC (GS out)	SBA DME	2.0	3.0	4.0	5.0	6.0
	ALTITUDE	777'	1148'	1520'	1892'	2263'



Gnd speed-Kts		70	90	100	120	140	160
ILS GS or LOC Descent angle	3.50°	434	557	619	743	867	991
MAP at D0.7 SBA							

HIALS REIL PAPI

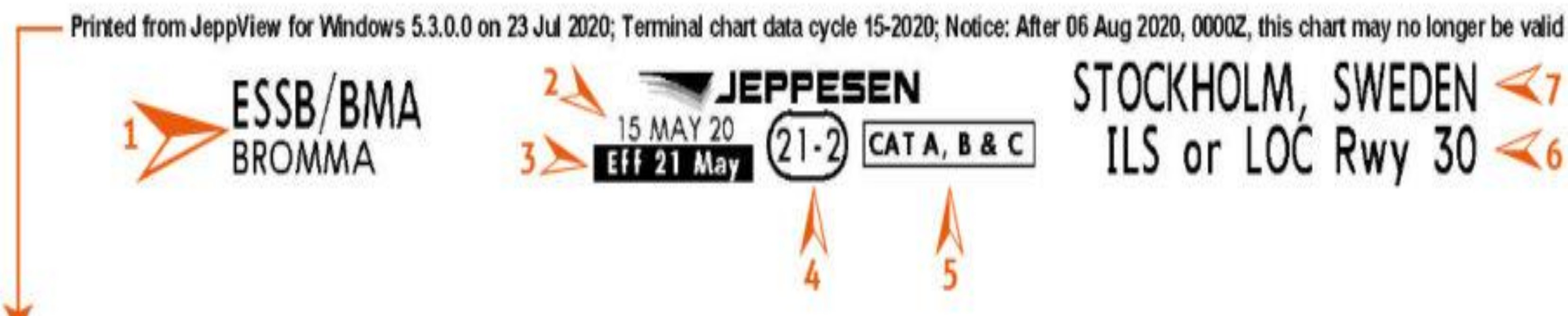
D5.2 SBA

Standard

	STRAIGHT-IN LANDING RWY 30 ILS DA(H) 243'(200') FULL	ALS out	LOC (GS out) CDFA DA/MDA(H) 550'(507')	ALS out	Max Kts	CIRCLE-TO-LAND Not authorized NE of rwy MDA(H)	VIS
A	RVR 750m	RVR 1200m	RVR 1500m		100	620' (573')	1500m
B	RVR 750m	RVR 1200m	RVR 1500m		135	670' (623')	1600m
C	RVR 750m	RVR 1200m	RVR 1900m	RVR 2400m	180	920' (873')	2400m
D	NOT APPLICABLE				D	NOT APPLICABLE	

BRIEFING STRIP | PANS OPS

CHANGES: New procedure.



Informations légales.
Date d'impression du document.
Cycle des données AIRAC dont est tirée la carte (ici 15-2020).
Date de validité du document (ici 6 Août 2020 minuit).

1: Identifiants de l'aéroport de Stockholm BROMMA

En premier le code **OACI** : ***ESSB***

En second le code **IATA** : ***BMA***

Le nom de l'aéroport **BROMMA**

2: Date de publication de la carte : **Le 15 Mai 2020**

3: Date de mise en vigueur de la carte : **Le 21 Mai 2020**

(Eff = effectivité de la carte)

4: Numéros de classification de la carte : **21-2**

5: Les catégories d'avions auxquels s'adresse cette carte:

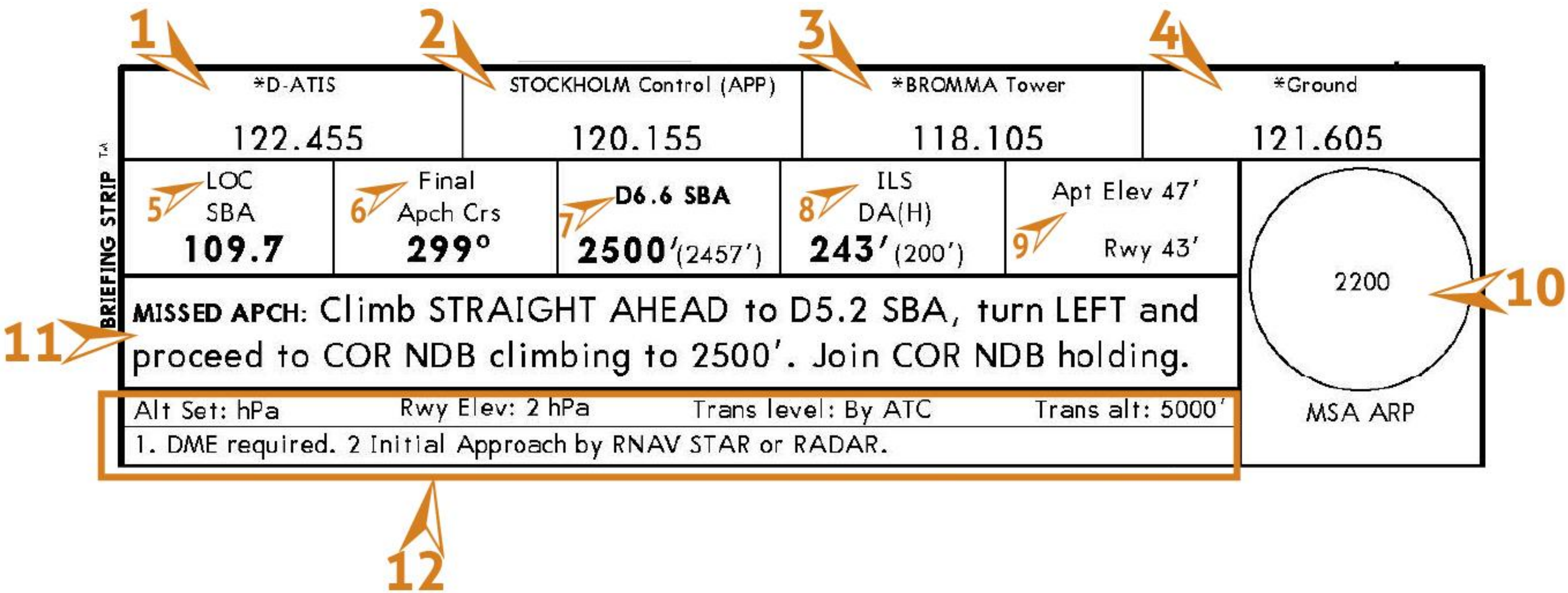
A, B et C

6: Procédure d'approche décrite par cette carte : **ILS ou LOC**

pour la piste **30**

7: Localisation de l'aéroport : **Stockholm, Suède**

CARTOUCHE D'INFORMATIONS POUR LE BRIEFING D'APPROCHE



1: **Informations pour l'approche La fréquence de l’l'ATIS : 122.455 (D-ATIS ou Data link-ATIS signifiant que cet aéroport dispose de la diffusion de donnée ATIS via les systèmes ACARS des avions)**

2: **La fréquence de STOCKHOLM Control (APP) :** ***120.155***

(APP pour Approche, fréquence à contacter en premier à l'approche de l'aéroport, le plus souvent c'est le service de contrôle avec lequel vous êtes en communication qui vous informera quand passer avec l'approche)

3: **La fréquence tour de BROMMA :** ***118.105*** **(Comme précédemment, c'est le service de contrôle avec lequel vous êtes en communication qui vous dira quand passer avec la Tour).**

4: **La féquence sol :** ***121.605*** **(Comme précédemment, c'est le service de contrôle avec lequel vous êtes en communication qui vous dira quand passer avec le sol)**

5: **La fréquence de la balise ILS (Localizer) : 109.7**

Ainsi que son identifiant : ***SBA (Serria Bravo Alpha)***

6: **La course en approche finale :** ***299°***

7: **Distance en Milles Nautiques de capture du GS**

(Glide Slope) au niveau du FAF (Final Approach Fix) : ***6.6***

NM ***DME de la balise SBA*** **Altitude minimale**

d'interception : ***2500'*** ***(ou une hauteur de 2457'***

8: **Type d'approche :** ***ILS*** **Altitude minimum de décision**

(Decision Altitude) : ***243' (ou 200' de hauteur)***

9: Altitude de l'aéroport : *47’*Altitude de la piste : *43’* (Ce sont des altitudes moyennes, car rien n'est jamais parfaitement plat.)

10: Cercle de 25 Milles Nautiques (sauf indication contraire) de l'Altitude Minimum de Secteur (Minimum Sector Altitude) : *2200’* (L'altitude minimale de secteur (MSA) est l'altitude la plus basse qui peut être utilisée afin d'avoir un dégagement minimum de 300 m (1000 pieds) au-dessus de tous les obstacles situés dans la zone contenue dans un cercle de 46 km (25 NM) de rayon.)

11: **Description des procédures en cas de remise de gaz (Missed Approach) :** ***Monter droit devant jusqu'à une distance de 5.2 de la balise SBA, tourner à gauche et se diriger vers le NDB COR en montant à 2500 pieds. Arrivé au NDB COR attendre.*** **(Attendre sur un point sous-entend faire un "Hold", procédure que nous allons voir plus bas)**

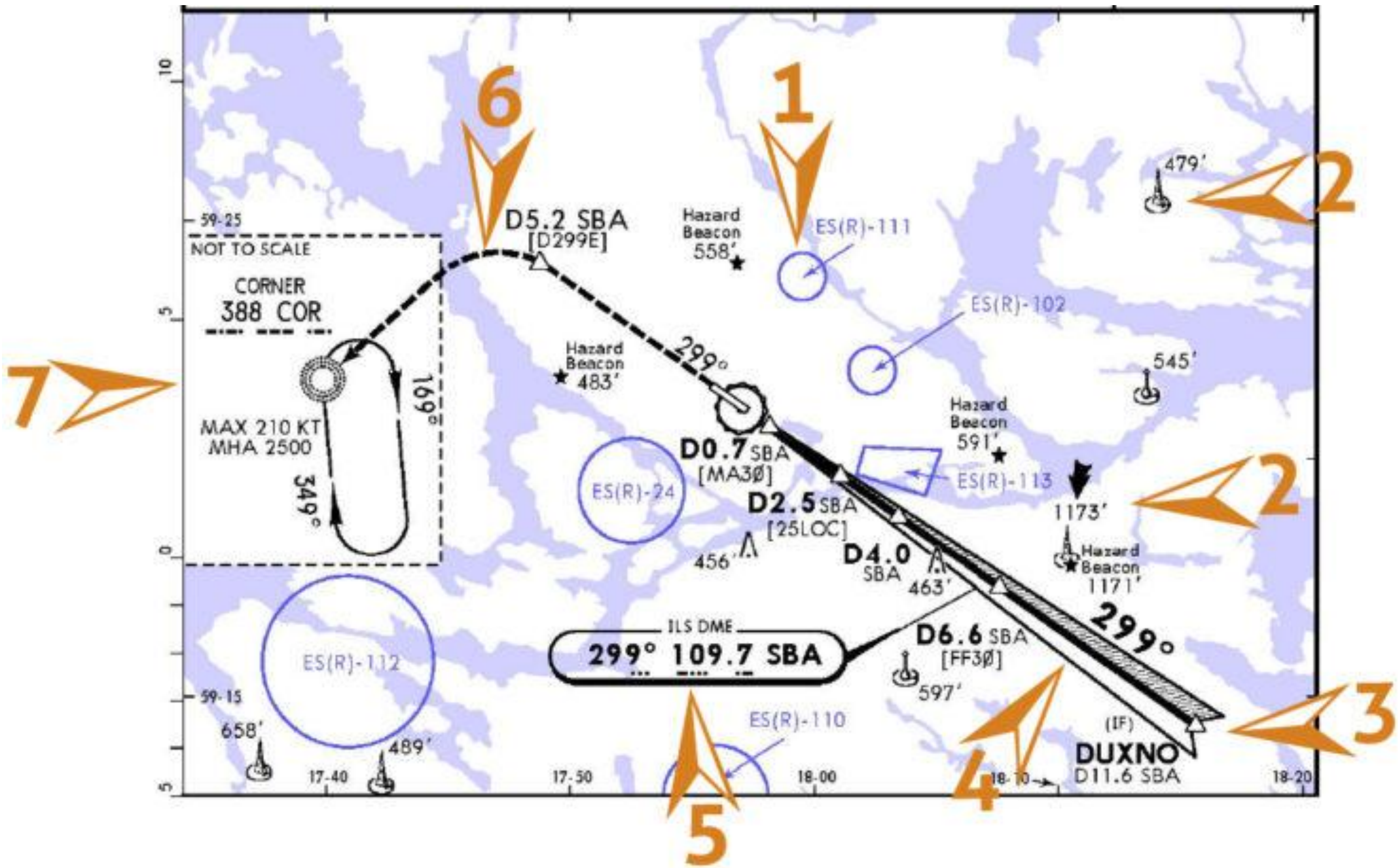
12: **Cartouche d'informations pratiques Le calage altimétrique se fait en :** ***hPa*** **L'élévation de la piste est de :** ***2 hPa*** **Le niveau de transition se fait par :** ***ATC*** **L'altitude de transition est de :** ***5000'***

1. Un équipement DME à bord de l'appareil est requis.

2. L'approche initiale s'effectue par RNAV ou RADAR.

(Cet aéroport se trouve en Europe, donc le calcul de la pression atmosphérique s'effectue en hectoPascal - hPa -, contrairement aux USA où elle s'effectue en pouce de mercure - inHg -)

CARTOUCHE DU PLAN D'APPROCHE



1: **Zone à caractère particulier :** ***ES(R)-111*** **(Toutes les zones repérées en bleu, sont des zones à caractère particulier, comme par exemple interdit de survol, ou au survol sous c ertaines conditions...)**

2: **Obstacles à proximité immédiate de l'aéroport :** ***Une antenne culminant à 479'*** **Obstacle le plus haut est représenté avec une flèche noire le pointant.**

(les obstacles sont représentés de façon graphique quand cela est possible)

3: Point d'entrée dans l'axe ILS appelé IF (IntermediateFix) :

Point DUXNO à 11.6 NM de la balise SBA

4: Représentation de l'axe ILS (Il est représenté par cet axe en forme de triangle allongé avec un coté blanc, l'autre gris et au centre un trait noir symbolisant la course de l'ILS, ici au 299°)

5: **Informations d'aide à la navigation pour l'approche finale avec : Le type d'approche** *ILS DME* **La course de l'axe de piste :** *299°* **La fréquence de la balise ILS :** *109.7* **L'identification de la balise associée :** *SBA* **(Aide de navigation latérale pour le segment d'approche finale déterminée)**

6: **Représentation graphique de la trajectoire de Remise de gaz (Missed Approach) avec :** *Un virage à gauche direction le NDB COR une fois arrivé à 5.2 NM de SBA*

7: Cartouche circuit d'attente Avec les informations :

L'identification de la balise NDB : *Le nom complet est CORNER, COR qui pour fréquence 388* **L'altitude Minimum d'attente (MHA pour Minimum Hold Altitude) :** *2500'* **La vitesse IAS maximum du circuit d'attente :** *210 kt* **La course d'éloignement :** *169°* **La course de rapprochement :** 349° **(Il est précisé en haut a gauche dans ce cartouche que la vue n'est pas à l'échelle)**

TABLEAU DES DESCENTES RECOMMANDÉES

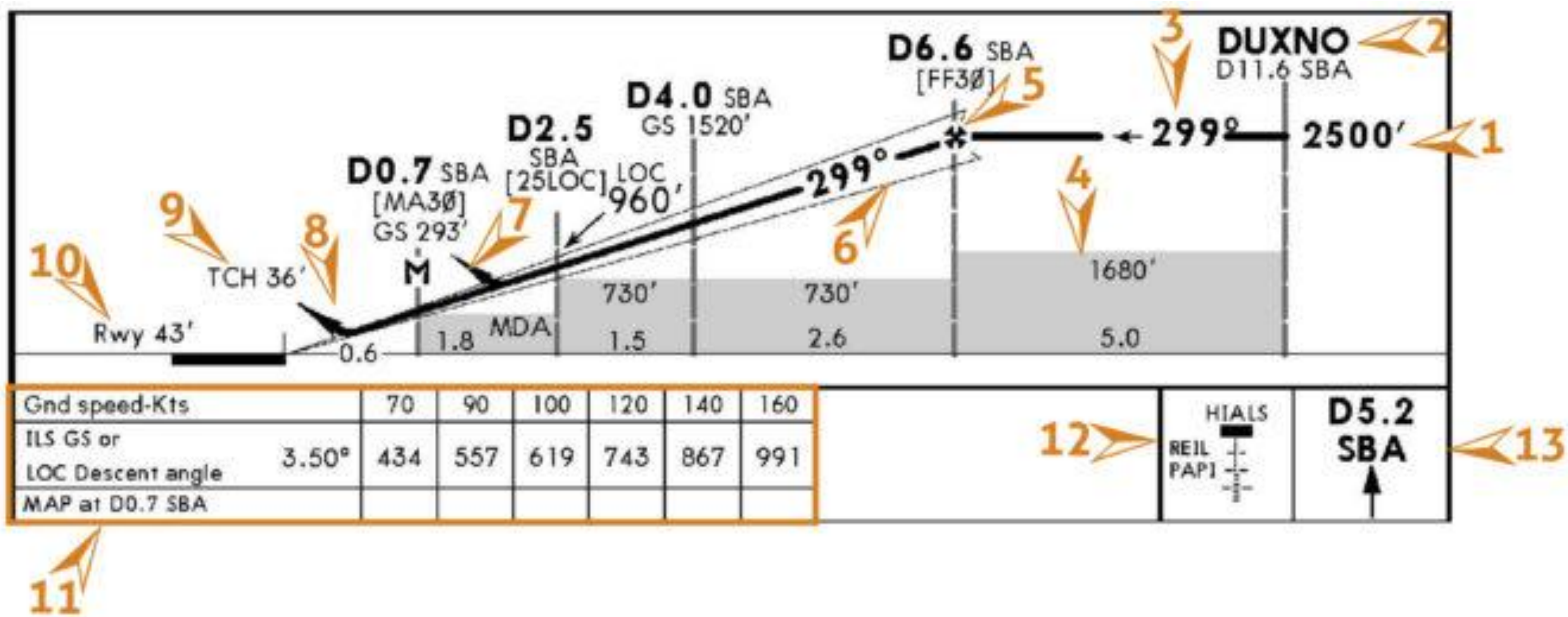
LOC (GS out)	SBA DME	2.0	3.0	4.0	5.0	6.0
	ALTITUDE	777′	1148′	1520′	1892′	2263′

(Uniquement utilisé en cas de non fonctionnement du Glide Slope)

1: Distance de la balise : SBA DME (Exemple, à 4 NM de SBA, vous devez être à 1520 pieds QNH)

2: Altitude associée à la distance : Minimum en pieds (Le début de descente se situe au FAF.)

VUE DE PROFILE DE LA TRAJECTOIRE DE DESCENTE



1: **Altitude minimum pour effectuer la procédure d'approche : *2500'***

2: **Point RNAV identifié par ses 5 lettres : DUXNO**

3: **Cap entrant (Inbound) de la piste pour effectuer la procédure d'approche :** *299°*

4: **Altitude Minimum de Segment (SMA - Segment Minimum Altiude) :** *1680’ durant 5NM* **(Altitude permettant d'avoir une marge de franchissement d'obstacles)**

5: **Point de mise en descente :** *à 6.6NM de a balise SBA*

6: **Triangle transparent représentant le Glide.**

7: **Point de remise de gaz en cas d'approche interrompue :** *Dans le cas d'une procédure d'approche LOC*

8: **Point de remise de gaz en cas d'approche interrompue : *Dans le cas d'une procédure d'approche ILS***

9: **Hauteur de franchissement du seuil associée à la pente de descente ou à l'angle de descente verticale : *36'***

10: **Altitude de la piste : *43'***

11: Tableau de conversion pour connaître votre taux de descente en fonction de votre vitesse sol en cas de panne du Glide. (Exemple : Si vous volez à 120 kt, vous devrez descendre à un taux de 743 pieds par minutes, soit à un angle de 3.50°) Rappel du Point d'Approche interrompue (MAP - Missed Approach Point) : *0.7NM de la balise SBA*

12: **Types de balisages disponibles:**

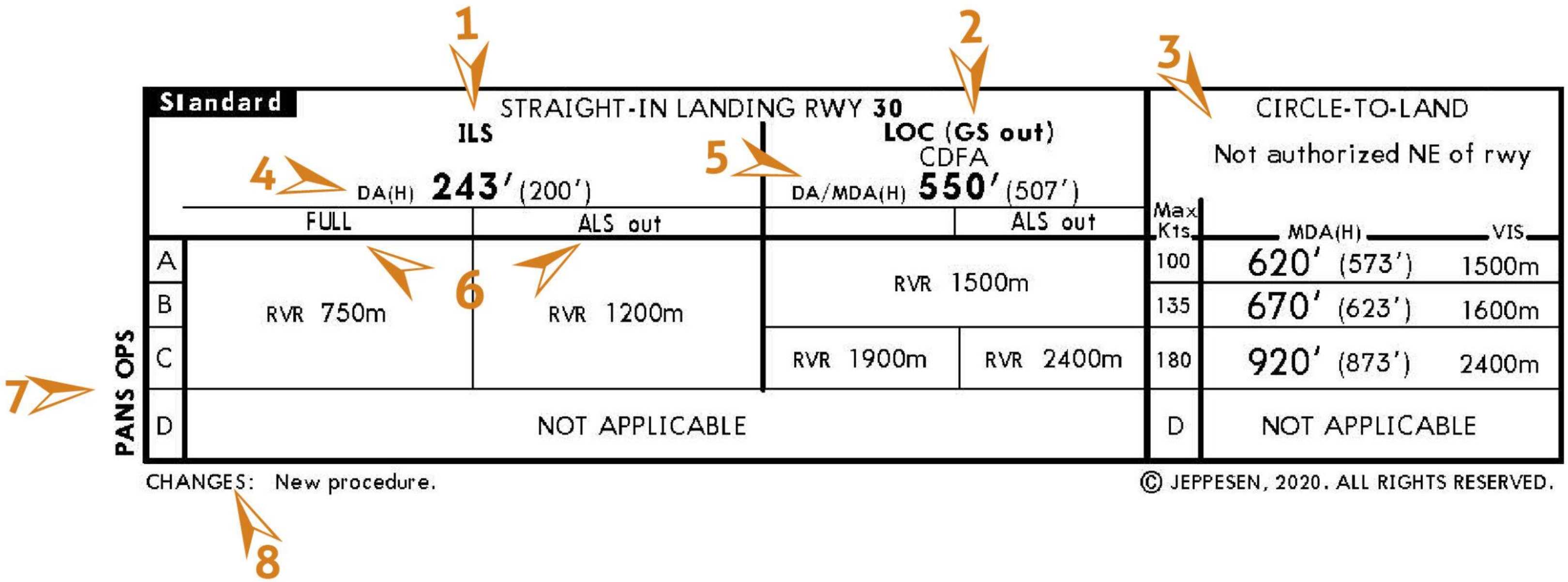
HIALS pour High Intensity Approach Landing System

REIL pour Runway End Identifier Lights

PAPI pour Precision Approach Path Indicator

13: **Rappel de la première étape à réaliser en cas de remise de gaz : Monter tout droit jusqu'a 5.2NM de la balise SBA**

LES MINIMAS OPÉRATIONNELS



1: **Minimas pour une procédure : ILS**

2: **Minimas pour une procédure : LOC**
(Ou dans le cas de panne du Glide pour une procédure ILS)

3: **Procédure de manœuvre à vue pour se poser dans le sens opposé de la piste. Le minimas sont exprimés en MDA/H (Minimum Descent Altitude/Height) et VIS (Visibility). En fonction de la catégorie de l'avion il y a des vitesses maximum à respecter. (Manœuvre interdite au Nord Est de la piste)**

4: **Pour la procédure ILS les minimas sont exprimés en DA (Descision Altitude),H (Height) et en RVR (Runway Visual Range). (Ici la DA est de 243'et la H de 200')**

5: Pour la procédure LOC les minimas sont exprimés en **DA** (Descision Altitude), H (Hight) et en **RVR** (Runway Visual Range) si l'approche est effectuée avec la technique dite de la descente continue **CDFA** (Continuous Descent Final Approach). Ou les minimas sont MDA/H (Minimum Descent Altitude/Height) et RVR (Runway Visual Range) (Ici la DA est de **550’** et la H de **507’**)

6: **Il y a deux types de RVR (Runway Visual Rage) :**

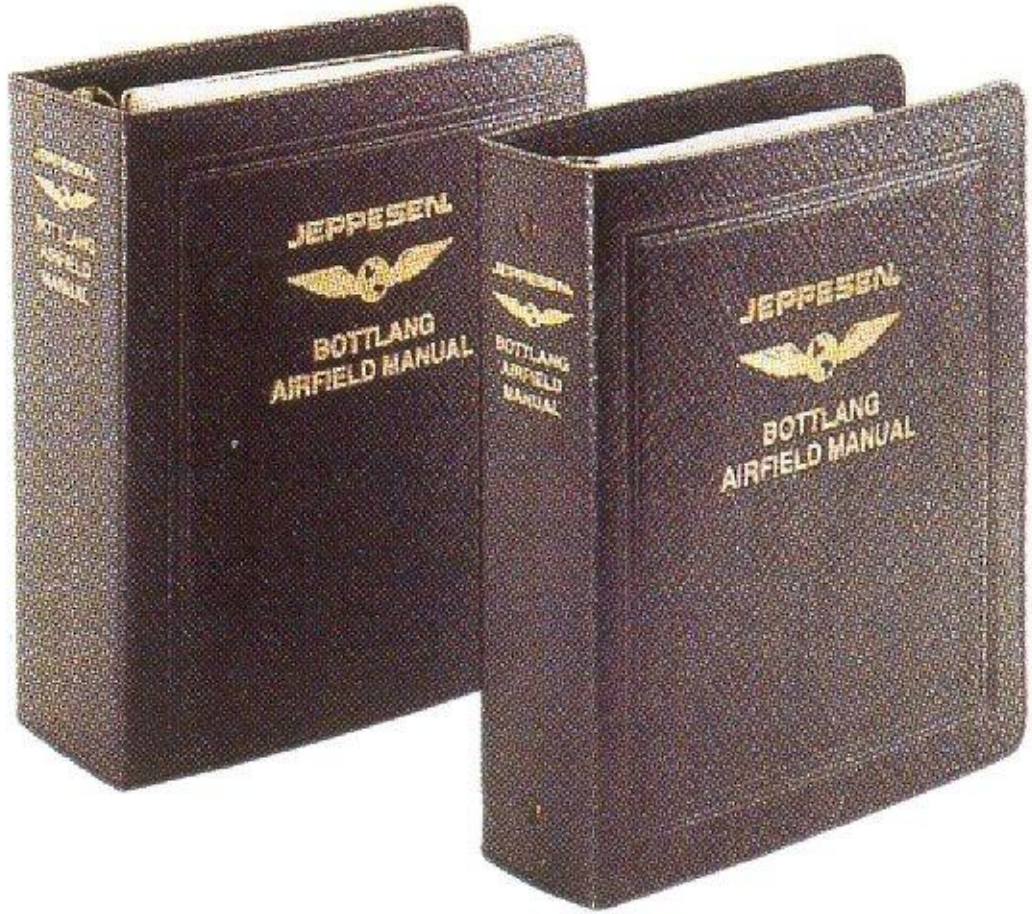
Le premier appelé FULL (Tous les systemes lumineux de la piste sont opérants)

Le second appelé ALS out (Approach Light System signifiant que la rampe d'approche est hors service)

7: **PANS-OPS : Procedures for Air Navigation Services – Aircraft OPerationS (Informe sur le type de données)**

8: **Changes : Quels ont été les changements effectués depuis la version précédente du document (Ici on nous informe que la procédure d'approche a été modifiée)**

JEPPESEN

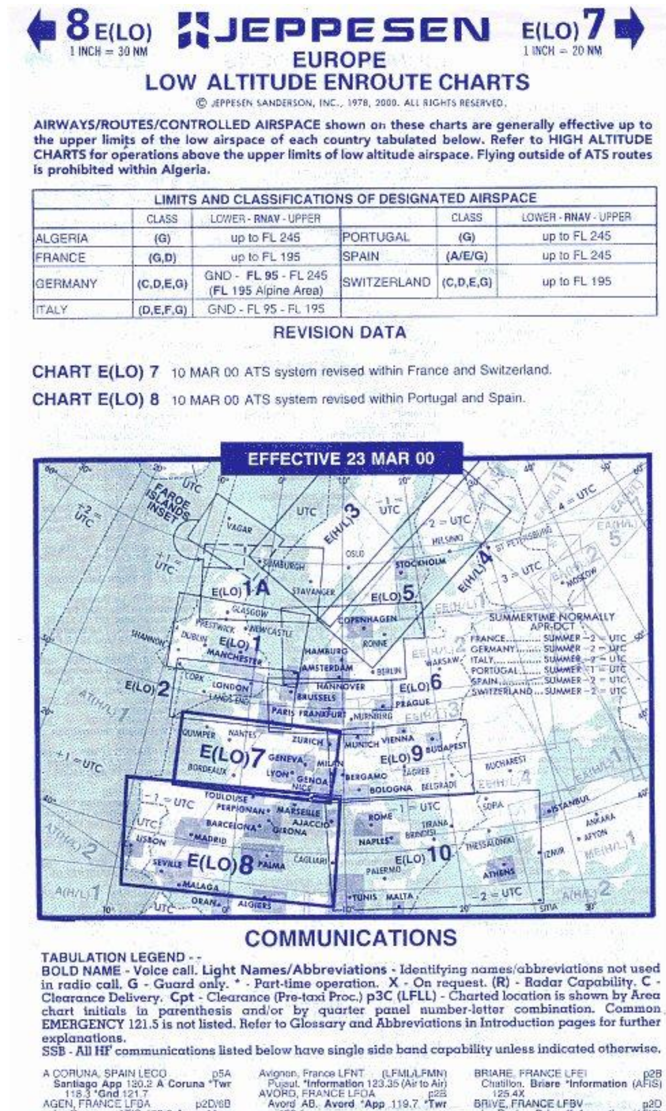

JEPPESEN
BOTTLANG
AIRFIELD MANUAL
JEPPESEN
BOTTLANG
AIRFIELD
MANUAL
JEPPESEN
BOTTLANG
AIRFIELD MANUAL
JEPPESEN
BOTTLANG
AIRFIELD
MANUAL

PRESENTATION

2 manuals:

1. First manual: you can find EN ROUTE charts,and different information about Meteorology, the airport, the radio aids,etc
2. Second manual: You can find TERMINAL CHARTS (departure, arrival,different procedures…)

DÉSIGNATION ET PRÉSENTATION DES CARTES



8 E(LO) 1 INCH = 30 NM — JEPPESEN — E(LO) 7 1 INCH = 20 NM

EUROPE
LOW ALTITUDE ENROUTE CHARTS

AIRWAYS/ROUTES/CONTROLLED AIRSPACE shown on these charts are generally effective up to the upper limits of the low airspace of each country tabulated below. Refer to HIGH ALTITUDE CHARTS for operations above the upper limits of low altitude airspace. Flying outside of ATS routes is prohibited within Algeria.

LIMITS AND CLASSIFICATIONS OF DESIGNATED AIRSPACE					
	CLASS	LOWER - RNAV - UPPER		CLASS	LOWER - RNAV - UPPER
ALGERIA	(G)	up to FL 245	PORTUGAL	(G)	up to FL 245
FRANCE	(G,D)	up to FL 195	SPAIN	(A/E/G)	up to FL 245
GERMANY	(C,D,E,G)	GND - FL 95 - FL 245 (FL 195 Alpine Area)	SWITZERLAND	(C,D,E,G)	up to FL 195
ITALY	(D,E,F,G)	GND - FL 95 - FL 195			

REVISION DATA

CHART E(LO) 7 10 MAR 00 ATS system revised within France and Switzerland.

CHART E(LO) 8 10 MAR 00 ATS system revised within Portugal and Spain.

EFFECTIVE 23 MAR 00

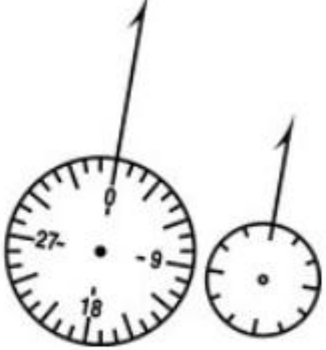
COMMUNICATIONS

TABULATION LEGEND - -
BOLD NAME - Voice call. **Light Names/Abbreviations** - Identifying names/abbreviations not used in radio call. **G** - Guard only. * - Part-time operation. **X** - On request. **(R)** - Radar Capability. **C** - Clearance Delivery. **Cpt** - Clearance (Pre-taxi Proc.) **p3C (LFLL)** - Charted location is shown by Area chart initials in parenthesis and/or by quarter panel number-letter combination. Common EMERGENCY 121.5 is not listed. Refer to Glossary and Abbreviations in Introduction pages for further explanations.
SSB - All HF communications listed below have single side band capability unless indicated otherwise.

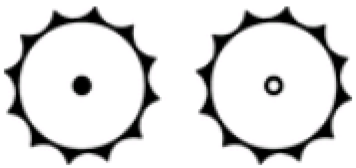
A CORUNA, SPAIN LECO p5A
Santiago App 120.2 **A Coruna** *Twr 118.3 *Gnd 121.7
AGEN, FRANCE LFBA p2D/6B
La Garenne. ATIS 129.6 **Agen** *App

Avignon, France LFNT (LFML/LFMN)
Pujaut. *Information 123.35 (Air to Air)
AVORD, FRANCE LFOA p2B
Avord AB. **Avord** *App 119.7 *Twr 122.1

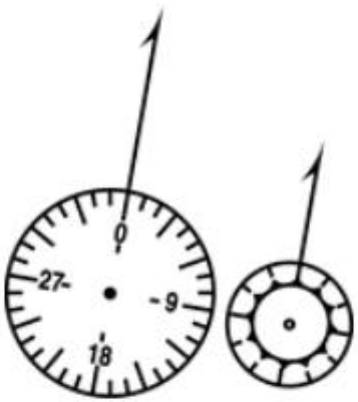
BRIARE, FRANCE LFEI p2B
Chatillon. **Briare** *Information (AFIS) 125.4X
BRIVE, FRANCE LFBV p2D
La Roche. **Brive** *Information (AFIS)



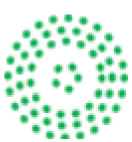
VOR (VHF Omnidirectional Range)



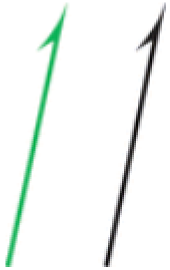
TACAN (Tactical Air Navigation) or DME (Distance Measuring Equipment)



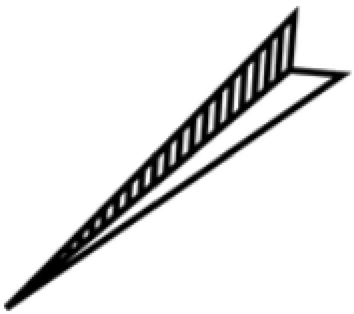
VORTAC/VORDME



NDB (Nondirectional Radio Beacon)



Magnetic north ticks on navigational facilities fit
compass roses on IFR Enroute Chart



LOC,

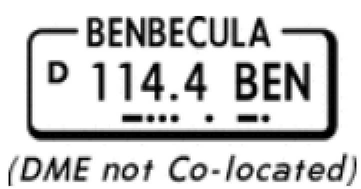
NAVAID IDENTIFICATION

Navaid identification is given in shadow box when navaid is airway or route component, with frequency, identifier, and Morse Code. DME capability is indicated by a small “D” preceding the VOR frequency at frequency paired navaids. VOR and VORTAC navaid operational ranges are identified (when known) within the navaid

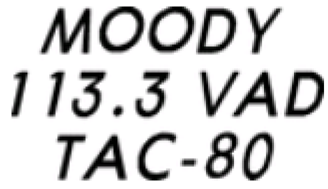
Armed Forces Radio Station

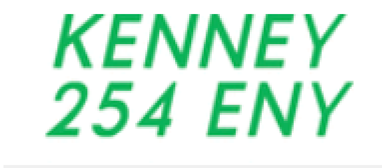
Fréquences: VOR, ADF + coordonnées
Indicatif et indicatif morse

When VOR and TAC/DME antennas are not co-located, a notation “DME not Co-located” is shown below the navaid box.



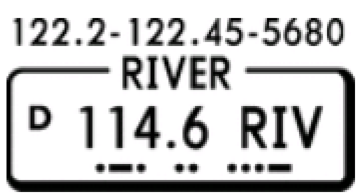
Off-airway navaids are unboxed on Low and High/Low charts. TACAN/DME channel is shown when VOR navaid has a frequency paired DME capability. When an L/MF navaid performs an enroute function, the Morse Code of its identification letters are shown



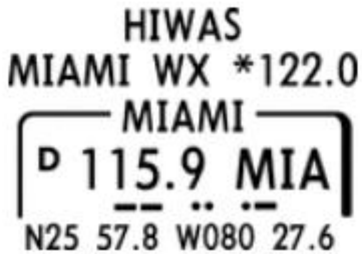


Off-airway navaids are unboxed on Low and High/Low charts. TACAN/DME channel is shown when VOR navaid has a frequency paired DME capability. When an L/MF navaid performs an enroute function, the Morse Code of its identification letters are shown.

When TACAN or DME are not frequency paired with the VOR, the TACAN is identified separately. The “Ghost” VOR frequency, shown in parentheses, enables civilian tuning of DME facility.



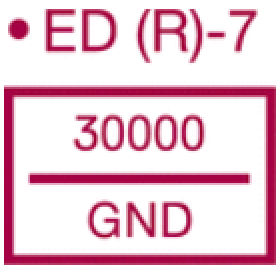
River Radio transmits on 114.6 and transmits and receives on 122.2, 122.45 MHz and HF frequency 5680.



HIWAS — Service d'avis météorologiques dangereux en vol. Diffusions SIGMETS, AIRMETS en continu sur la fréquence VOR.

Restricted airspace. The accompanying label indicates it as prohibited, restricted, danger, etc.

Training, Alert, Caution, and Military Operations Areas.



Dot indicates permanent activation on some chart series.

RESTRICTED AIRSPACE DESIGNATION

A:	**Alert**
T:	**Training**
C:	**Caution**
W:	**Warning**
D:	**Danger**
TRA:	**Temporary Reserved Airspace**
P:	**Prohibited**
TSA :	**Temporary Segregated Area**
R:	**Restricted**
MOA:	**Military Operations Area**

AIRPORTS

(LAA) : LAA Local Airport Advisory
(AFIS : AFIS (Aerodrome Flight Information Service)
(ALA) : Authorized Landing Area

AIRWAY AND ROUTE COMPONENTS
AIRWAY AND ROUTES CENTER LINES

Airway/Route

Diversionary Route

Overlying High Altitude Airway/Route

OTR 12

Oceanic Transition Route

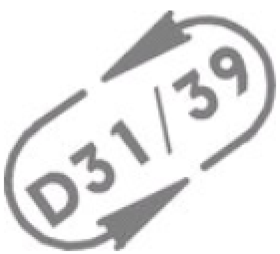
RNAV Airway/Route

Compulsory Reporting Point

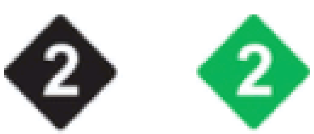
Non-Compulsory Reporting Point

Low Altitude Compulsory Reporting Point

Low Altitude non- Compulsory Reporting Point



Holding Pattern. DME figures, when provided, give the DME distance of the fix as the first figure followed by the outbound limit as the second figure.

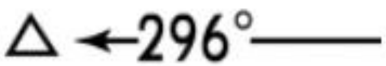


Length of holding pattern in minutes when other than standard

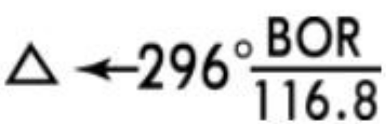
LIMON
V-8 7500 NW
(MRA 7000)

Fix name with Minimum Crossing Altitude (MCA) showing airway, altitude, and direction, and Minimum Reception Altitude (MRA).

LF bearings forming a fix are to the navaid



VHF radials forming a fix are from the navaid.



VHF frequency and identifier included when off chart or remoted.

LF frequency, identifier and Morse Code included when off chart or remoted.

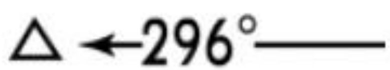
Arrow along airway points from the navaid designating the reporting point.

Fix formed by 55 DME from MAZ navaid.

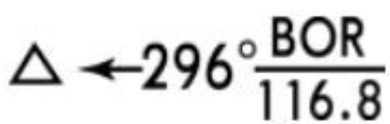
LIMON
V-8 7500 NW
(MRA 7000)

Fix name with Minimum Crossing Altitude (MCA) showing airway, altitude, and direction, and Minimum Reception Altitude (MRA).

LF bearings forming a fix are to the navaid



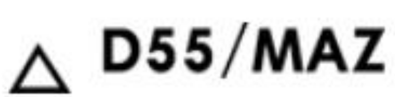
VHF radials forming a fix are from the navaid.



VHF frequency and identifier included when off chart or remoted.

LF frequency, identifier and Morse Code included when off chart or remoted.

Arrow along airway points from the navaid designating the reporting point.



Fix formed by 55 DME from MAZ navaid.

V 168 — Airway and route designators. Negative (white letters in black) designators are used for distinction.

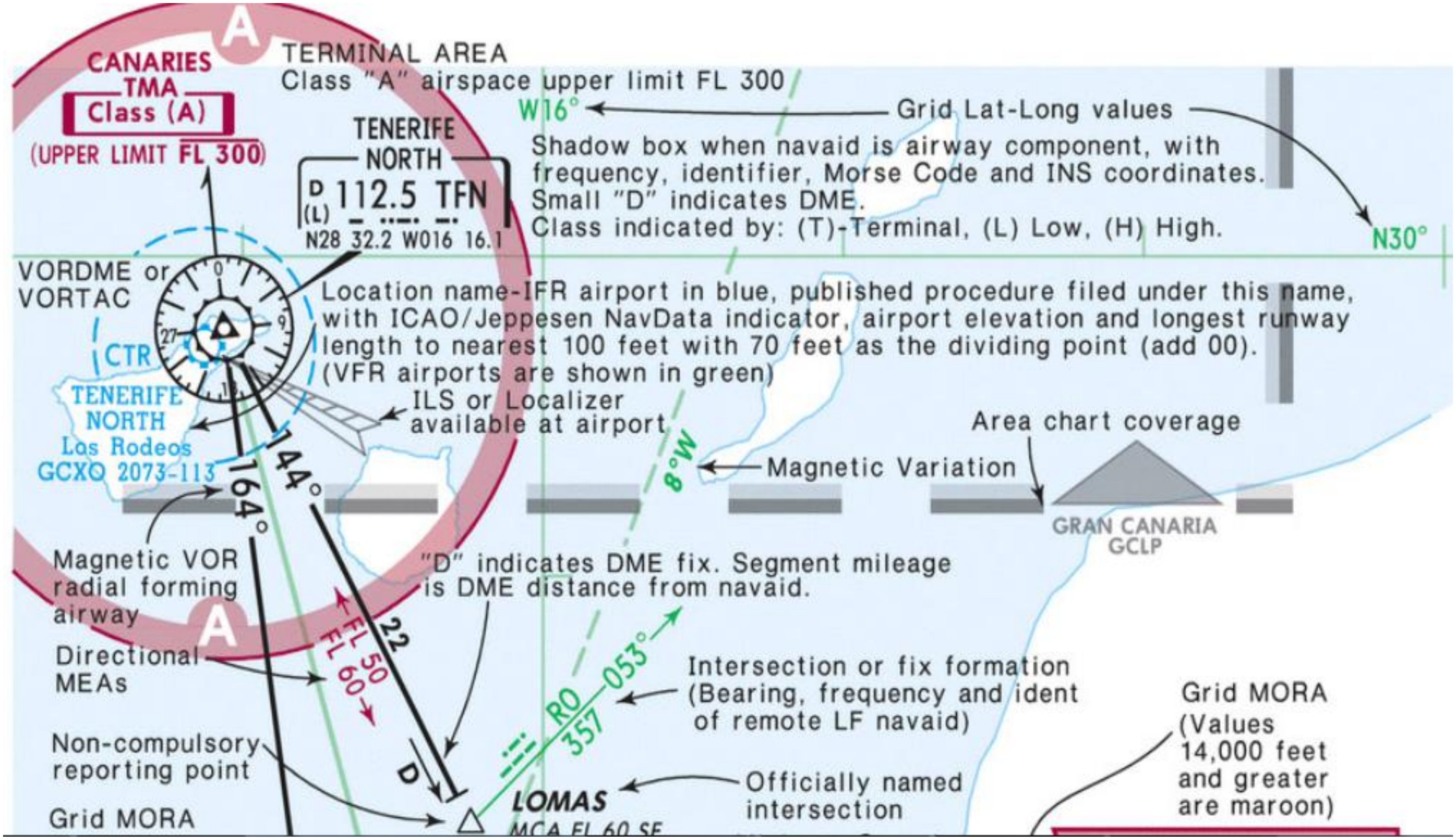
ATS — ATS-Designated route without published identifier

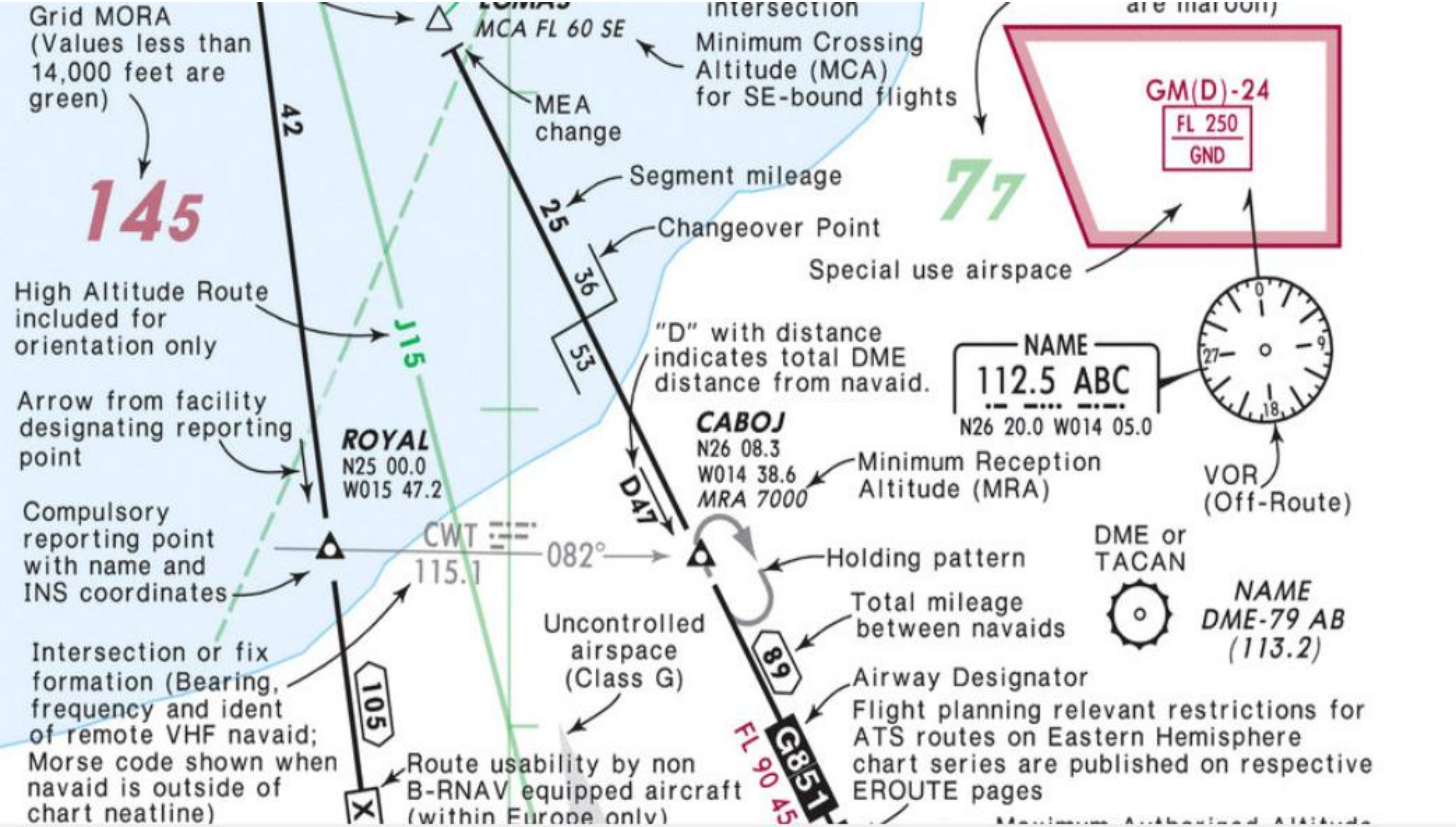
AWY 4 — AWY-Airway

BR 7 — BR-Bahama Route, Canada Bravo Route

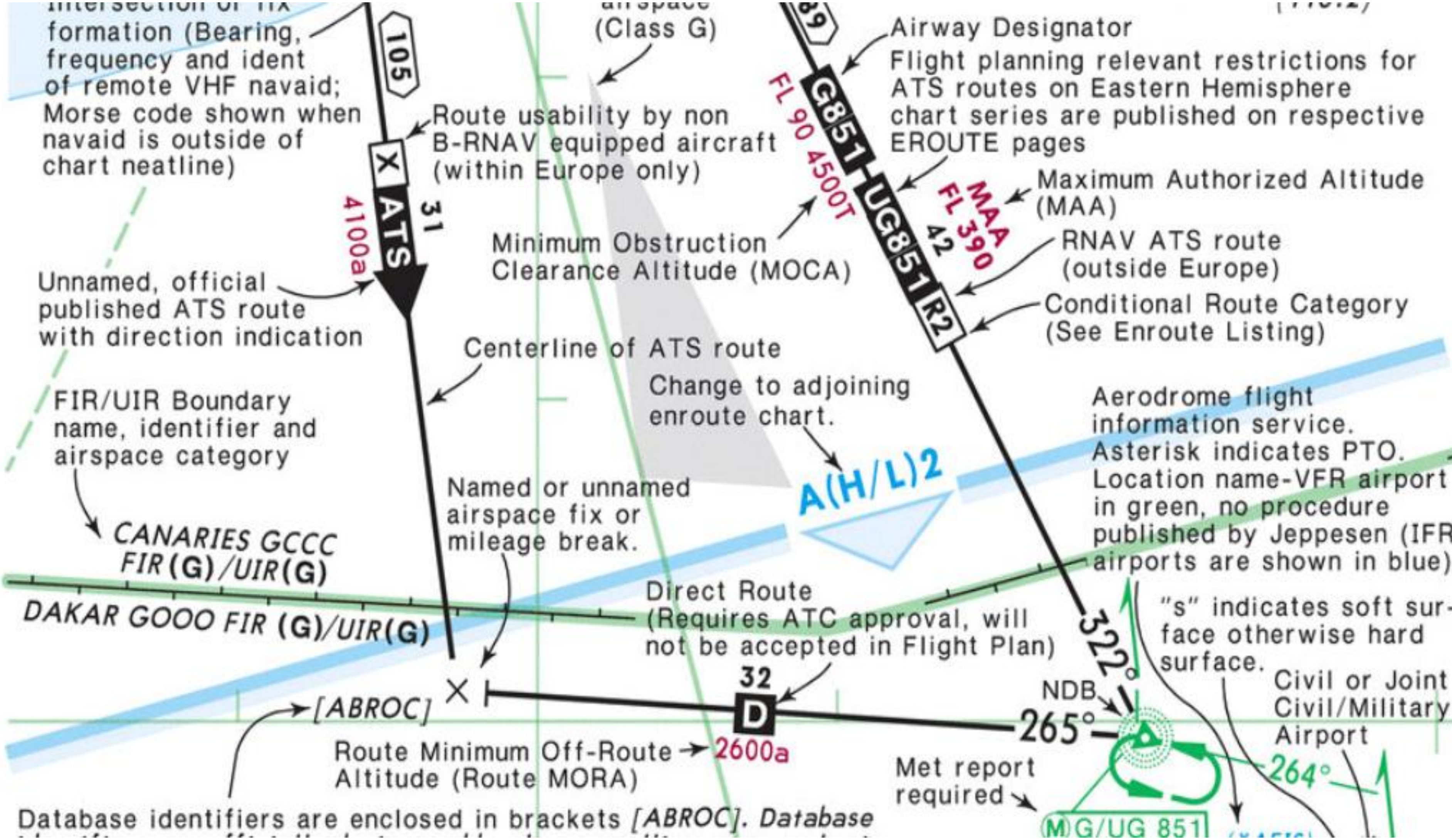
D — Direct Route

DOM — DOM-Domestic Route. Use by foreign operators requires special authorization.

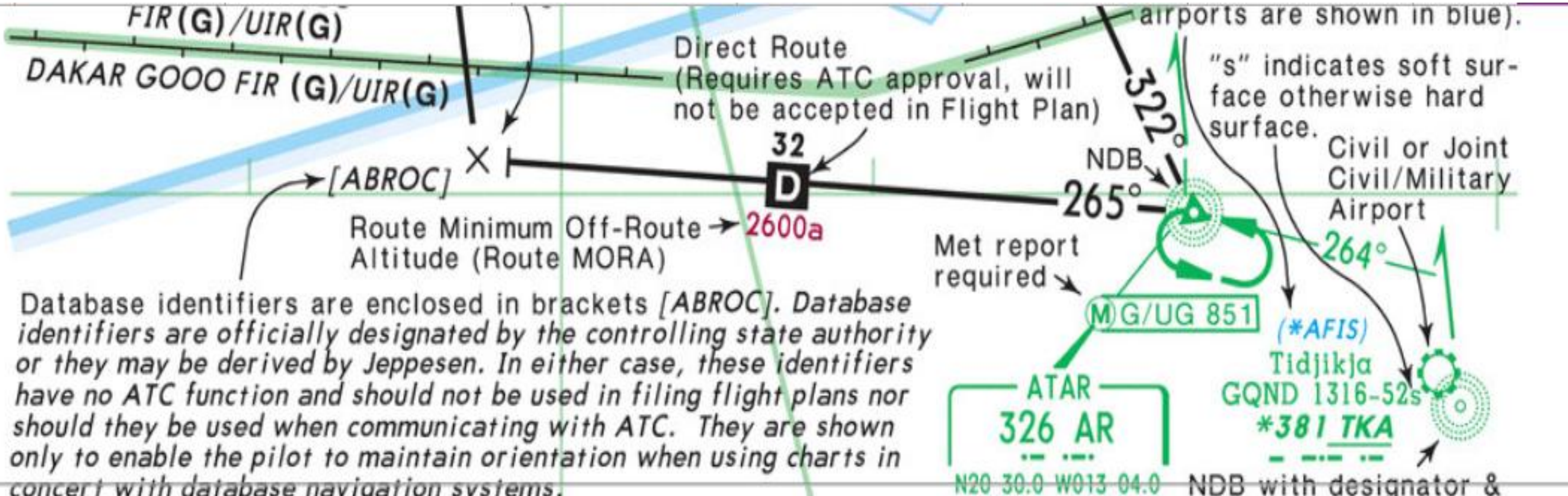

CANARIES
TMA
Class (A)
(UPPER LIMIT FL 300)
TERMINAL AREA
Class "A" airspace upper limit FL 300
W16°
Grid Lat-Long values
TENERIFE
NORTH
D 112.5 TFN
(L)
N28 32.2 W016 16.1
Shadow box when navaid is airway component, with frequency, identifier, Morse Code and INS coordinates.
Small "D" indicates DME.
Class indicated by: (T)-Terminal, (L) Low, (H) High.
N30°
VORDME or
VORTAC
Location name-IFR airport in blue, published procedure filed under this name, with ICAO/Jeppesen NavData indicator, airport elevation and longest runway length to nearest 100 feet with 70 feet as the dividing point (add 00).
(VFR airports are shown in green)
CTR
TENERIFE
NORTH
Los Rodeos
GCXO 2073-113
ILS or Localizer
available at airport
Area chart coverage
8°W
Magnetic Variation
144°
164°
GRAN CANARIA
GCLP
Magnetic VOR
radial forming
airway
"D" indicates DME fix. Segment mileage is DME distance from navaid.
22
FL 50
FL 60
Directional
MEAs
RO 053°
357
Intersection or fix formation
(Bearing, frequency and ident
of remote LF navaid)
Grid MORA
(Values
14,000 feet
and greater
are maroon)
Non-compulsory
reporting point
D
LOMAS
Officially named
intersection
Grid MORA

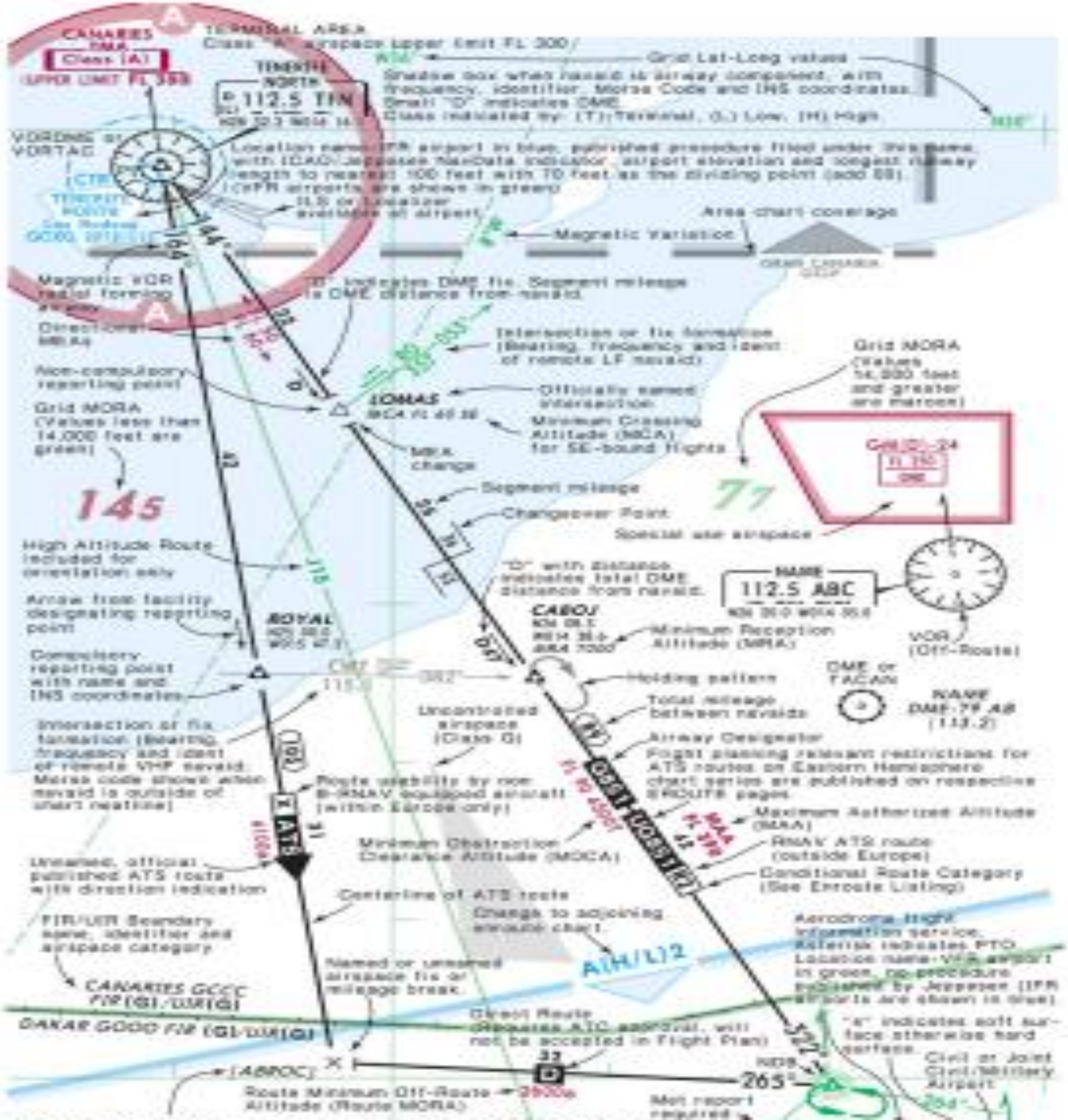


Grid MORA
(Values less than
14,000 feet are
green)
145
42
MCA FL 60 SE
Minimum Crossing
Altitude (MCA)
for SE-bound flights
MEA
change
Segment mileage
25
Changeover Point
36
53
77
GM(D)-24
FL 250
GND
Special use airspace
High Altitude Route
included for
orientation only
J15
"D" with distance
indicates total DME
distance from navaid.
NAME
112.5 ABC
N26 20.0 W014 05.0
Arrow from facility
designating reporting
point
ROYAL
N25 00.0
W015 47.2
CABOJ
N26 08.3
W014 38.6
MRA 7000
D47
Minimum Reception
Altitude (MRA)
VOR
(Off-Route)
Compulsory
reporting point
with name and
INS coordinates
CWT
115.1
082°
Holding pattern
DME or
TACAN
NAME
DME-79 AB
(113.2)
Total mileage
between navaids
89
Uncontrolled
airspace
(Class G)
Intersection or fix
formation (Bearing,
frequency and ident
of remote VHF navaid;
Morse code shown when
navaid is outside of
chart neatline)
105
Airway Designator
G851
FL 90
Flight planning relevant restrictions for
ATS routes on Eastern Hemisphere
chart series are published on respective
EROUTE pages
Route usability by non
B-RNAV equipped aircraft
X

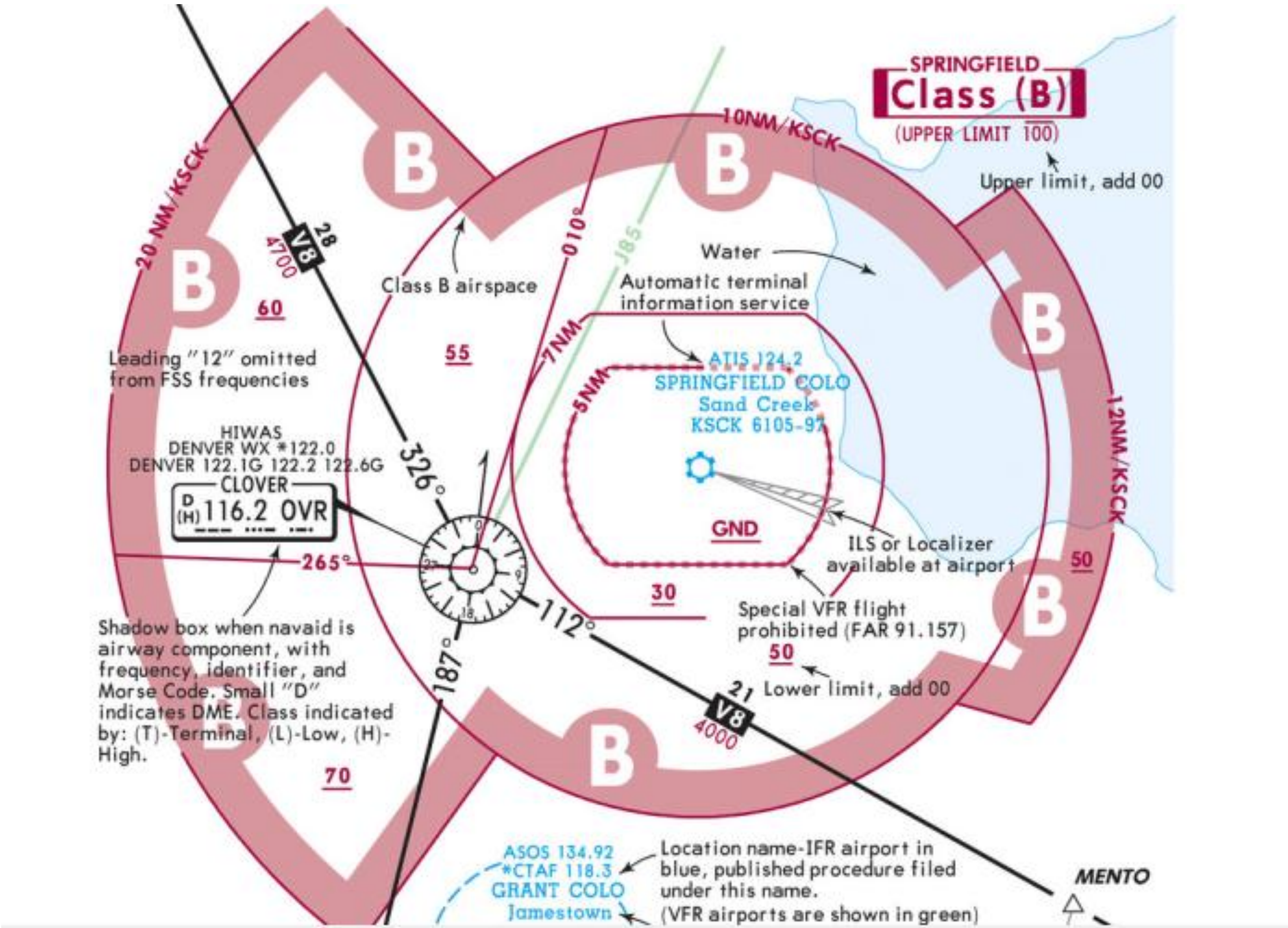


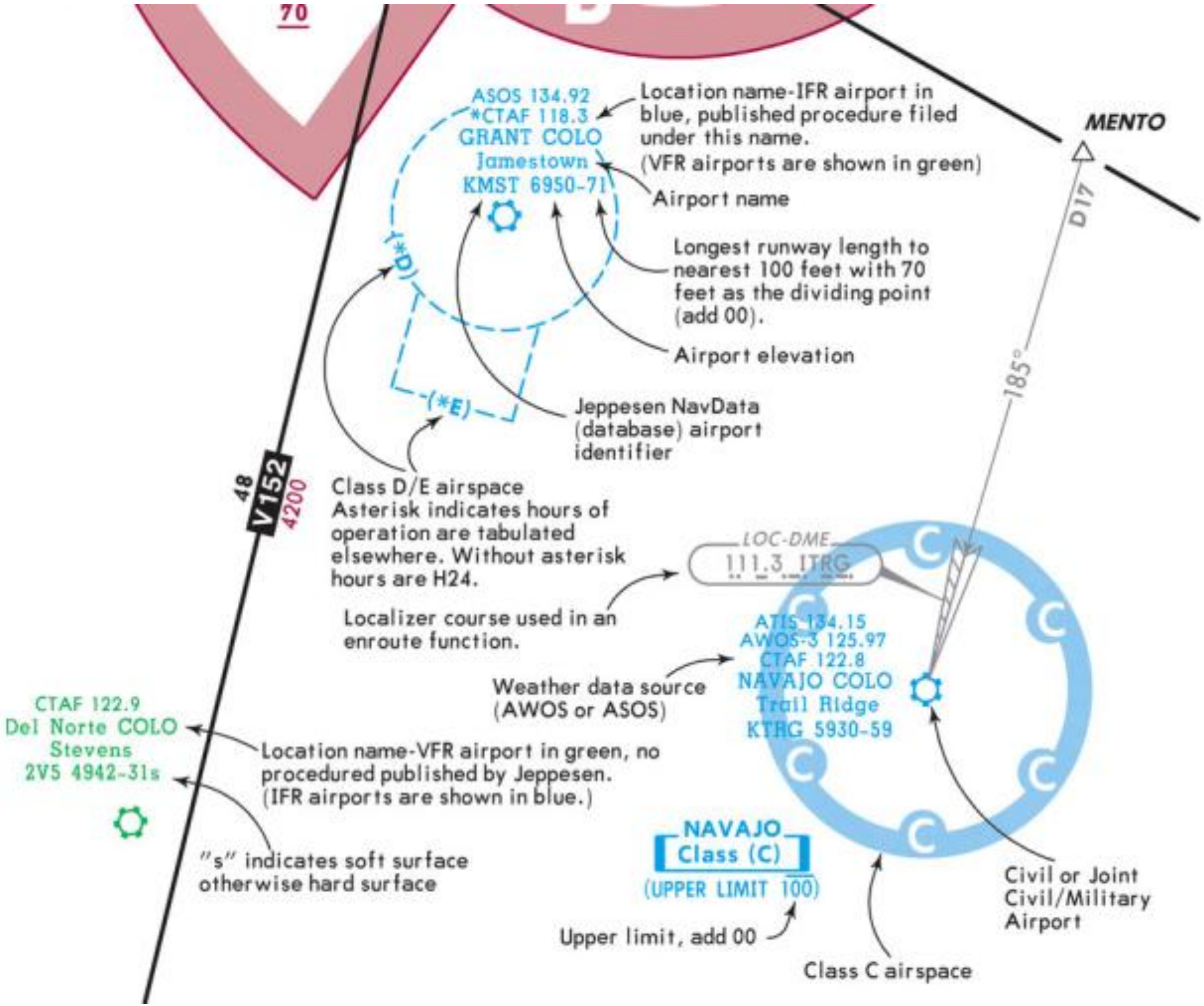
formation (Bearing,
frequency and ident
of remote VHF navaid;
Morse code shown when
navaid is outside of
chart neatline)
105
(Class G)
Airway Designator
Flight planning relevant restrictions for
ATS routes on Eastern Hemisphere
chart series are published on respective
EROUTE pages
Route usability by non
B-RNAV equipped aircraft
(within Europe only)
X
ATS
31
4100a
FL 90 4500T
G851
UG851
42
MAA
FL 390
R2
Maximum Authorized Altitude
(MAA)
Minimum Obstruction
Clearance Altitude (MOCA)
RNAV ATS route
(outside Europe)
Conditional Route Category
(See Enroute Listing)
Unnamed, official
published ATS route
with direction indication
Centerline of ATS route
Change to adjoining
enroute chart.
A(H/L)2
FIR/UIR Boundary
name, identifier and
airspace category
Aerodrome flight
information service.
Asterisk indicates PTO.
Location name-VFR airport
in green, no procedure
published by Jeppesen (IFR
airports are shown in blue).
Named or unnamed
airspace fix or
mileage break.
CANARIES GCCC
FIR (G)/UIR (G)
DAKAR GOOO FIR (G)/UIR (G)
Direct Route
(Requires ATC approval, will
not be accepted in Flight Plan)
322°
"s" indicates soft sur-
face otherwise hard
surface.
32
D
[ABROC]
NDB
265°
Civil or Joint
Civil/Military
Airport
Route Minimum Off-Route
Altitude (Route MORA)
2600a
264°
Met report
required
M G/UG 851
Database identifiers are enclosed in brackets [ABROC]. Database


FIR (G)/UIR(G)
DAKAR GOOO FIR (G)/UIR(G)
airports are shown in blue).
Direct Route (Requires ATC approval, will not be accepted in Flight Plan)
"s" indicates soft surface otherwise hard surface.
Civil or Joint Civil/Military Airport
322°
32
D
NDB
265°
264°
[ABROC]
Route Minimum Off-Route Altitude (Route MORA)
2600a
Met report required
M G/UG 851
(*AFIS)
Tidjikja
GQND 1316-52s
*381 TKA
ATAR
326 AR
N20 30.0 W013 04.0
Database identifiers are enclosed in brackets [ABROC]. Database identifiers are officially designated by the controlling state authority or they may be derived by Jeppesen. In either case, these identifiers have no ATC function and should not be used in filing flight plans nor should they be used when communicating with ATC. They are shown only to enable the pilot to maintain orientation when using charts in concert with database navigation systems.
NDB with designator & Morse code (Of
JEPPESEN

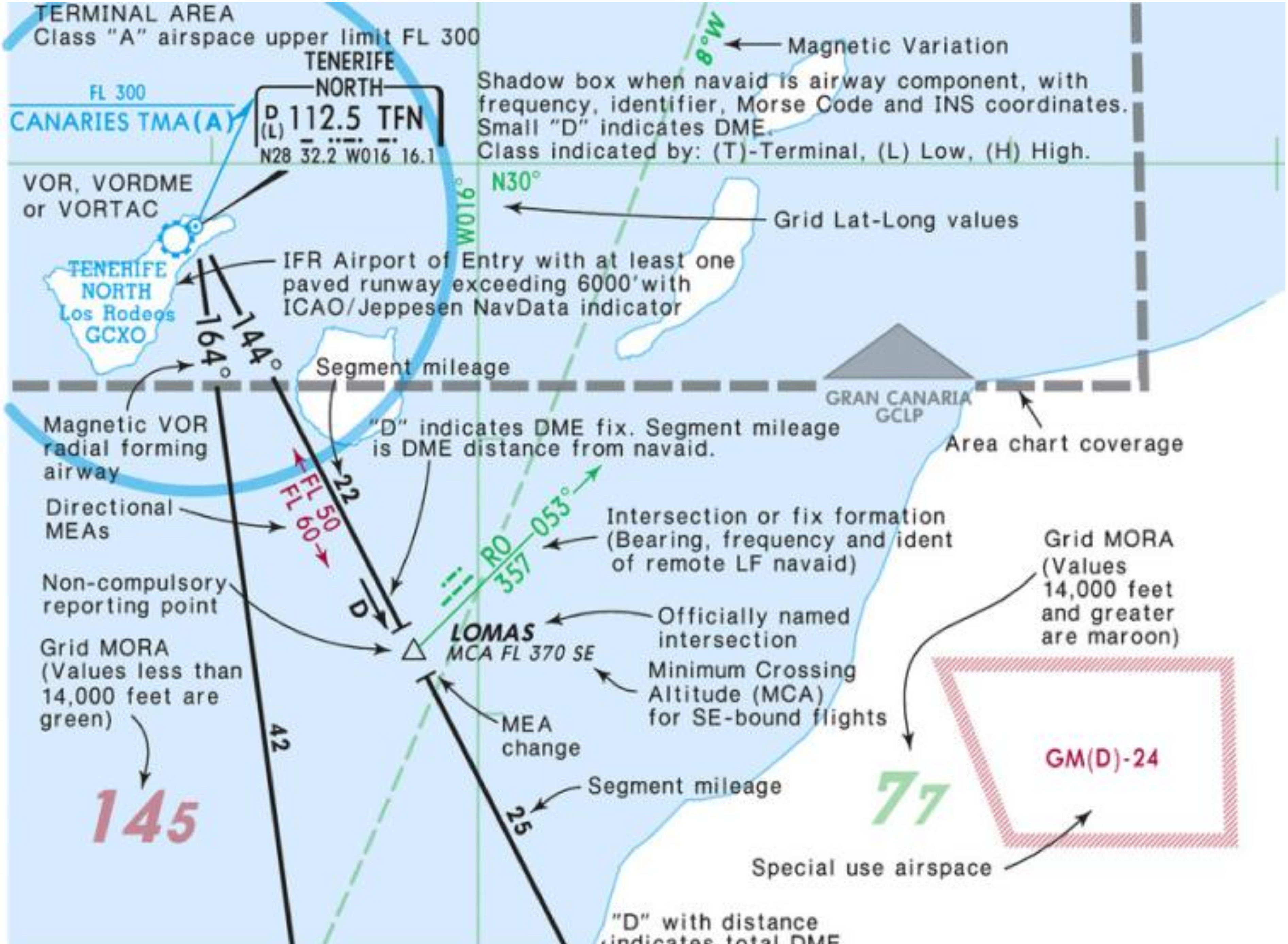


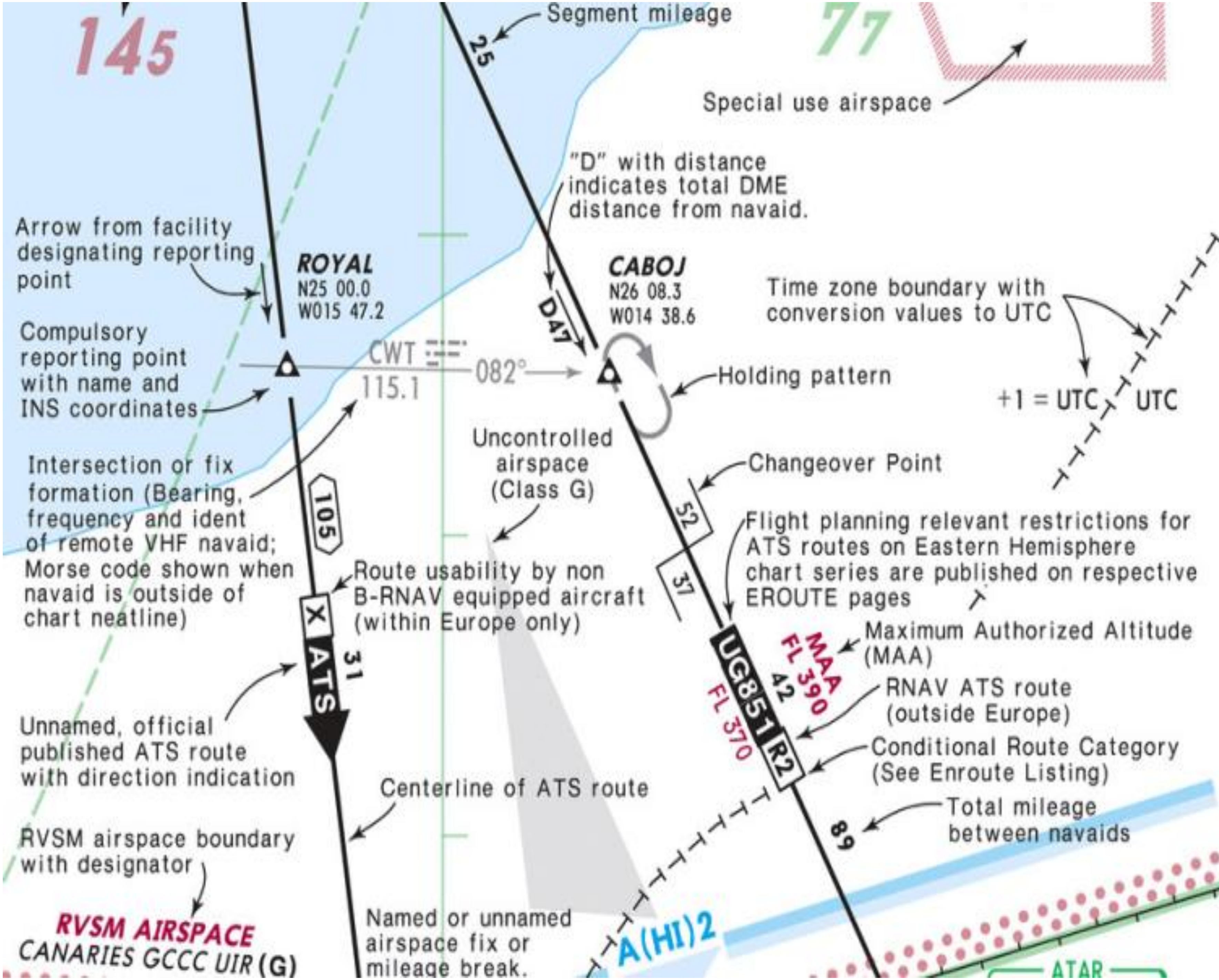
TERMINAL AREA
Grid Lat-Long values
Shadow box when navaid is airway component, with frequency, identifier, Morse Code and INS coordinates.
Small "D" indicates DME.
Class indicated by: (T) Terminal, (L) Low, (H) High
112.5 TFN
VOR/DME or VORTAC
ILS or Localizer available at airport
Magnetic Variation
Area chart coverage
Magnetic VOR radial forming airway
"D" indicates DME fix. Segment mileage is DME distance from navaid.
Directional MEAs
Intersection or fix formation (Bearing, frequency and ident of remote LF navaid)
Grid MORA
Non-compulsory reporting point
Officially named intersection
LOMAS
Minimum Crossing Altitude (MCA) for SE-bound flights
MEA change
Segment mileage
Changeover Point
145
77
Special use airspace
High Altitude Route included for orientation only
"D" with distance indicates total DME distance from navaid.
112.5 ABC
Arrow from facility designating reporting point
ROYAL
CABOJ
Minimum Reception Altitude (MRA)
VOR (Off-Route)
Compulsory reporting point with name and INS coordinates
Holding pattern
DME or TACAN
Total mileage between navaids
Uncontrolled airspace (Class G)
Airway Designator
Route usability by non B-RNAV equipped aircraft (within Europe only)
Maximum Authorized Altitude (MAA)
RNAV ATS route (outside Europe)
Minimum Obstruction Clearance Altitude (MOCA)
Conditional Route Category (See Enroute Listing)
Unnamed, official published ATS route with direction indication
Centerline of ATS route
Change to adjoining enroute chart
A(H/L)2
FIR/UIR Boundary name, identifier and airspace category
CANARIES GCCC FIR(G)/UIR(G)
DAKAR GOOO FIR (G)/UIR(G)
Named or unnamed airspace fix or mileage break
Direct Route requires ATC approval, will not be accepted in Flight Plan
[ABROC]
265°
Route Minimum Off-Route Altitude (Route MORA)
Met report required
Civil or Joint Civil/Military Airport


SPRINGFIELD
Class (B)
(UPPER LIMIT 100)
Upper limit, add 00
10NM/KSCK
20 NM/KSCK
12NM/KSCK
B
28
V8
4700
60
55
010°
J85
7NM
5NM
Class B airspace
Water
Automatic terminal information service
ATIS 124.2
SPRINGFIELD COLO
Sand Creek
KSCK 6105-97
Leading "12" omitted from FSS frequencies
HIWAS
DENVER WX *122.0
DENVER 122.1G 122.2 122.6G
CLOVER
D (H) 116.2 OVR
326°
265°
112°
187°
GND
ILS or Localizer available at airport
30
50
Special VFR flight prohibited (FAR 91.157)
50
Lower limit, add 00
21
V8
4000
Shadow box when navaid is airway component, with frequency, identifier, and Morse Code. Small "D" indicates DME. Class indicated by: (T)-Terminal, (L)-Low, (H)-High.
70
ASOS 134.92
*CTAF 118.3
GRANT COLO
Jamestown
Location name-IFR airport in blue, published procedure filed under this name.
(VFR airports are shown in green)
MENTO

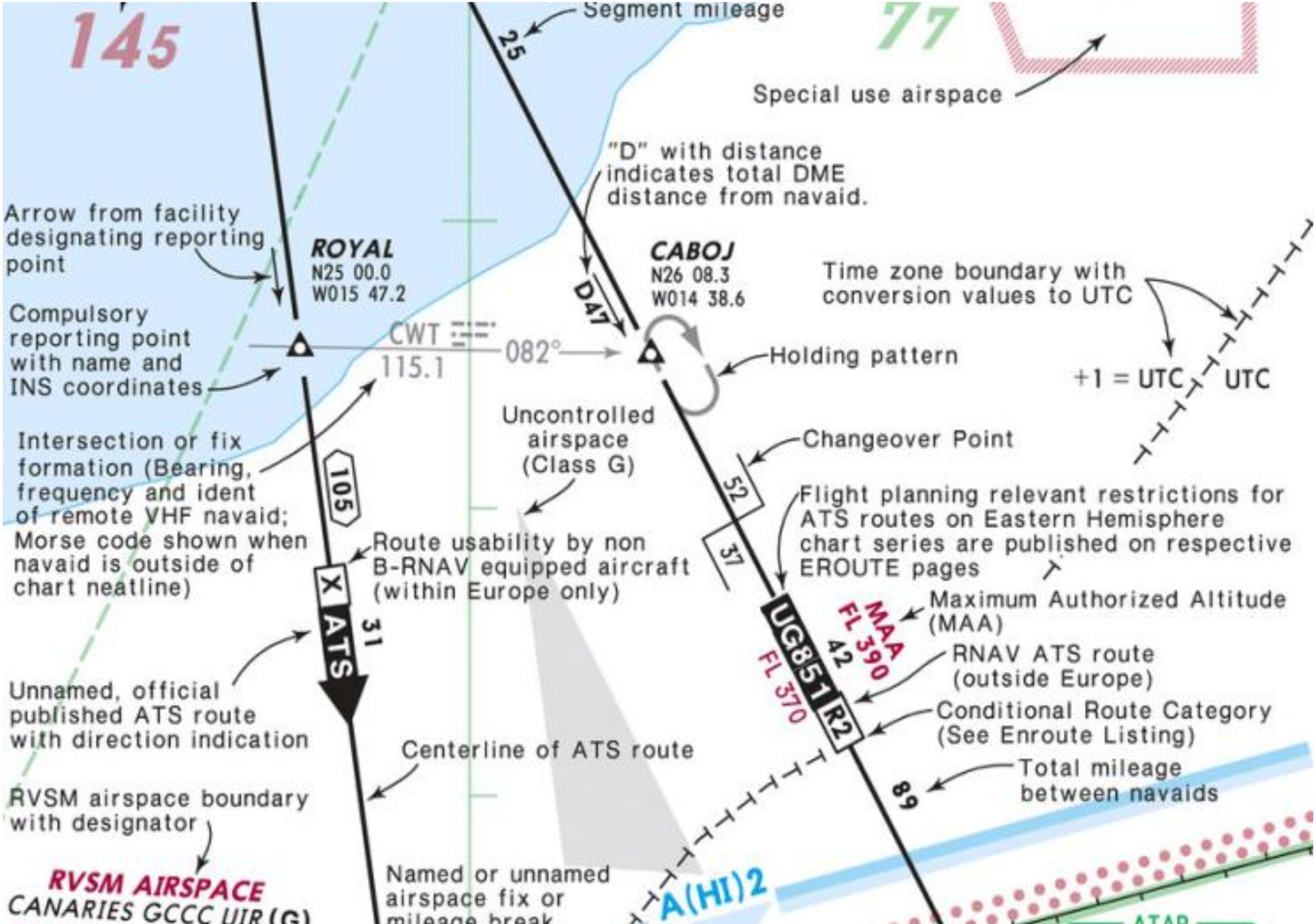


70
MENTO
D17
185°
ASOS 134.92
*CTAF 118.3
GRANT COLO
Jamestown
KMST 6950-71
(D*)
(*E)
Location name-IFR airport in blue, published procedure filed under this name.
(VFR airports are shown in green)
Airport name
Longest runway length to nearest 100 feet with 70 feet as the dividing point (add 00).
Airport elevation
Jeppesen NavData (database) airport identifier
Class D/E airspace Asterisk indicates hours of operation are tabulated elsewhere. Without asterisk hours are H24.
48
V152
4200
LOC-DME
111.3 ITRG
Localizer course used in an enroute function.
Weather data source (AWOS or ASOS)
ATIS 134.15
AWOS-3 125.97
CTAF 122.8
NAVAJO COLO
Trail Ridge
KTRG 5930-59
CTAF 122.9
Del Norte COLO
Stevens
2V5 4942-31s
Location name-VFR airport in green, no procedured published by Jeppesen.
(IFR airports are shown in blue.)
"s" indicates soft surface otherwise hard surface
NAVAJO
Class (C)
(UPPER LIMIT 100)
Upper limit, add 00
Class C airspace
Civil or Joint Civil/Military Airport


TERMINAL AREA
Class "A" airspace upper limit FL 300
FL 300
CANARIES TMA (A)
TENERIFE NORTH
D (L) 112.5 TFN
N28 32.2 W016 16.1
VOR, VORDME or VORTAC
8°W
Magnetic Variation
Shadow box when navaid is airway component, with frequency, identifier, Morse Code and INS coordinates.
Small "D" indicates DME.
Class indicated by: (T)-Terminal, (L) Low, (H) High.
W016°
N30°
Grid Lat-Long values
TENERIFE NORTH Los Rodeos GCXO
IFR Airport of Entry with at least one paved runway exceeding 6000' with ICAO/Jeppesen NavData indicator
164°
144°
Segment mileage
GRAN CANARIA GCLP
Area chart coverage
Magnetic VOR radial forming airway
"D" indicates DME fix. Segment mileage is DME distance from navaid.
22
FL 50
FL 60
Directional MEAs
RO 053°
357
Intersection or fix formation (Bearing, frequency and ident of remote LF navaid)
Grid MORA (Values 14,000 feet and greater are maroon)
Non-compulsory reporting point
D
LOMAS
MCA FL 370 SE
Officially named intersection
Grid MORA (Values less than 14,000 feet are green)
Minimum Crossing Altitude (MCA) for SE-bound flights
MEA change
42
GM(D)-24
Segment mileage
145
77
25
Special use airspace
"D" with distance indicates total DME

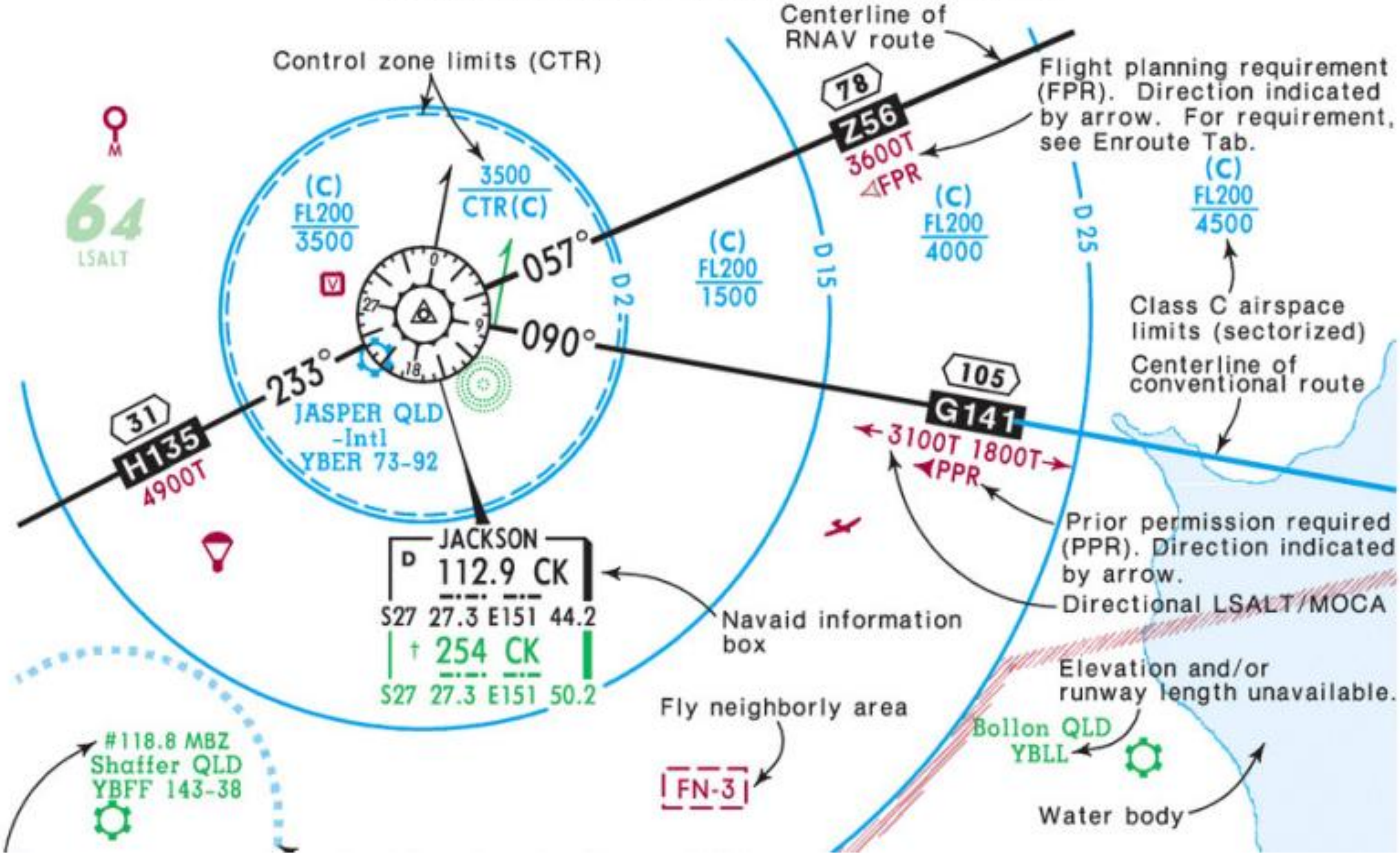


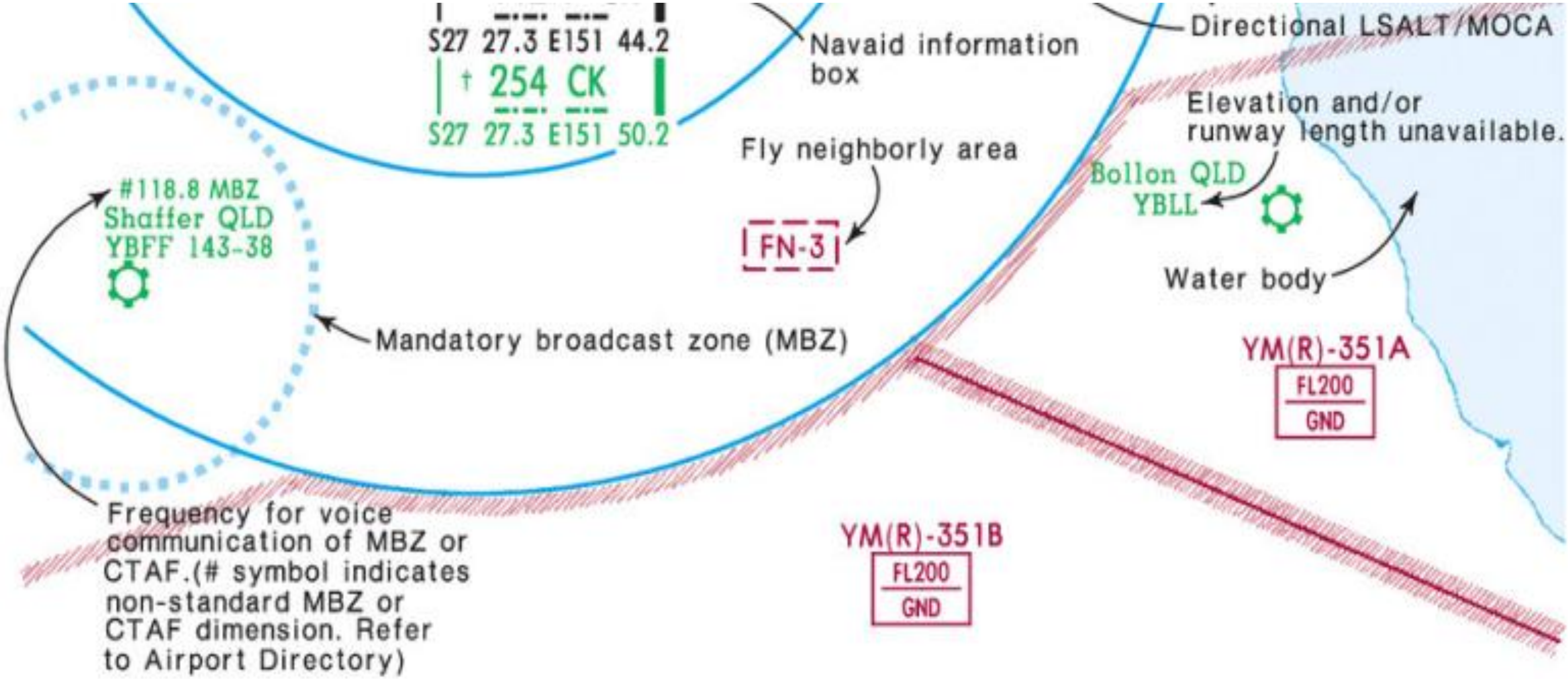
145
Segment mileage
25
77
Special use airspace
"D" with distance indicates total DME distance from navaid.
Arrow from facility designating reporting point
ROYAL
N25 00.0
W015 47.2
CABOJ
N26 08.3
W014 38.6
D47
Time zone boundary with conversion values to UTC
Compulsory reporting point with name and INS coordinates
CWT
115.1
082°
Holding pattern
+1 = UTC
UTC
Uncontrolled airspace (Class G)
Changeover Point
Intersection or fix formation (Bearing, frequency and ident of remote VHF navaid; Morse code shown when navaid is outside of chart neatline)
105
52
37
Flight planning relevant restrictions for ATS routes on Eastern Hemisphere chart series are published on respective EROUTE pages
Route usability by non B-RNAV equipped aircraft (within Europe only)
X
ATS
31
Maximum Authorized Altitude (MAA)
MAA
FL 390
UG851
42
FL 370
R2
RNAV ATS route (outside Europe)
Conditional Route Category (See Enroute Listing)
Unnamed, official published ATS route with direction indication
Centerline of ATS route
Total mileage between navaids
89
RVSM airspace boundary with designator
RVSM AIRSPACE
CANARIES GCCC UIR (G)
Named or unnamed airspace fix or mileage break.
A(HI)2
ATAR



145
77
25
Segment mileage
Special use airspace
"D" with distance indicates total DME distance from navaid.
Arrow from facility designating reporting point
ROYAL
N25 00.0
W015 47.2
CABOJ
N26 08.3
W014 38.6
D47
Time zone boundary with conversion values to UTC
Compulsory reporting point with name and INS coordinates
CWT
115.1
082°
Holding pattern
+1 = UTC
UTC
Uncontrolled airspace (Class G)
Intersection or fix formation (Bearing, frequency and ident of remote VHF navaid; Morse code shown when navaid is outside of chart neatline)
Changeover Point
105
52
37
Flight planning relevant restrictions for ATS routes on Eastern Hemisphere chart series are published on respective EROUTE pages
Route usability by non B-RNAV equipped aircraft (within Europe only)
X
ATS
31
Maximum Authorized Altitude (MAA)
MAA
FL 390
42
UG851
FL 370
R2
RNAV ATS route (outside Europe)
Conditional Route Category (See Enroute Listing)
Unnamed, official published ATS route with direction indication
Centerline of ATS route
Total mileage between navaids
89
RVSM airspace boundary with designator
RVSM AIRSPACE
CANARIES GCCC UIR (G)
Named or unnamed airspace fix or mileage break.
A(HI)2
ATAR

The symbology explained on these pages pertain specifically to Australia Enroute and Area charts.





SPECIAL ACTIVITY AREAS

- Ultra-light activity above 500′ AGL.
- Hang glider activity above 5000′ AGL.
- Model aircraft activity above 300′ AGL.
- Meteorology balloon ascents.
- Manned balloon ascents.
- Parachute jumping area.

AIR TRAFFIC SERVICE UNITS & BOUNDARIES

Hang glider activity above 5000′ AGL.

Model aircraft activity above 300′ AGL.

Meteorology balloon ascents.

M Manned balloon ascents.

Parachute jumping area.

Glider Operations.

W Gliders Launching.

V Airport within VHF range of responsible ATS unit.

\# Non-standard CTAF and MBZ, see airport directory for dimensions.

† † Navaid limitation, see Radio Aids page AU-37 (applicable only for Australia domestic services).

REPORTING POINTS (AUSTRALIA)

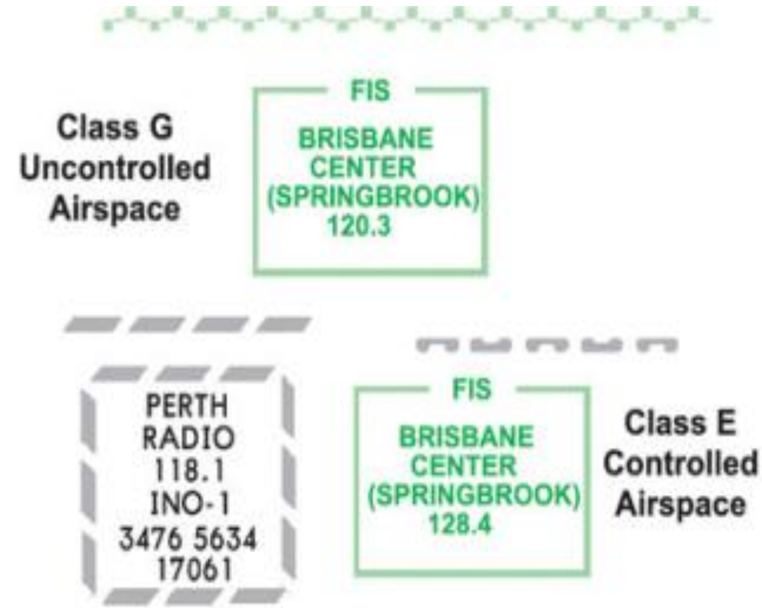
COMPULSORY for all aircraft.

All alititude

Low altitude

ON-REQUEST 300 KT TAS or more.
COMPULSORY Under 300 KT TAS.

All altitude



ROUTE DESIGNATORS

Conventional Routes:

A,B,G,R: Regional

H (one-way), J (two-way): Domestic

V (one-way), W (two-way):
Predominantly low-level domestic

RNAV Routes:

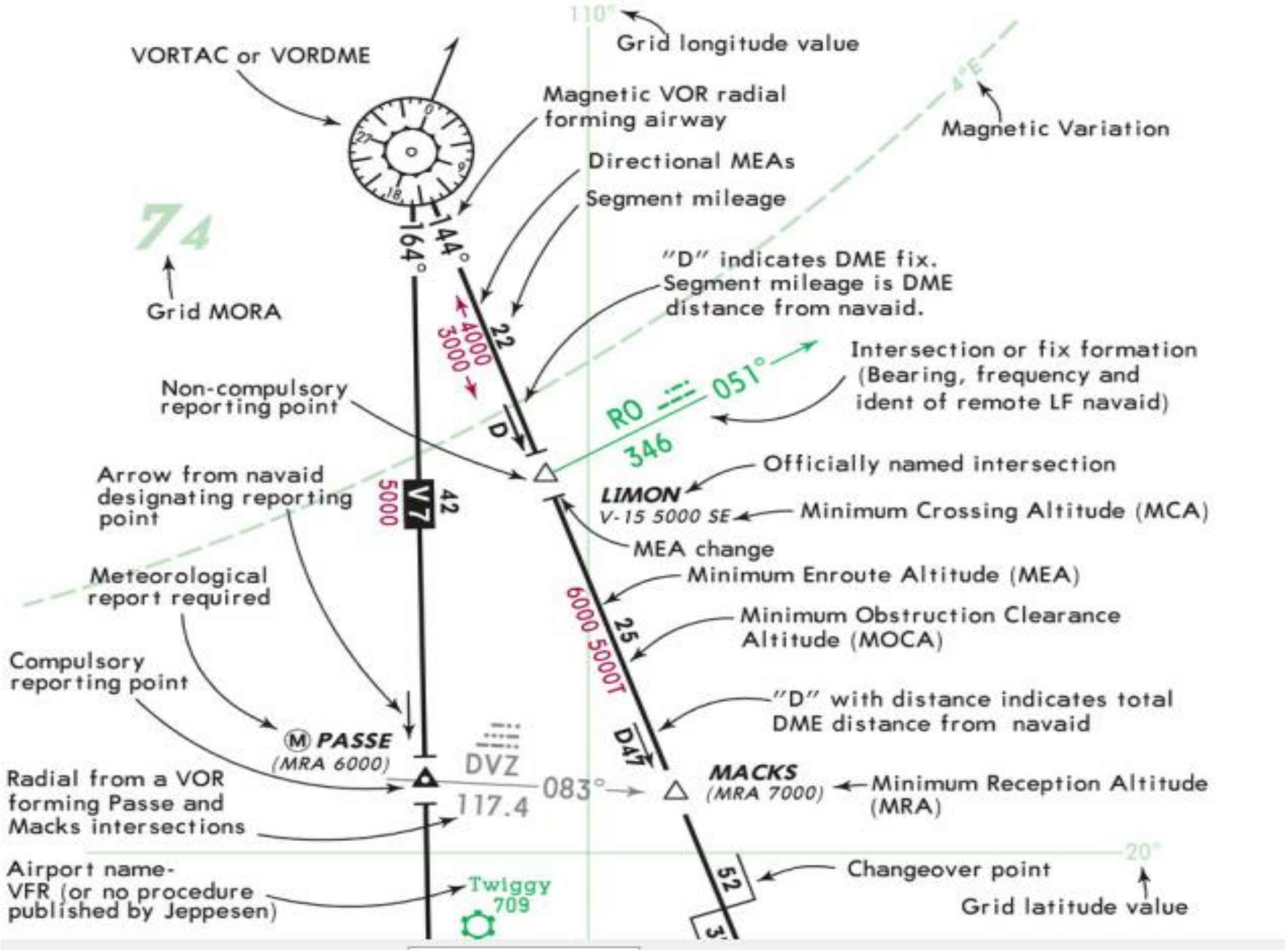
L,M,N: Regional (Tasman)

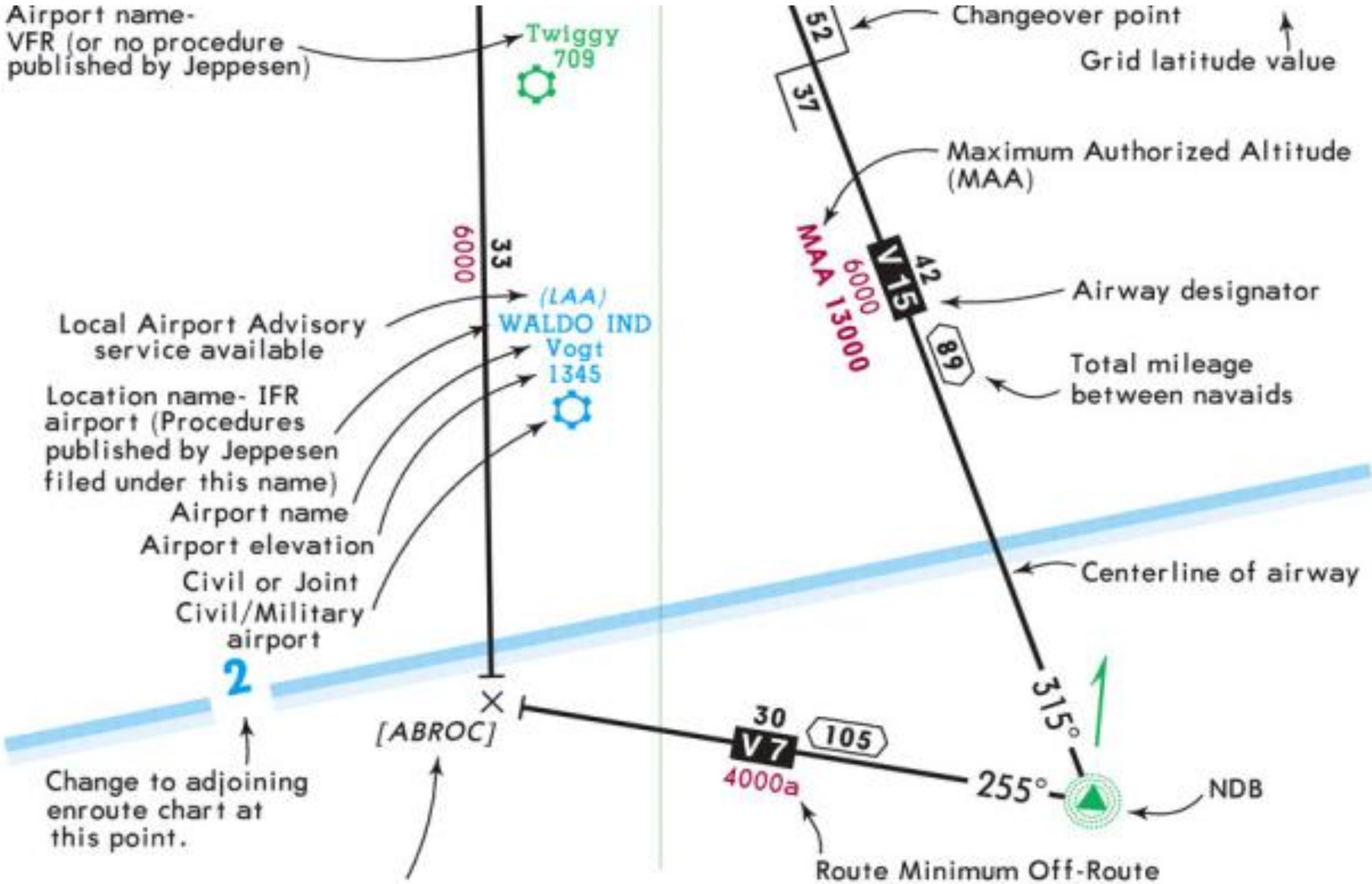
Q: 180°-359° domestic

Y: 360°-179° domestic

T: Two-way domestic

Z: Two-way low-level domestic


110°
Grid longitude value
VORTAC or VORDME
Magnetic VOR radial
forming airway
4°E
Magnetic Variation
Directional MEAs
Segment mileage
74
Grid MORA
164°
144°
"D" indicates DME fix.
Segment mileage is DME
distance from navaid.
22
4000
3000
Intersection or fix formation
(Bearing, frequency and
ident of remote LF navaid)
RO
051°
346
Non-compulsory
reporting point
D
Officially named intersection
LIMON
V-15 5000 SE
Minimum Crossing Altitude (MCA)
Arrow from navaid
designating reporting
point
5000
V7
42
MEA change
Minimum Enroute Altitude (MEA)
Meteorological
report required
6000
5000T
25
Minimum Obstruction Clearance
Altitude (MOCA)
Compulsory
reporting point
"D" with distance indicates total
DME distance from navaid
M PASSE
(MRA 6000)
DVZ
D47
MACKS
(MRA 7000)
Minimum Reception Altitude
(MRA)
Radial from a VOR
forming Passe and
Macks intersections
083°
117.4
20°
Changeover point
Airport name-
VFR (or no procedure
published by Jeppesen)
Twiggy
709
52
Grid latitude value



Database identifiers are enclosed in brackets [ABROC]
Database identifiers are officially designated by the controlling state authority or they may be derived by Jeppesen. In either case, these identifiers have no ATC function and should not be used in filing flight plans nor should they be used when communicating with ATC. They are shown only to enable the pilot to maintain orientation when using charts in concert with database navigation systems.

-

-

-
-

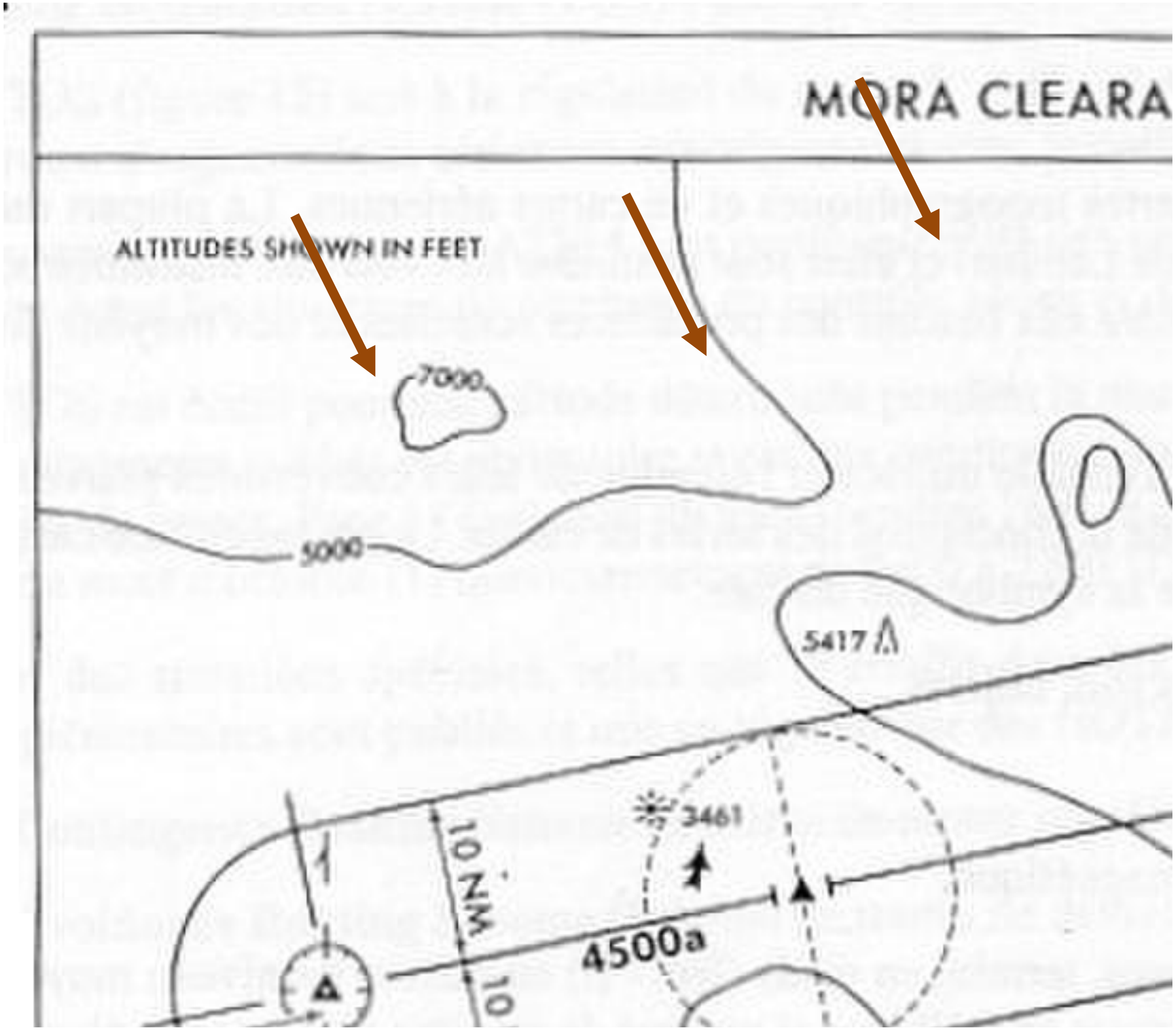
MEA

- MFO 1 000 ft
- 5 NM de part et d’autre de la voie aérienne.
- <u>Assure la réception des Moyens radio.</u>

MORA
MINIMUM OF ROUTE ALTITUDE

- MFO 1 000 ft (obstacles inférieurs à 5000 ft)
- MFO 2 000 ft (obstacles > 5 000 ft)
- 10 NM de part et d’autre de la voie aérienne

MORA (suite)



MORA CLEARA
ALTITUDES SHOWN IN FEET
7000
5000
5417
3461
10 NM
4500a
10

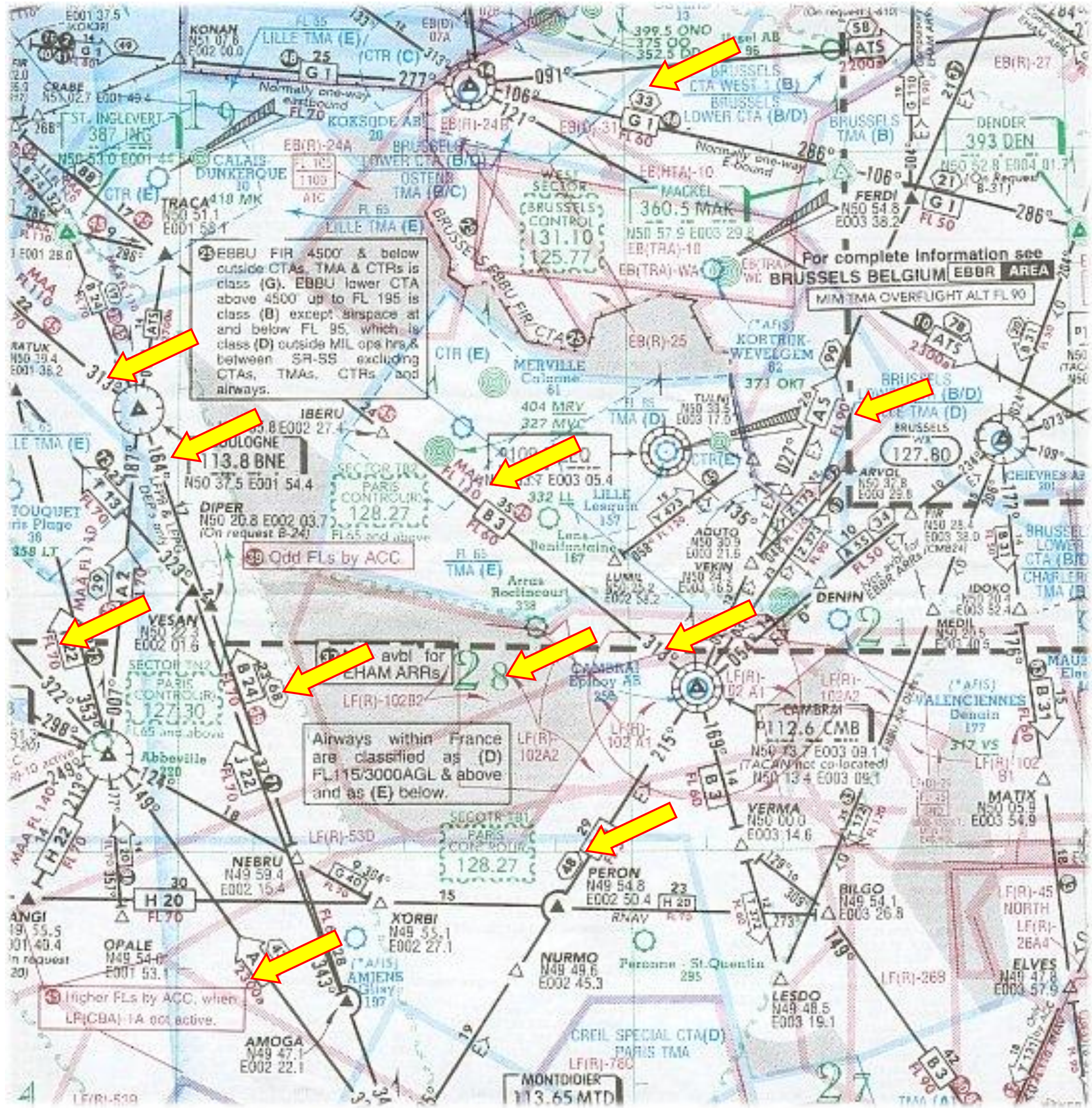
MOCA

Minimum Obstruction Clearance altitude

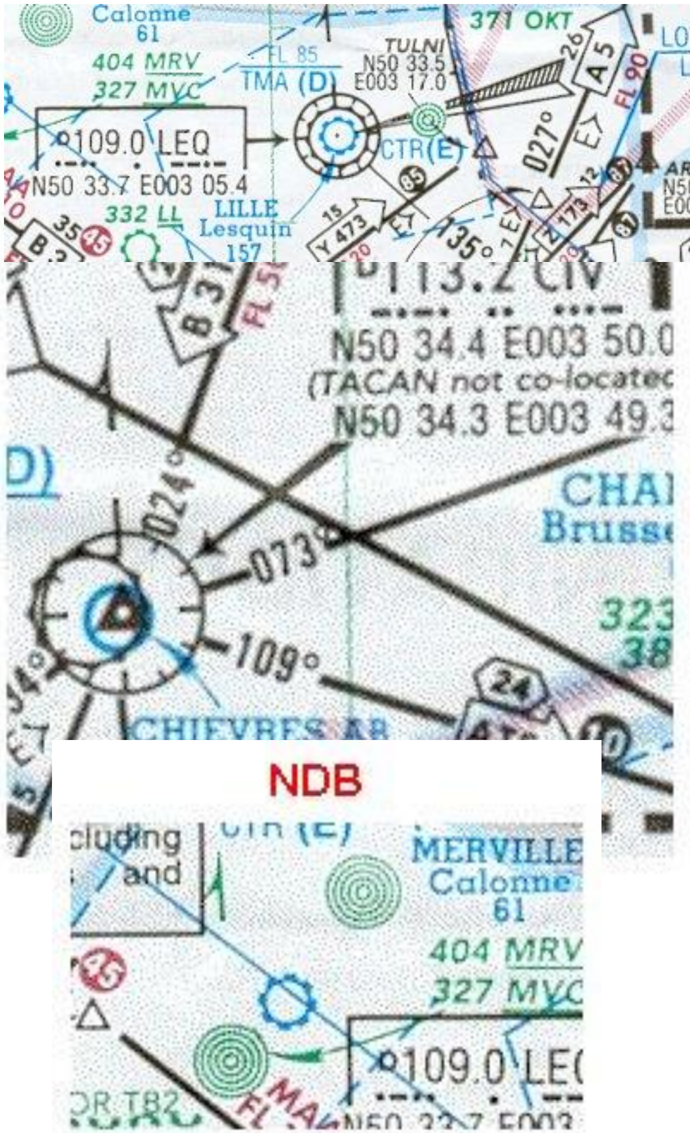
- MFO # MEA
- Ecriture 3000T
- ± 5 NM
- n’assure pas la réception parfaite des liaisons radio.

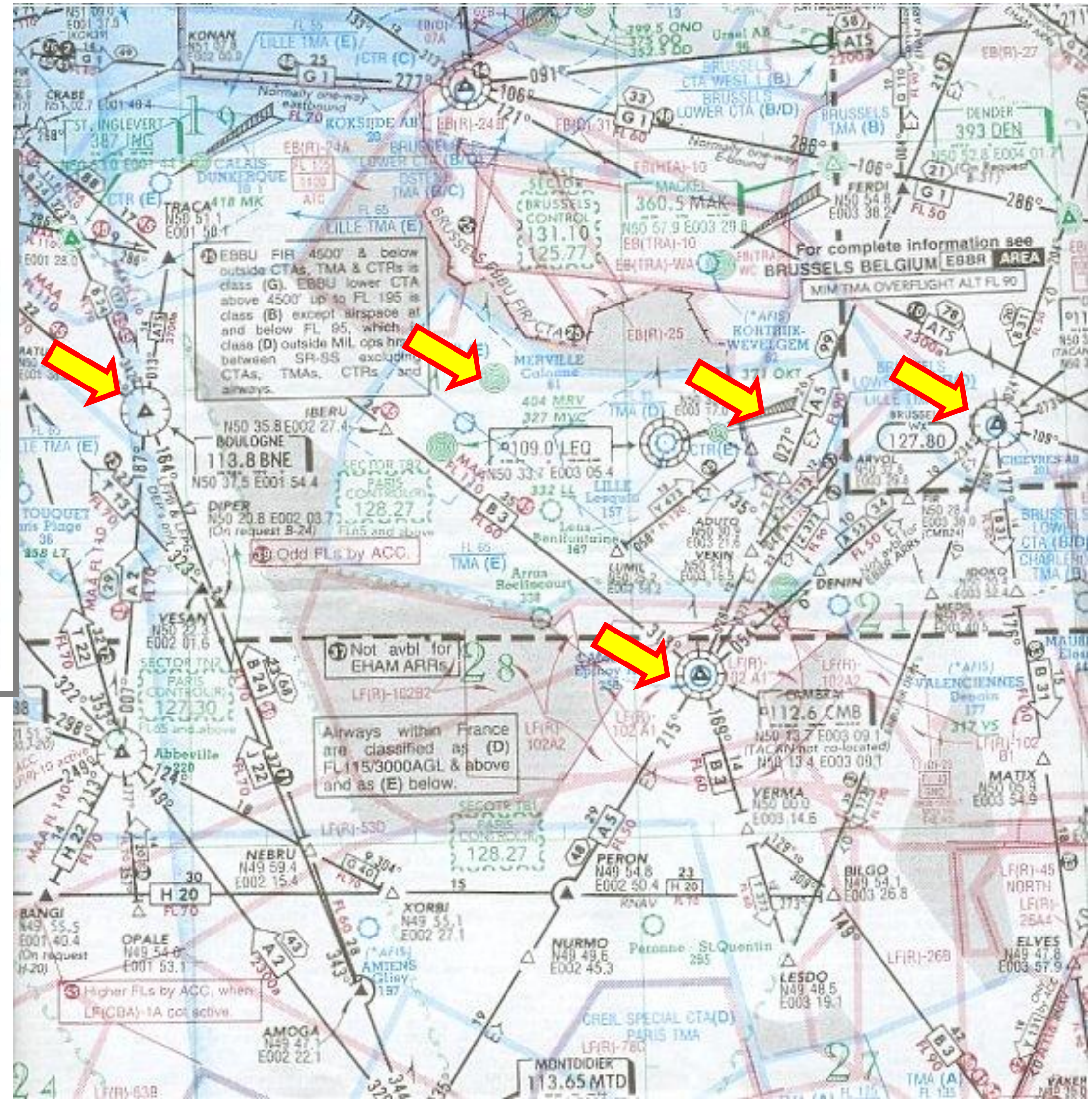
Informations sur les routes et les relèvements

- •Relèvements magnétiques
- •Information de distances
- •MEA
- •MORA
- •Grid Mora
- •MAA

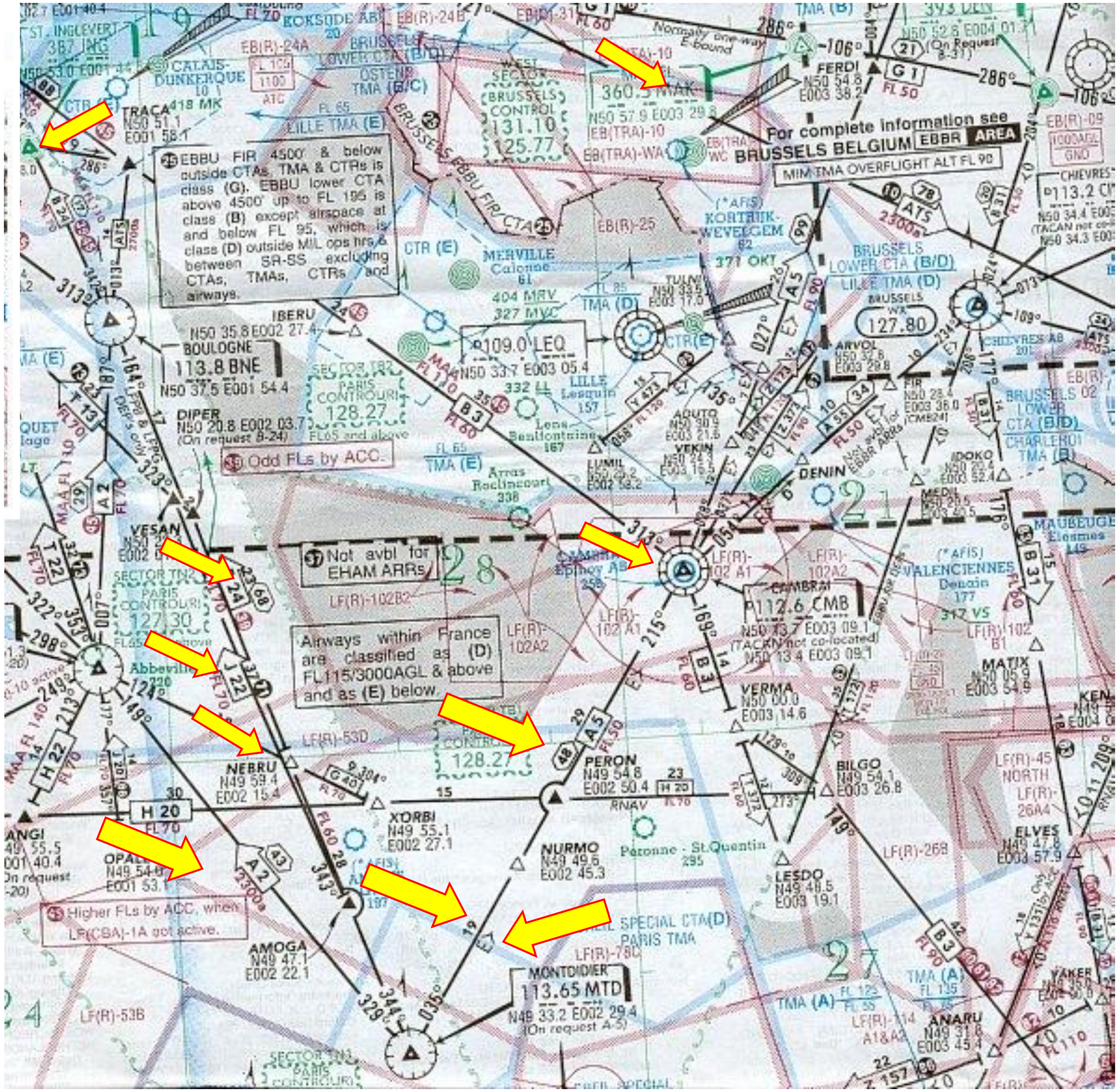


Moyens radioélectriques



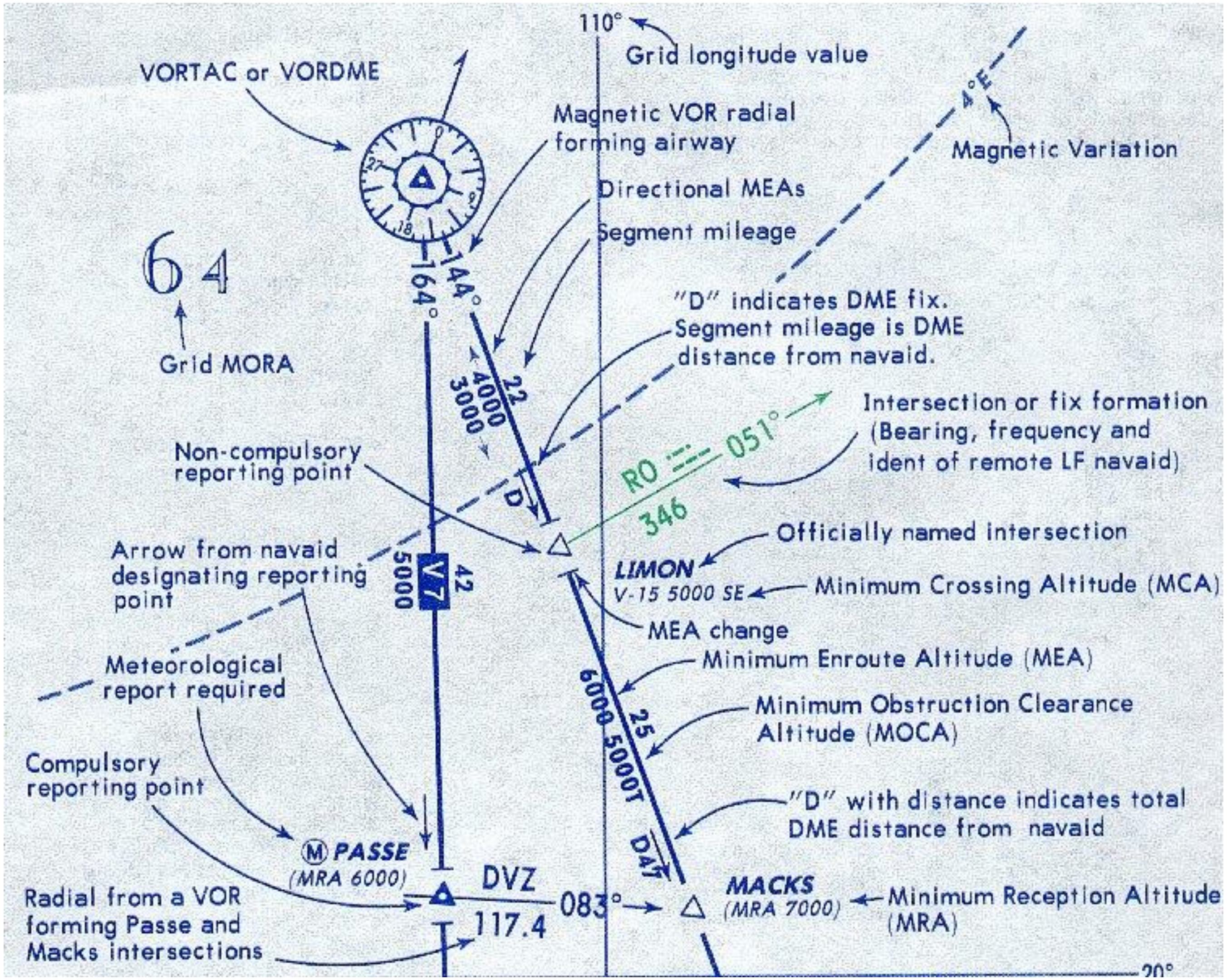


Informations sur les routes et les points de report

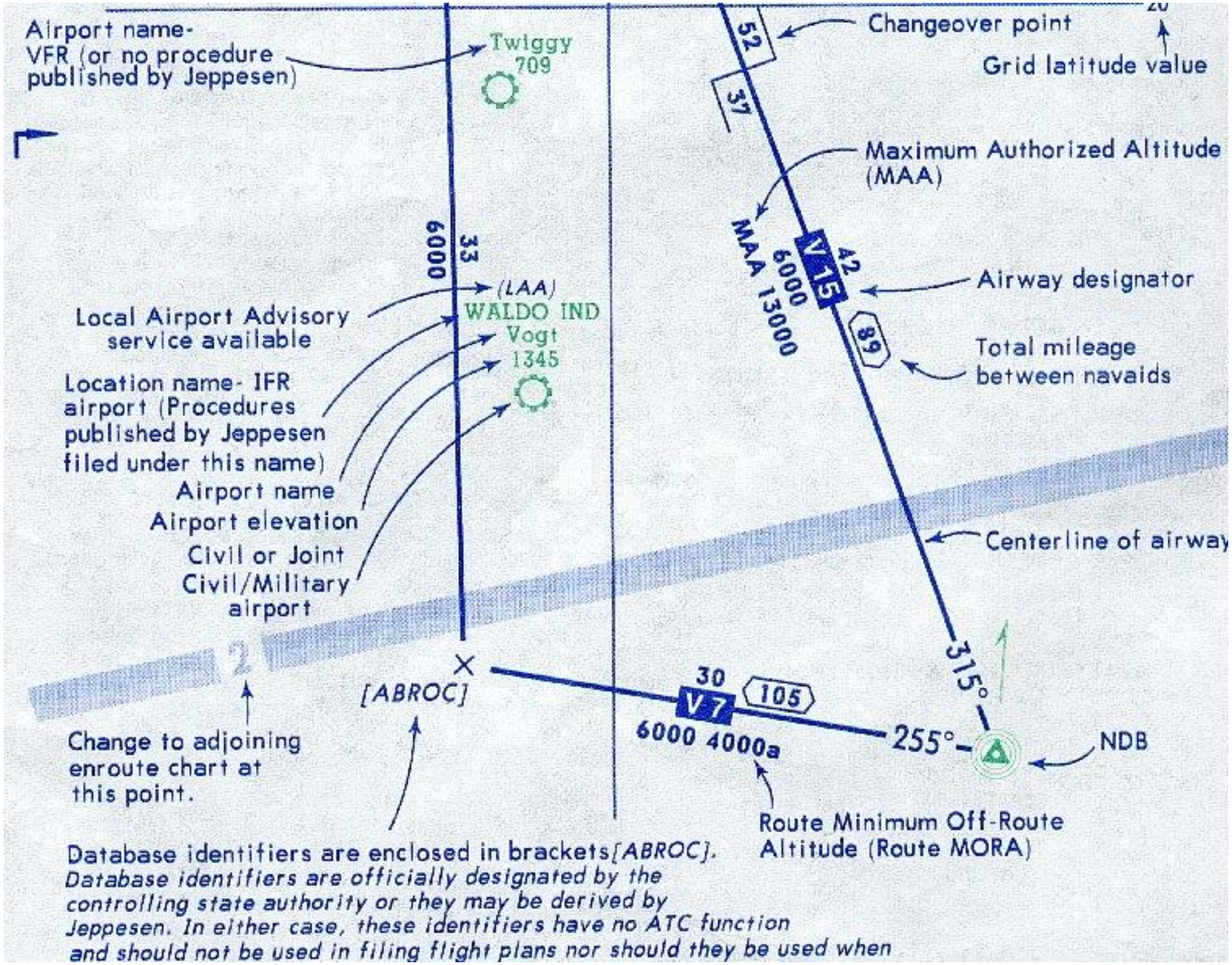


Zone interdite

ENROUTE CHART LEGEND



ENROUTE CHART LEGEND



Aérodromes

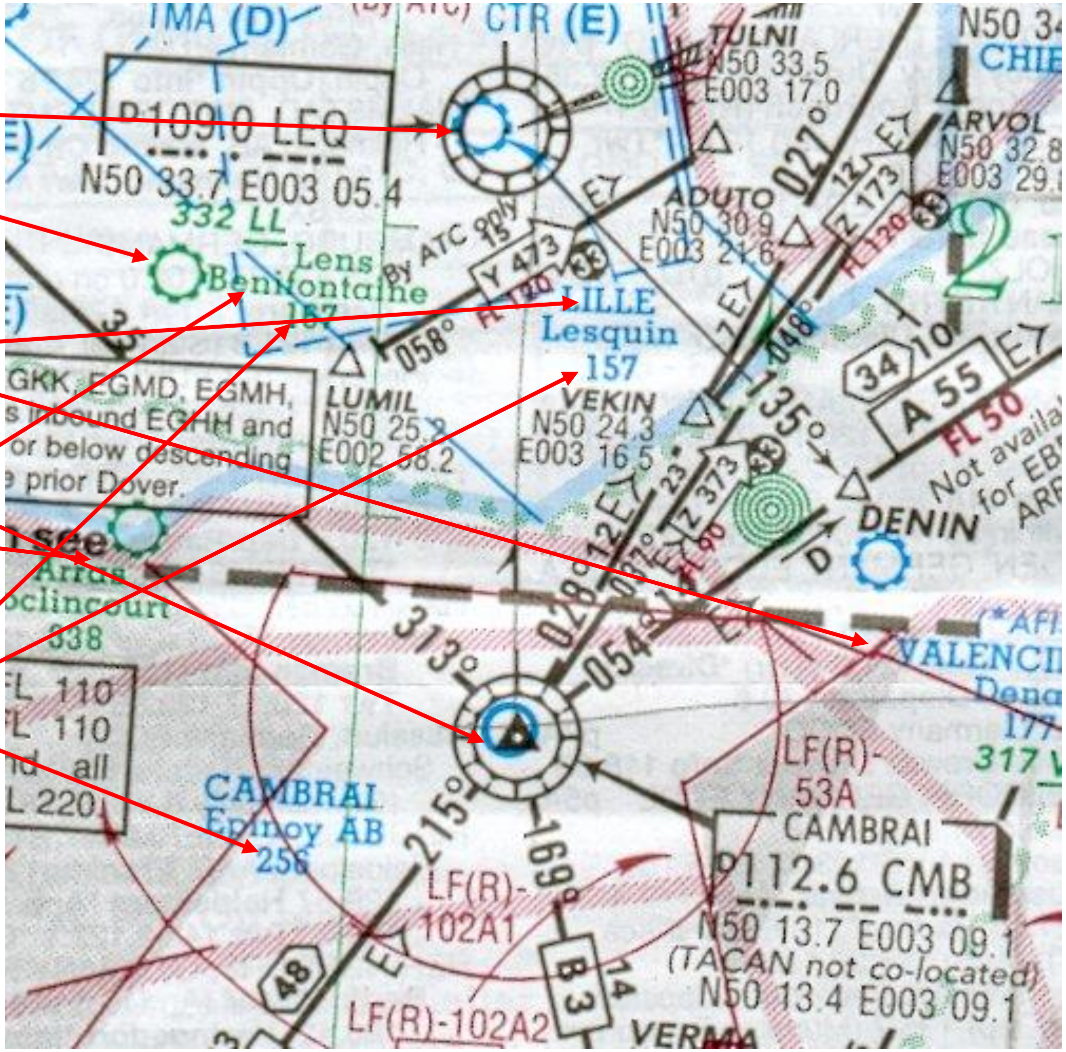
•AERODROME CIVIL

•Nom de ville de l 'aérodrome en majuscules pour lequel les cartes Jeppesen sont publiées

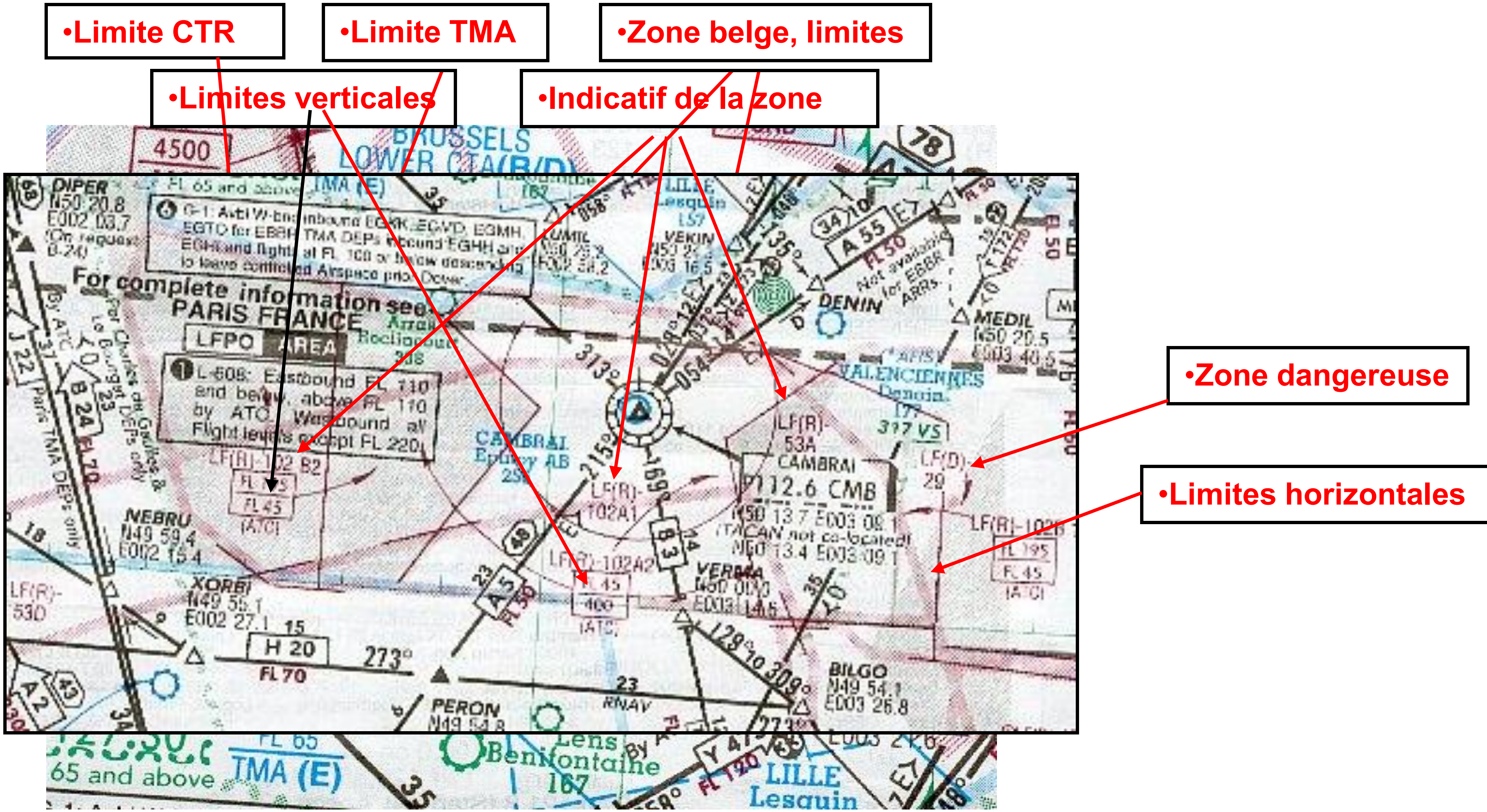
•Nom de l 'aérodrome en minuscules : pas de cartes d 'approches Jeppesen disponibles

•Altitude du point de référence de l 'aéroport

•Aérodrome militaire



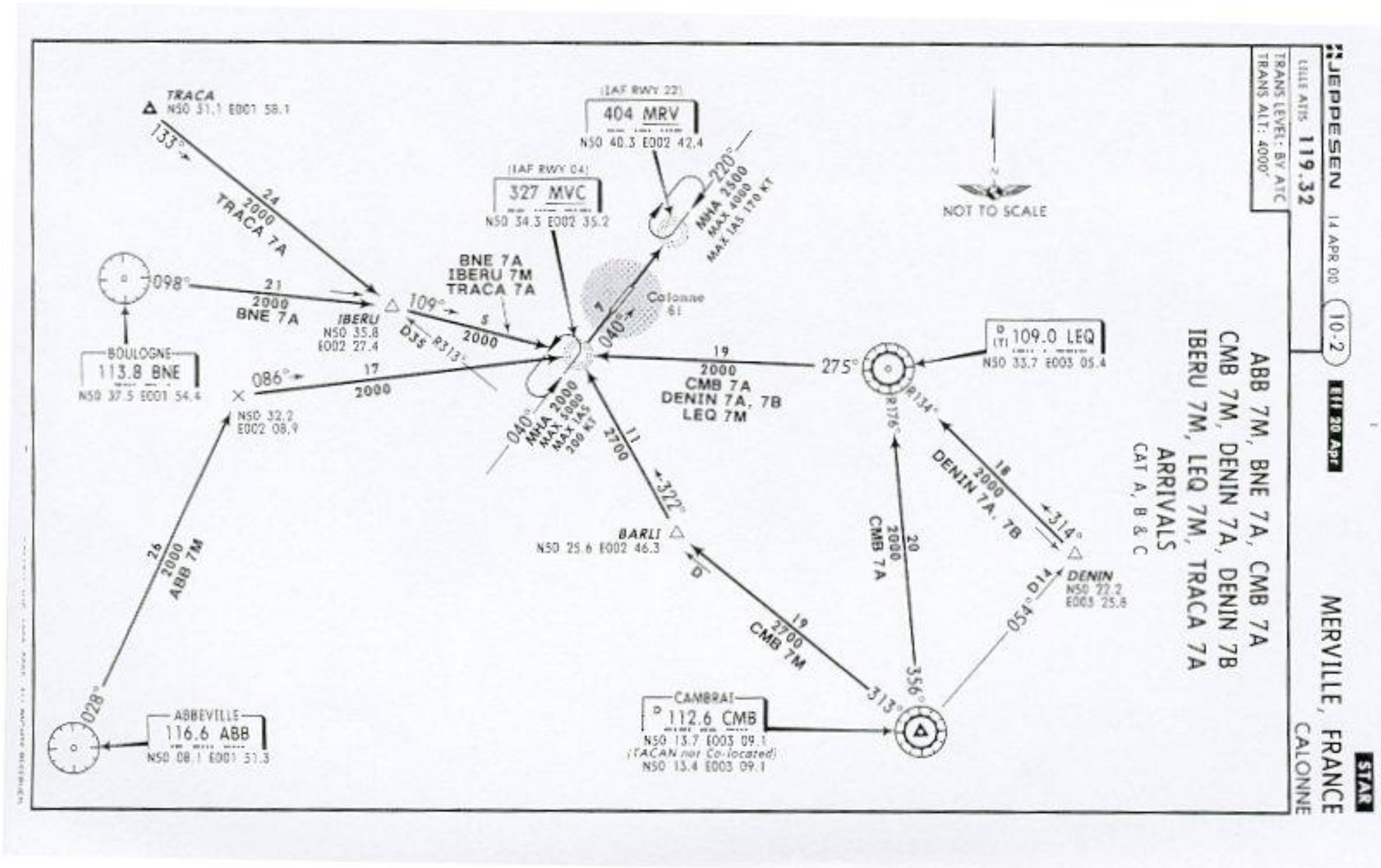
Espaces aériens réglementés



Area


ATS
282°
2300a
Only avbl for LFLA and LFQB DEPs inbound LFPO, LFPN and LFPV
SEINE CTR(E)
093°
112°
135°
160°
175°
28
ATS
3000
UA 1 UR 11
28
G 4
FL 60
23
273°
347°
014°
TROYES Barberey 388
TROYES
292°
268°
UH 37
UM 733
PARIS TMA
SEINE TMA(E)
SECTOR AO
PARIS CONTROL(R)
125.07
Pont-Sur-Yonne 236
UM 976
TROYES
116.0 TRO
N48 15.1 E003 57.8
100
34
A 1
FL 60
191°
174°
316°
G 40
88
FL 70
24
UH 10
A 27
FL 60
25
103
A 2
FL 70
29
104
CHAILLEY
353 CHY
UH 37
94
14
B 373
FL 60
UT 15
LUREN
N48 01.6
E003 54.8
54
T 11
280°
100°
229°
Joigny 728
OKRIX
N47 58.0
E003 34.1
226°
St. Florentin Cheu 351
27
T 11
FL 60
66
354°
315°
30
ROLAMPONT RLP
TMA
N48 00.0 E003 22.7
[TMA11]
78
3°30'
30
FL 65
SEINE TMA(E)
417 AX
355°
D29/34
26 FROM AVLON
ATN AUTUN
UH 10
UH 47
049°
28 FROM AVLON
011°
MOU MOULINS
26
ATN AUTUN
DIJON DIJ
3°
MOULINS MOU
D29/38 OKRIX
340°
OSKIN
© JEPPESEN SANDERSON, INC., 1982, 2000. ALL RIGHTS RESERVED.
21 JAN 00
10-1
Eff 27 Jan
PARIS, FRANCE
AREA

STAR



JEPPESEN
MERVILLE, FRANCE
CALONNE
10-2
ABB 7M, BNE 7A, CMB 7A
CMB 7M, DENIN 7A, DENIN 7B
IBERU 7M, LEQ 7M, TRACA 7A
ARRIVALS
CAT A, B & C
NOT TO SCALE
TRACA
327 MVC
404 MRV
113.8 BNE
116.6 ABB
112.6 CMB
109.0 LEQ
IBERU
BARLI
DENIN

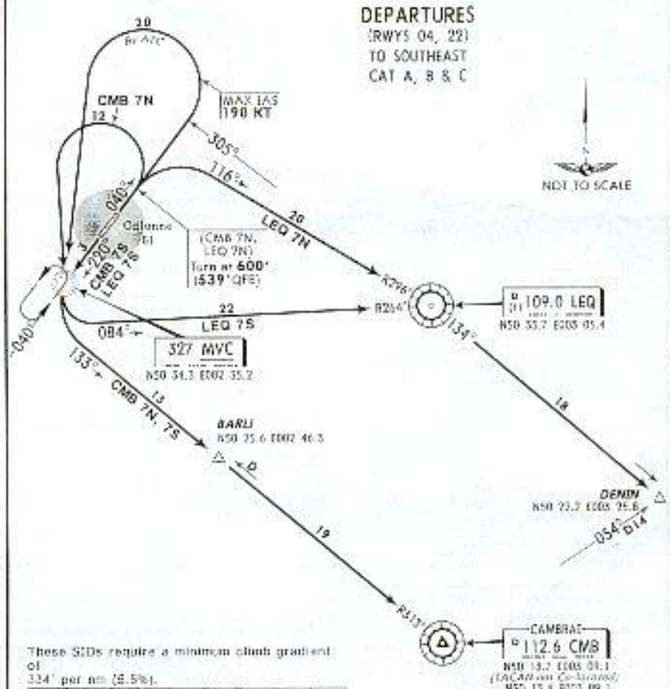
SID

JEPPESEN 14 APR 00 (10-3) Eff 20 Apr

MERVILLE, FRANCE
CALONNE

TRANS LEVEL: BY ATC
TRANS ALT: 4000'

CMB 7N, CMB 7S
LEQ 7N, LEQ 7S
DEPARTURES
(RWYS 04, 22)
TO SOUTHEAST
CAT A, B & C



These SIDs require a minimum climb gradient of
334' per nm (5.5%).

Gnd speed-Kts	75	100	150	200	250	300
334' per nm	418	557	835	1114	1392	1671

If unable to comply inform ATC prior to start-up.

SID	RWY	ROUTING
CMB 7N	04	Climb on 040° track, at **600'** (**539'** QFE) turn **LEFT** to MVC Lctr. Intercept CMB R-313 inbound to CMB VORTAC.
CMB 7S	22	To MVC Lctr, intercept CMB R-313 inbound to CMB VORTAC.
LEQ 7N	04	Climb on 040° track, at **600'** (**539'** QFE) turn RIGHT, intercept LEQ R-296 inbound to LEQ VORDME.
LEQ 7S	22	To MVC Lctr, intercept LEQ R-264 inbound to LEQ VORDME.

RWY 04 ATC TAKE-OFF

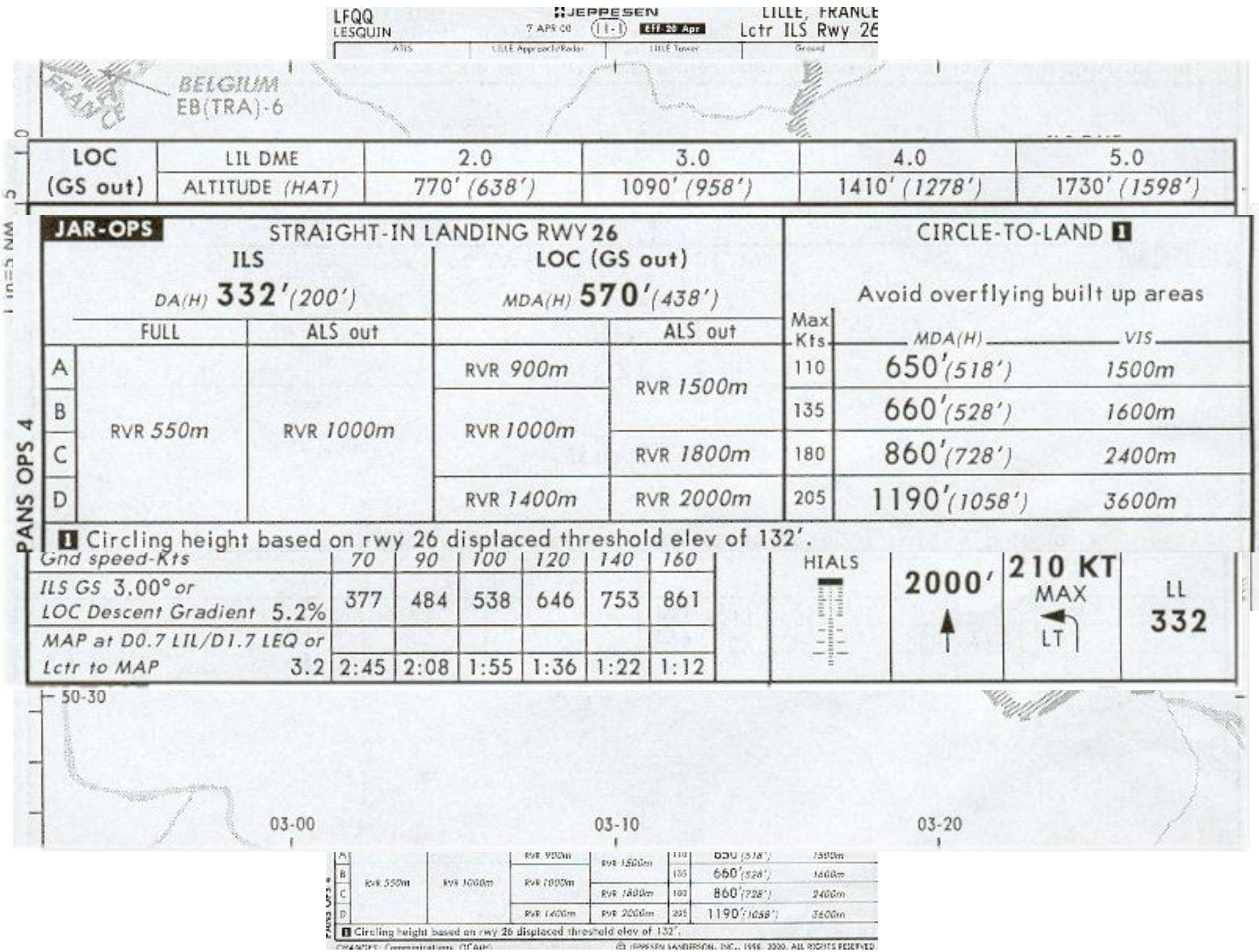
In order to separate traffic in the MVC holding from departing traffic:
On 040° track, when passing LEQ R-305 turn LEFT (MAX IAS 190 KT) to MVC Lctr climbing to **3000'** (**2939'** QFE).

This departure requires a minimum climb gradient of
365' per nm (6%)

Gnd speed-Kts	75	100	150	200	250	300
365' per nm	456	608	911	1215	1519	1823

CHANGES: SIDs renumbered.

Procédure d 'approche aux instruments



LFQQ
LESQUIN

JEPPESEN

7 APR 00 (11-1) Eff 20 Apr

LILLE, FRANCE
Lctr ILS Rwy 26

BELGIUM
EB(TRA)-6

FRANCE

LOC (GS out)	LIL DME	2.0	3.0	4.0	5.0
	ALTITUDE *(HAT)*	770' *(638')*	1090' *(958')*	1410' *(1278')*	1730' *(1598')*

JAR-OPS	STRAIGHT-IN LANDING RWY 26				CIRCLE-TO-LAND 1		
	ILS DA(H) **332'** *(200')*		LOC (GS out) MDA(H) **570'** *(438')*		Avoid overflying built up areas		
	FULL	ALS out		ALS out	Max Kts	*MDA(H)*	*VIS*
A	RVR *550m*	RVR *1000m*	RVR *900m*	RVR *1500m*	110	650' *(518')*	*1500m*
B			RVR *1000m*		135	660' *(528')*	*1600m*
C				RVR *1800m*	180	860' *(728')*	*2400m*
D			RVR *1400m*	RVR *2000m*	205	1190' *(1058')*	*3600m*

1 Circling height based on rwy 26 displaced threshold elev of 132'.

PANS OPS 4

Gnd speed-Kts	*70*	*90*	*100*	*120*	*140*	*160*
ILS GS 3.00° *or LOC Descent Gradient* 5.2%	377	484	538	646	753	861
MAP at D0.7 LIL/D1.7 LEQ or Lctr to MAP 3.2	2:45	2:08	1:55	1:36	1:22	1:12

HIALS

2000'

210 KT MAX LT

LL 332

50-30

03-00 03-10 03-20

CHANGES: Communications. OCA(H).

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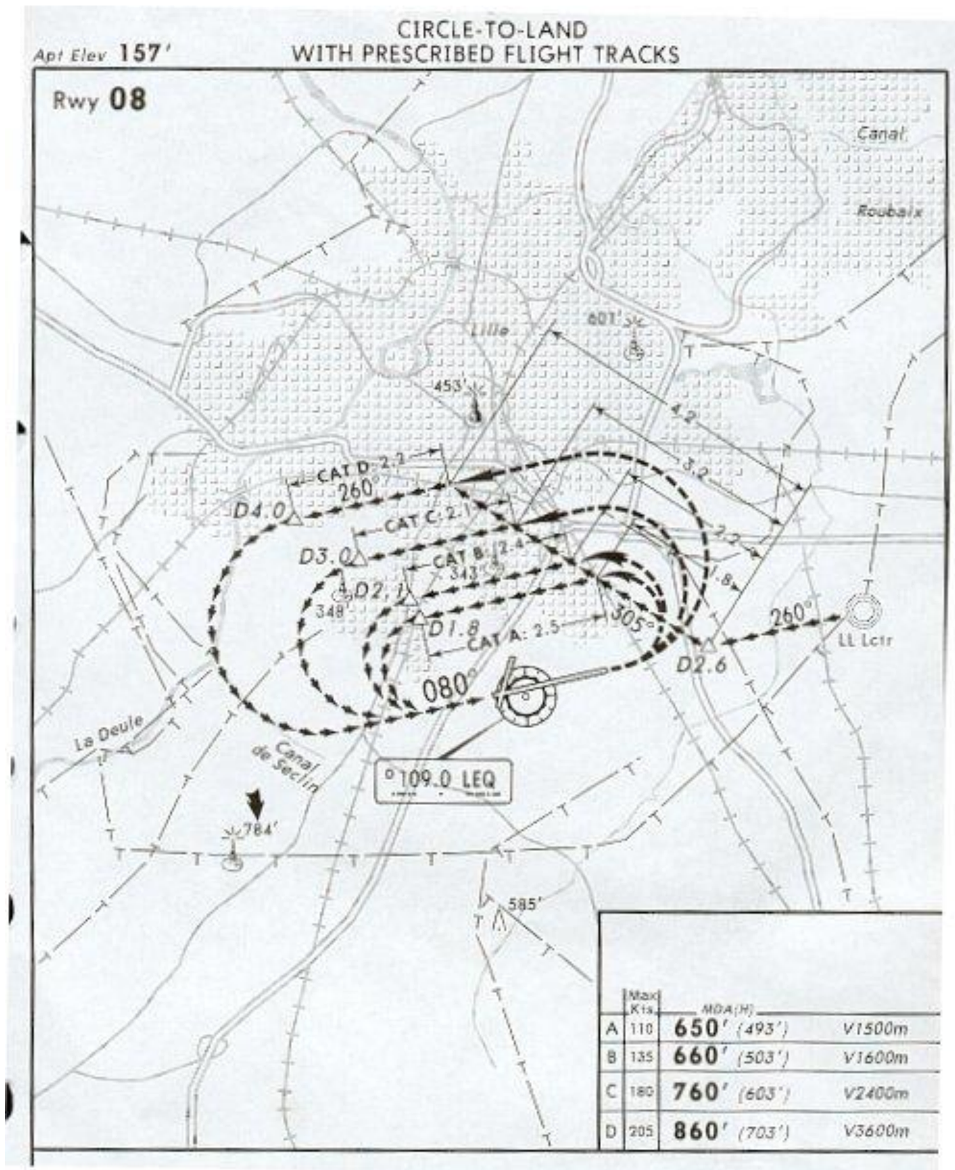
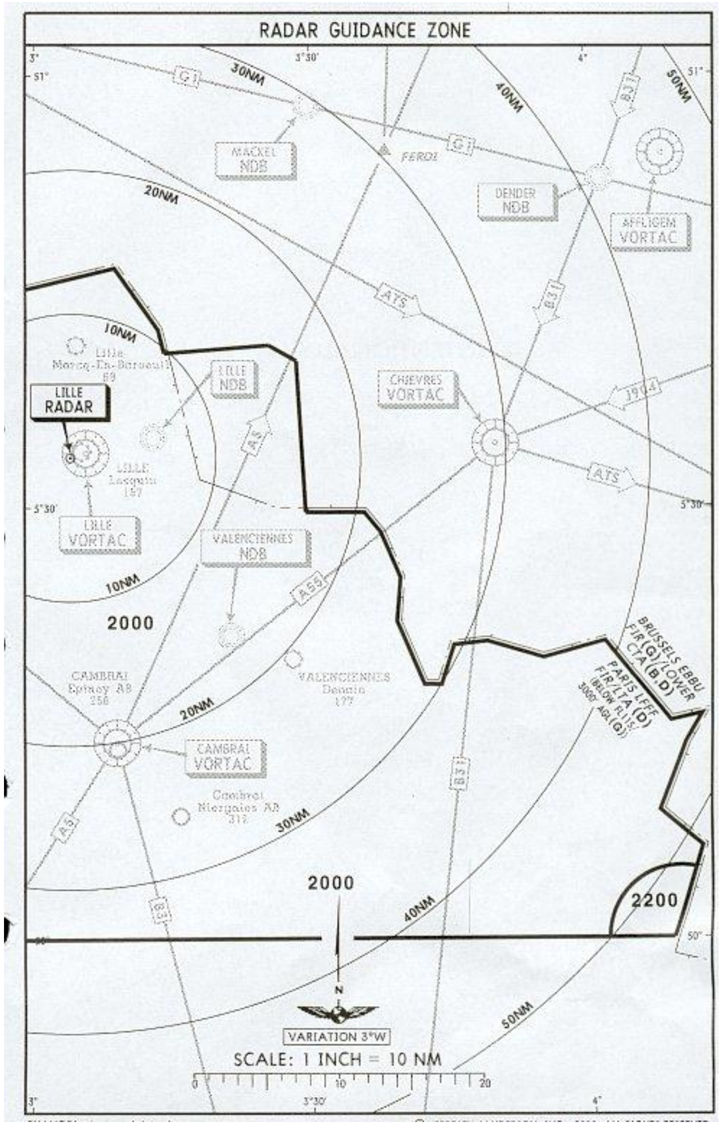
Approche indirecte



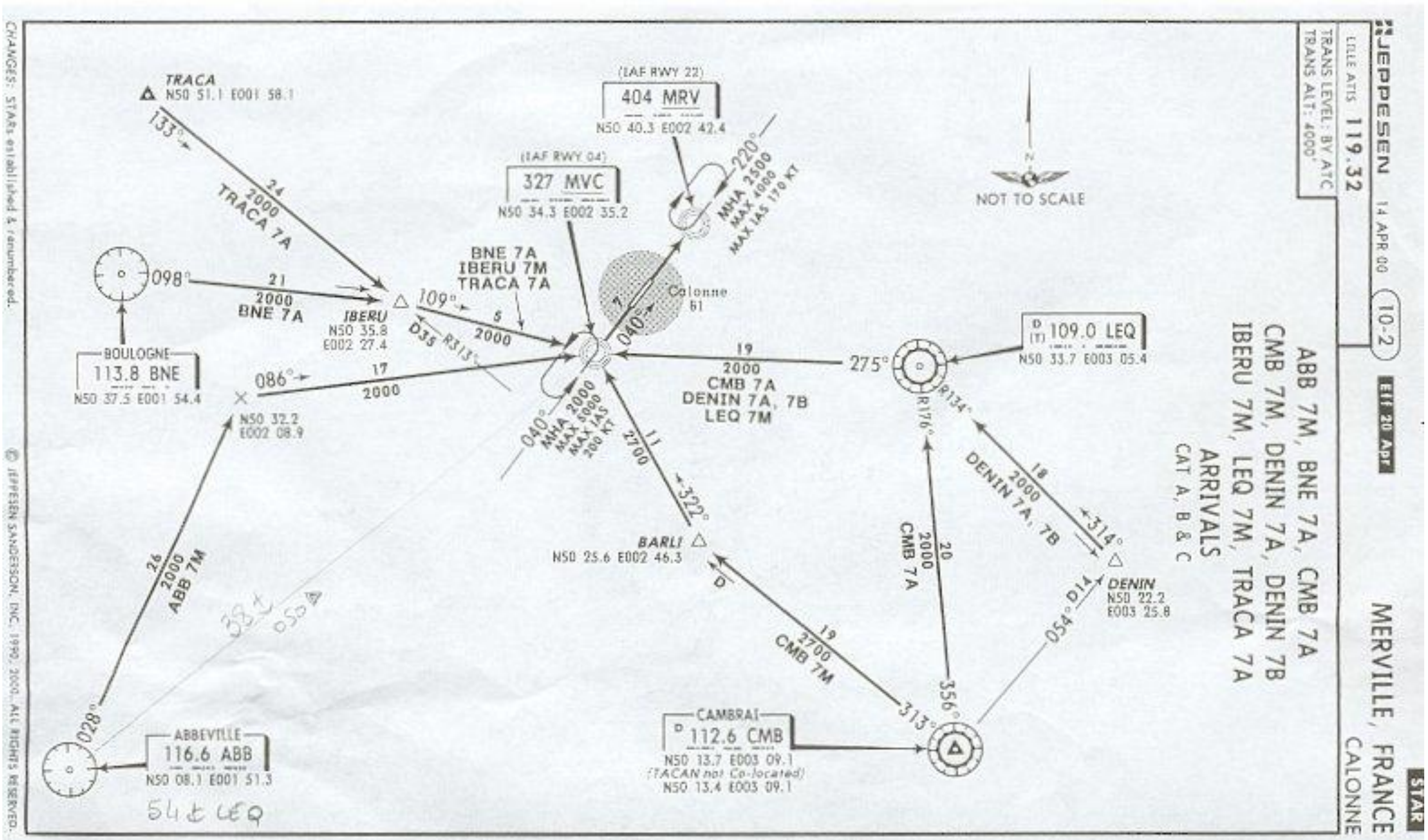
Radar landing minimum



Chapitre 3

- [Fiches d 'aérodrome Merville]
  - STAR
  - SID CMB LEQ
  - SID BNE TRACA
  - Aérodrome
  - Lctr RWY 04
  - Lctr RWY 22

STAR



MERVILLE, FRANCE
CALONNE
ABB 7M, BNE 7A, CMB 7A
CMB 7M, DENIN 7A, DENIN 7B
IBERU 7M, LEQ 7M, TRACA 7A
ARRIVALS
CAT A, B & C
JEPPESEN
LILLE ATIS 119.32
TRANS LEVEL: BY ATC
TRANS ALT: 4000'
14 APR 00
10-2
STAR
NOT TO SCALE
TRACA N50 51.1 E001 58.1
(IAF RWY 22) 404 MRV N50 40.3 E002 42.4
(IAF RWY 04) 327 MVC N50 34.3 E002 35.2
BOULOGNE 113.8 BNE N50 37.5 E001 54.4
IBERU N50 35.8 E002 27.4
109.0 LEQ N50 33.7 E003 05.4
N50 32.2 E002 08.9
BARLI N50 25.6 E002 46.3
DENIN N50 22.2 E003 25.8
CAMBRAI 112.6 CMB N50 13.7 E003 09.1 (TACAN not Co-located) N50 13.4 E003 09.1
ABBEVILLE 116.6 ABB N50 08.1 E001 51.3
BNE 7A IBERU 7M TRACA 7A
CMB 7A DENIN 7A, 7B LEQ 7M
MHA 2500 MAX 4000 MAX IAS 170 KT
MHA 2000 MAX 2000 MAX IAS 200 KT
Calonne 61
CHANGES: STARs established & renumbered.
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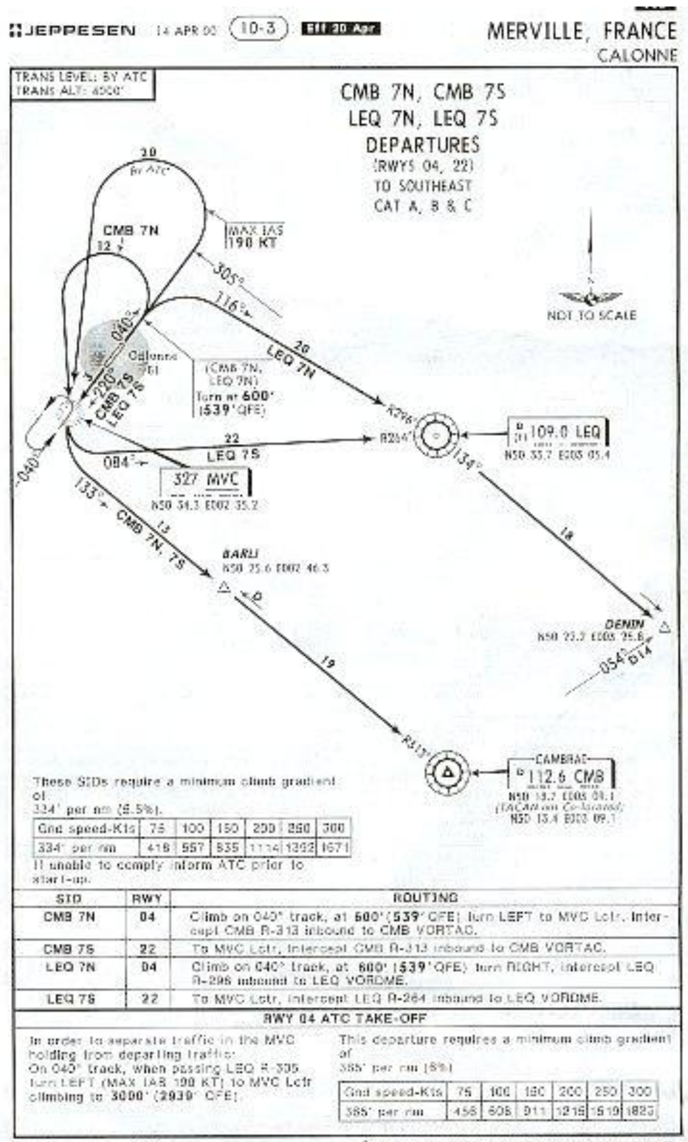
54 t LEQ

SID CMB LEQ


JEPPESEN
14 APR 00
10-3
Eff 20 Apr
MERVILLE, FRANCE
CALONNE
TRANS LEVEL: BY ATC
TRANS ALT: 4000'
CMB 7N, CMB 7S
LEQ 7N, LEQ 7S
DEPARTURES
(RWYS 04, 22)
TO SOUTHEAST
CAT A, B & C
NOT TO SCALE
CMB 7N
MAX IAS
190 KT
305°
116°
040°
Calonne
(CMB 7N,
LEQ 7N)
Turn at 600'
(539' QFE)
LEQ 7N
CMB 7S
LEQ 7S
084°
327 MVC
N50 34.3 E002 35.2
133°
CMB 7N, 7S
BARLI
N50 25.6 E002 46.3
R296°
R264°
109.0 LEQ
N50 33.7 E003 05.4
134°
DENIN
N50 23.2 E003 25.8
054°
D14
R313°
CAMBRAI
112.6 CMB
N50 13.7 E003 09.1
(TACAN on Co-located)
These SIDs require a minimum climb gradient of
334' per nm (5.5%).
Gnd speed-Kts | 75 | 100 | 150 | 200 | 250 | 300
334' per nm | 418 | 557 | 835 | 1114 | 1392 | 1671
If unable to comply inform ATC prior to start-up.
SID | RWY | ROUTING
CMB 7N | 04 | Climb on 040° track, at 600' (539' QFE) turn LEFT to MVC Lctr. Intercept CMB R-313 inbound to CMB VORTAC.
CMB 7S | 22 | To MVC Lctr, intercept CMB R-313 inbound to CMB VORTAC.
LEQ 7N | 04 | Climb on 040° track, at 600' (539' QFE) turn RIGHT, intercept LEQ R-296 inbound to LEQ VORDME.
LEQ 7S | 22 | To MVC Lctr, intercept LEQ R-264 inbound to LEQ VORDME.
RWY 04 ATC TAKE-OFF
In order to separate traffic in the MVC holding from departing traffic:
On 040° track, when passing LEQ R-305 turn LEFT (MAX IAS 190 KT) to MVC Lctr climbing to 3000' (2939' QFE).
This departure requires a minimum climb gradient of
385' per nm (6%)
Gnd speed-Kts | 75 | 100 | 150 | 200 | 250 | 300
385' per nm | 456 | 608 | 911 | 1215 | 1519 | 1823
CHANGES: SIDs renumbered.
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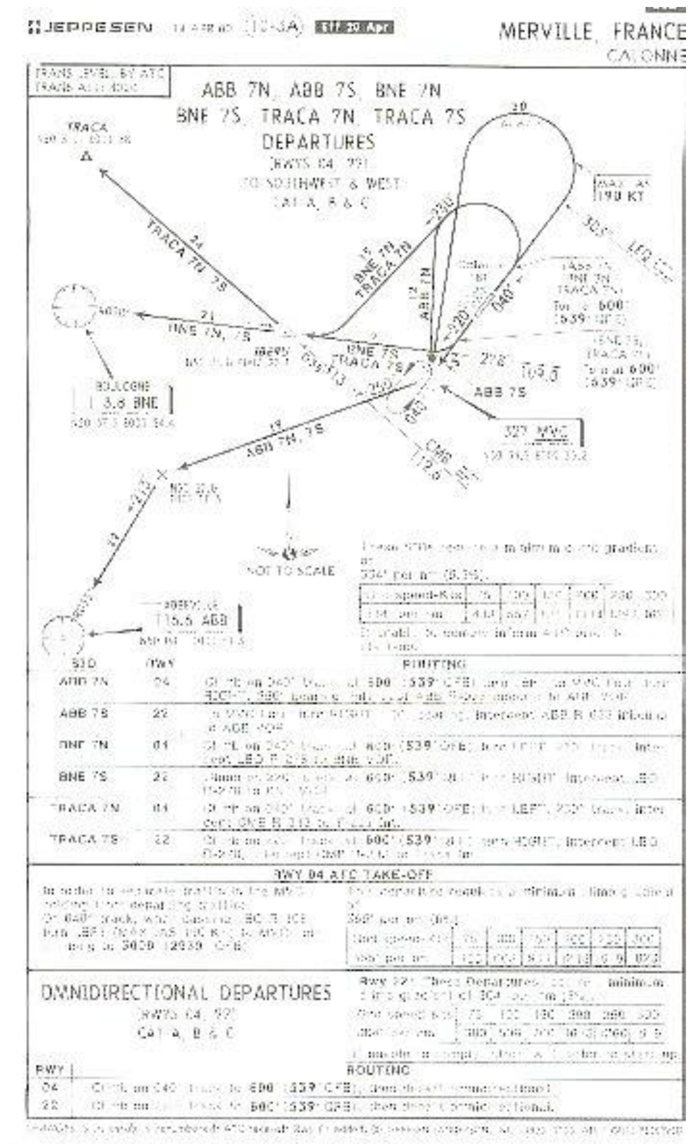
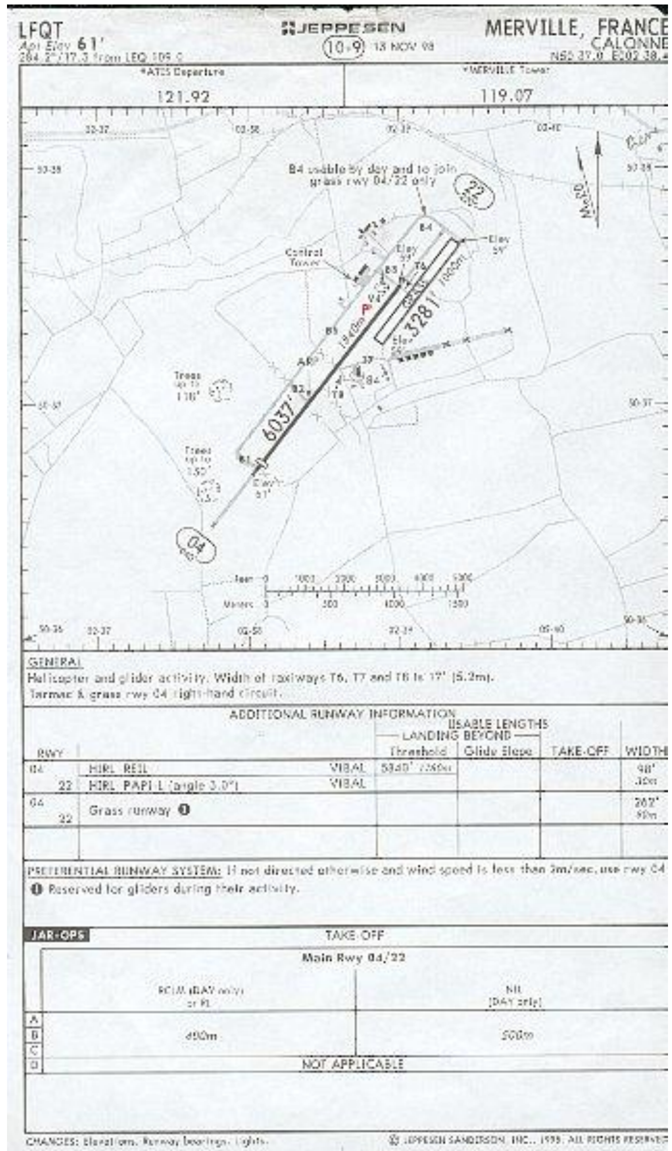
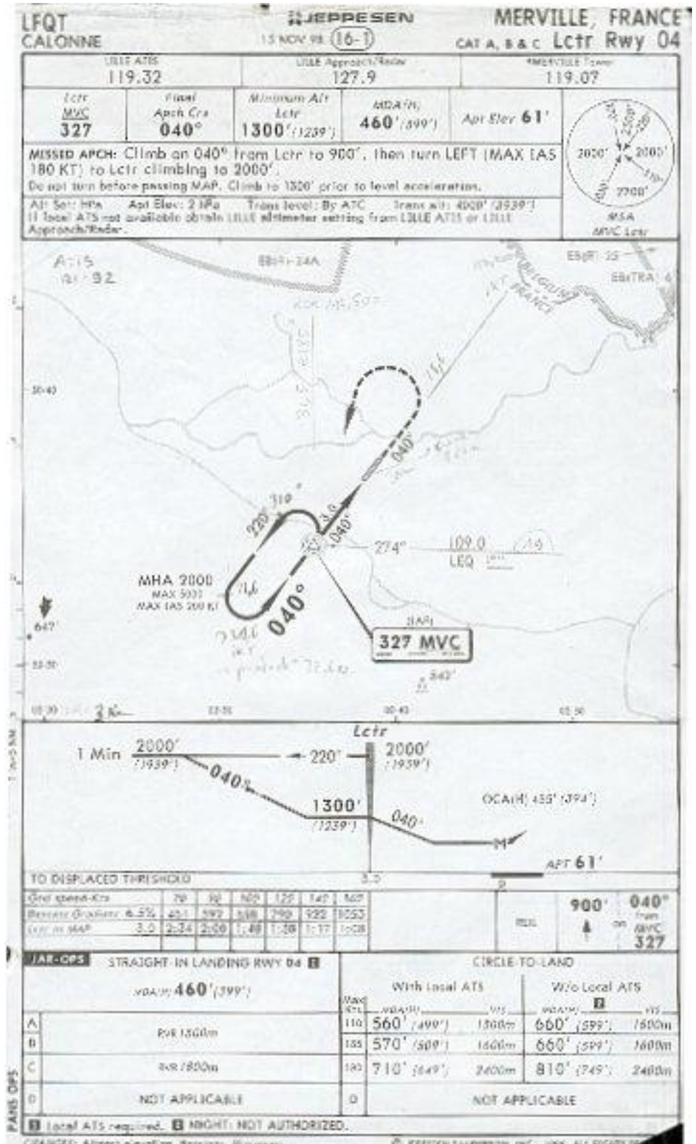
SID BNE TRACA



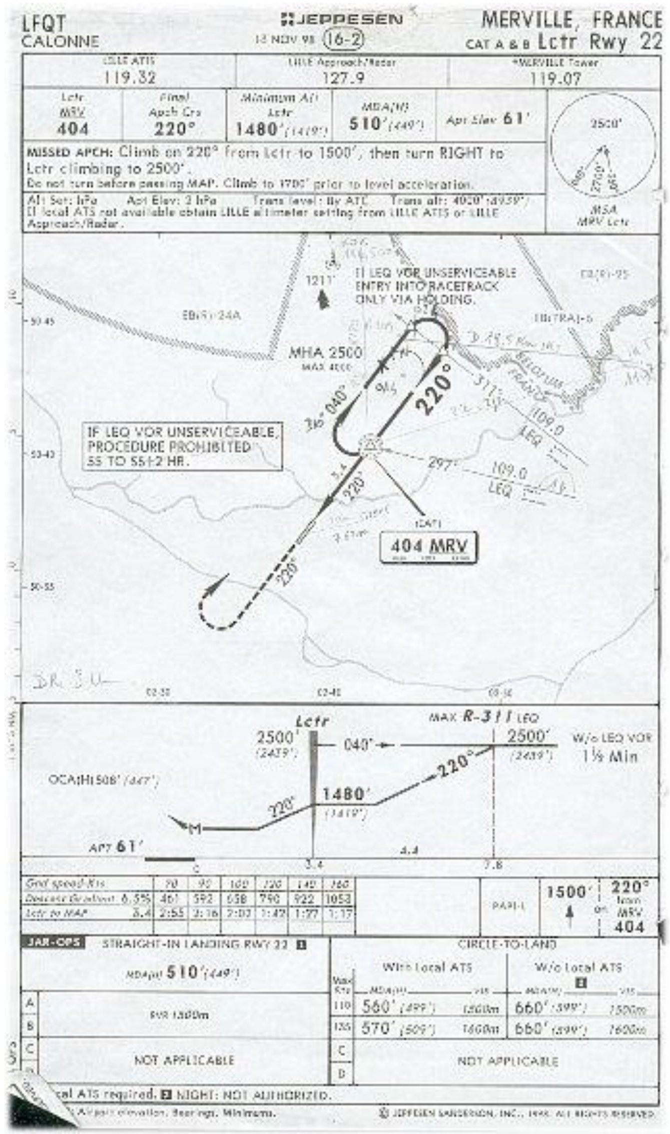
Aérodrome


LFQT
JEPPESEN
MERVILLE, FRANCE
CALONNE
10-9
ATIS Departure
121.92
MERVILLE Tower
119.07
B4 usable by day and to join grass rwy 04/22 only
22
Control Tower
Elev
6037'
3281'
04
Trees
GENERAL
Helicopter and glider activity. Width of taxiways T6, T7 and T8 is 17' (5.2m).
Tarmac & grass rwy 04 right-hand circuit.
ADDITIONAL RUNWAY INFORMATION
USABLE LENGTHS
LANDING BEYOND
Threshold
Glide Slope
TAKE-OFF
WIDTH
RWY
HIRL REIL
VIBAL
HIRL PAPI-L (angle 3.0°)
Grass runway
PREFERENTIAL RUNWAY SYSTEM:
Reserved for gliders during their activity.
JAR-OPS
TAKE-OFF
Main Rwy 04/22
NIL
(DAY only)
490m
500m
NOT APPLICABLE

Procédure Lctr 04



Procédure Lctr 22



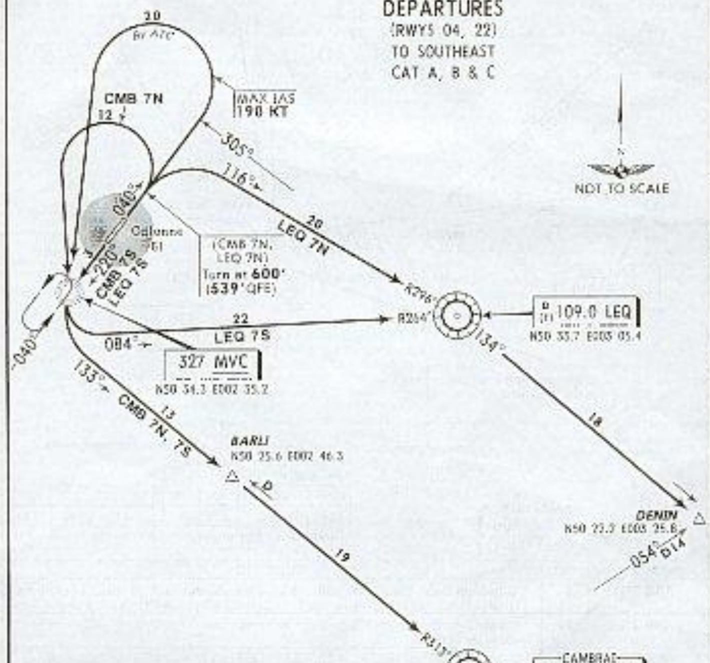
Merville SID CMB

JEPPESEN 14 APR 00 (10-3) Eff 20 Apr

MERVILLE, FRANCE
CALONNE

TRANS LEVEL: BY ATC
TRANS ALT: 4000'

CMB 7N, CMB 7S
LEQ 7N, LEQ 7S
DEPARTURES
(RWYS 04, 22)
TO SOUTHEAST
CAT A, B & C



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334' per nm (5.5%).

Gnd speed-Kts	75	100	150	200	250	300
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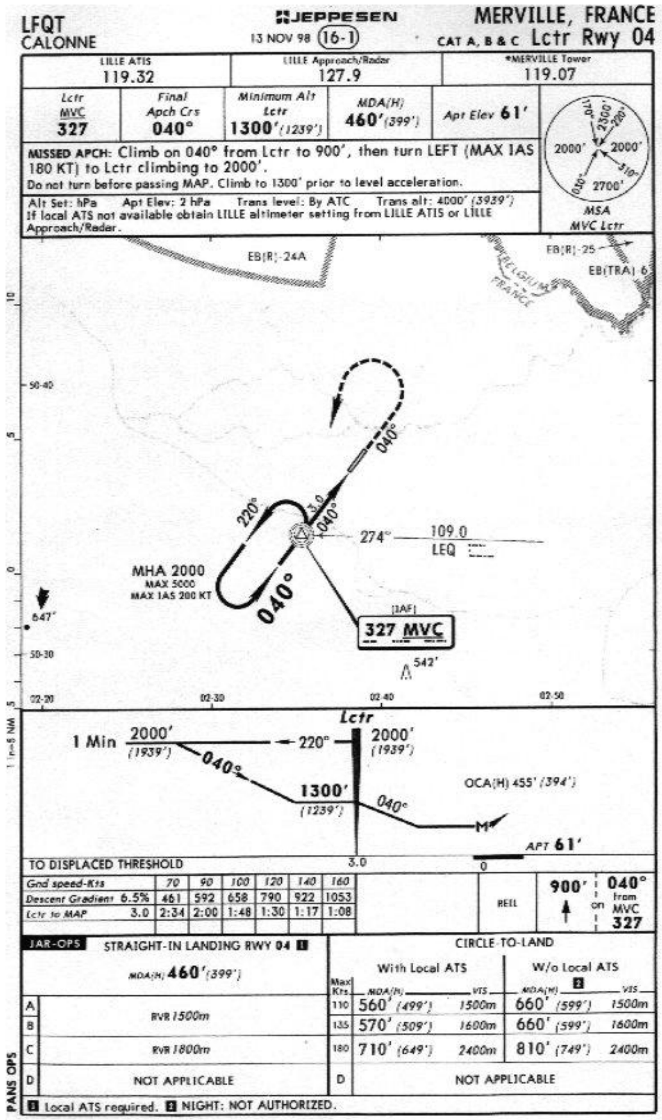
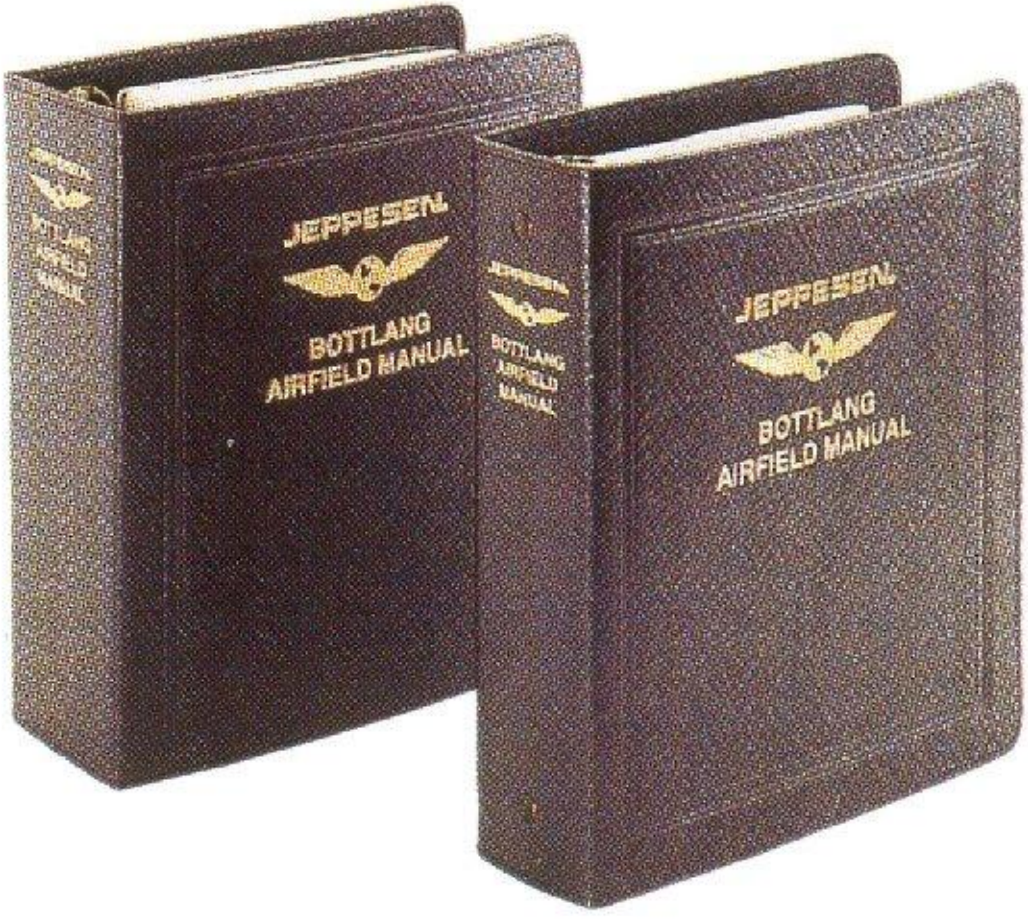
CHANGES: SIDs renumbered.

Merville Lctr Rwy 04



Jeppesen manuals



Presentation

- 2 manuals

- First manual:
- you can find EN ROUTE charts,and different information about Meteorology, the airport, the radio aids,etc…
- Second manual
- You can find TERMINAL CHARTS (departure, arrival,different procedures…)

Tome 1

Annual Update

JEPPESEN GMBH

FRANCE COV. FRA04

This a complete listing of all effective sheets appearing in your Jeppesen Airway Manual.

Record Sheets required on the Checklist Handling Form(s). Return only the Checklist Handling Form(s) with recorded pages. Do not return the checklist.

Please note tabs are not included as a checklist item.

Whenever you have a page with a more recent date than the date listed on the checklist, retain that chart and do not reorder. Manual sheets that are not shown on this checklist but that are in your manual should be removed and destroyed. If you need less than 40 sheets per binder, they may be obtained at no charge. *

*** If this completed checklist indicates that you require more than 40 sheets per binder, new contents priced at EURO 76.50 (Postage extra) per binder will be required.**

THIS CHECKLIST INCLUDES REVISION NO. F 26-02. PLEASE BE SURE TO COMPLETE THE CHECKLIST *BEFORE* INSERTING THE ENCLOSED REVISION NO. F 27-02.

NOTE: THIS CHECKLIST CAN NOT BE FILLED IF RECEIVED AFTER 02 AUG 02.

	TITLE PAGE (FRA VOLUME 1)	22 FEB 85	( )
	TITLE PAGE (FRA VOLUME 2)	22 FEB 85	( )
	RECORD OF REVISIONS (F)	04 JAN 02	( )
	BULLETIN FRA 98-A/98-A2	30 JAN 98	( )
	BULLETIN FRA 98-A3/98-A4	30 JAN 98	( )
	BULLETIN FRA 99-J/99-J2	12 NOV 99	( )
	BULLETIN FRA 01-A/01-A2	27 APR 01	( )
	BULLETIN FRA 01-A3/01-A4	27 APR 01	( )
	INTRODUCTION		
	INTRODUCTION I/II	03 AUG 01	( )
L	INTRODUCTION JAR-OPS 1/2 (JAA)	10 SEP 99	( )
	INTRODUCTION 1/2	26 MAR 99	( )
	INTRODUCTION 3/4	27 JUN 97	( )
	INTRODUCTION 5/6	04 FEB 00	( )
	INTRODUCTION 7/8	04 FEB 00	( )
	INTRODUCTION 9/10	27 JUN 97	( )
	INTRODUCTION 11/12	27 JUN 97	( )
	INTRODUCTION 13/14	14 MAY 99	( )
	INTRODUCTION 41/42	06 JUL 01	( )
	INTRODUCTION 43/44	06 JUL 01	( )
	INTRODUCTION 45/46	03 AUG 01	( )

Weekly Update

JEPPESEN®

Airway Manual Services Record of Revisions

FRANKFURT PRODUCED REVISIONS

CUSTOMER NUMBER: ____________

ASSIGNED TO: ____ ____________ LOCATION: ____________

REVISION NUMBER	REVISION DATE	REVISION NUMBER	REVISION DATE	REVISION NUMBER	REVISION DATE
1	04-01-02	24	14-06-02	47	
2	11-01-02	25	21-06-02.	48	
3	18-01-02	26		49	
4	25-01-02	27		50	
5	01-02-02	28	12-07-02	51	
6	08-02-02	29	19-07-02	52	
7	15-02-02	30	26-07-02	53	
8	22 02-02	31	02-08-02.	54	
9	01-03-02	32	09-08-02	55	
10	08-03-02	33	16-08-02	56	
11	15-03-02	34	23-08-02	57	
12	22-03-02	35	30-08-02	58	
13	29-03-02	36	06-09-02.	59	

Control slip

Customer Comment Slip

Thank you for purchasing Jeppesen's Flight Information Services, they are the best and most comprehensive service available. We keep it that way by listening to our customers and continuously improving our service.

We solicit your ideas and suggestions on all aspects of the service you are receiving and will commit to implement as many suggestions as feasible.

This comment slip is provided for your convenience, and may also be used for communicating with your Customer Service Representative.

Your comments:

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To enable us to identify your subscription, and to contact you where required, please copy the details from your mailing label and add your phone and/or fax numbers. Capital letters would be appreciated.

Name: ____________________ Customer #: ____________________

Street: ____________________ City: ____________________

Country: ____________________ Telephone: ____________________

Fax: ____________________ e-mail: ____________________

Briefing bulletin

JEPPESEN
BRIEFING BULLETIN

JEP 02-C

VERTICAL NAVIGATION (VNAV) - Concept Extension

Background

Beginning in December 1999, Jeppesen started to include Vertical Navigation (VNAV) descent information on selected non-precision instrument approach charts. VNAV information had previously been provided in Jeppesen's electronic navigation database (NavData) only.

VNAV path information illustrates the geometric descent path with a descent angle from the Final Approach Fix (FAF) to the Threshold Crossing Height (TCH). The inclusion of VNAV angles on non-precision approach charts was done on a limited basis.

For those non-precision approach procedures for which the State authority did not specify a descent gradient or did not provide a recommended DME/ Altitude table, a descent angle **derived** from the Jeppesen NavData database is to be shown on the corresponding approach chart.

This angle, if used by certified VNAV-capable avionics equipment, will ensure a stable, constant rate of descent clearing all intervening altitude restrictions (step-down fixes) established by the State authority.

Extending the Concept

As next step of the VNAV implementation plan, Jeppesen will start providing, on a routine basis, the VNAV elements for straight-in non-precision operations when a descent gradient and/or recommended DME/Altitude information table has been specified by the controlling State authority.

In these cases, a descent angle will be **converted** from the State specified

Introduction: Table of contents

TABLE OF CONTENTS

Below is a complete list of the standard contents of Airway Manual. *Limited or special coverages may not contain all items*, but that material which is included should be arranged in the order outlined.

Introduction: Chart glossary

CHART GLOSSARY

CHART NOTAMS — Jeppesen Chart NOTAMs include significant information changes affecting Enroute, Area, and Terminal charts. Entries are published until the temporary condition no longer exists, or until the permanent change appears on revised charts. Enroute chart numbers / panel numbers / letters and area chart identifiers are included for each entry in the enroute portion of the chart NOTAMs. To avoid duplication of information in combined Enroute and Terminal Chart NOTAMs, navaid conditions, except for ILS components, are listed only in the Enroute portion of the Chart NOTAMs. All times are local unless otherwise indicated. Arrows indicate new or revised information. Chart NOTAMs are only an *abbreviated* service. Always ask for pertinent NOTAMs prior to flight.

COMMON TRAFFIC ADVISORY FREQUENCY (CTAF) (USA) — A frequency designed for the purpose of carrying out airport advisory practices while operating to or from an uncontrolled airport. The CTAF may be a UNICOM, Multicom, FSS, or tower frequency.

COMMUNITY AERODROME RADIO STATION (CARS) — An aerodrome radio that provides weather, field conditions, accepts flight plans and position reports.

COMPULSORY REPORTING POINTS — Reporting points which must be reported to ATC. They are designated on aeronautical charts by solid triangles or filed in a flight plan as fixes selected to define direct routes. These points are geographical locations which are defined by navigation aids/fixes. Pilots should discontinue position reporting over compulsory reporting points when informed by ATC that their aircraft is in "radar contact."

CONDITIONAL ROUTES (CDR) – (Europe) — Category 1,2,3.

Category 1: Permanently plannable CDR during designated times.

Category 2: Plannable only during times designated in the Conditional Route Availability Message (CRAM) published at 1500Z for the 24 hour period starting at 0600Z the next day.

CONTROLLED AIRSPACE — An airspace of defined dimensions within which air traffic control service is provided to IFR flights and to VFR flights in accordance with the airspace classification.

NOTE: Controlled airspace is a generic term which covers ATS airspace Classes A, B, C, D, and E.

CONTROL ZONE (ICAO) — A controlled airspace extending upwards from the surface of the earth to a specified upper limit.

COURSE —

1. The intended direction of flight in the horizontal plane measured in degrees from north.
2. The ILS localizer signal pattern usually specified as front course or back course.
3. The intended track along a straight, curved, or segmented MLS path.

CRITICAL HEIGHT — Lowest height in relation to an aerodrome specified level below which an approach procedure cannot be continued in a safe manner solely by the aid of instruments.

DECISION ALTITUDE/HEIGHT (DA/H) (ICAO) — A specified altitude or height (A/H) in the precision approach at which a missed approach must be initiated if the required visual reference to continue the approach has not been established.

NOTE 1: Decision altitude (DA) is referenced to mean sea level (MSL) and decision height (DH) is referenced to the threshold elevation.

NOTE 2: The required visual reference means that section of the visual aids or of the approach area which should have been in view for sufficient time for the pilot to have made an assessment of the aircraft position and rate of change of position, in relation to the desired flight path.

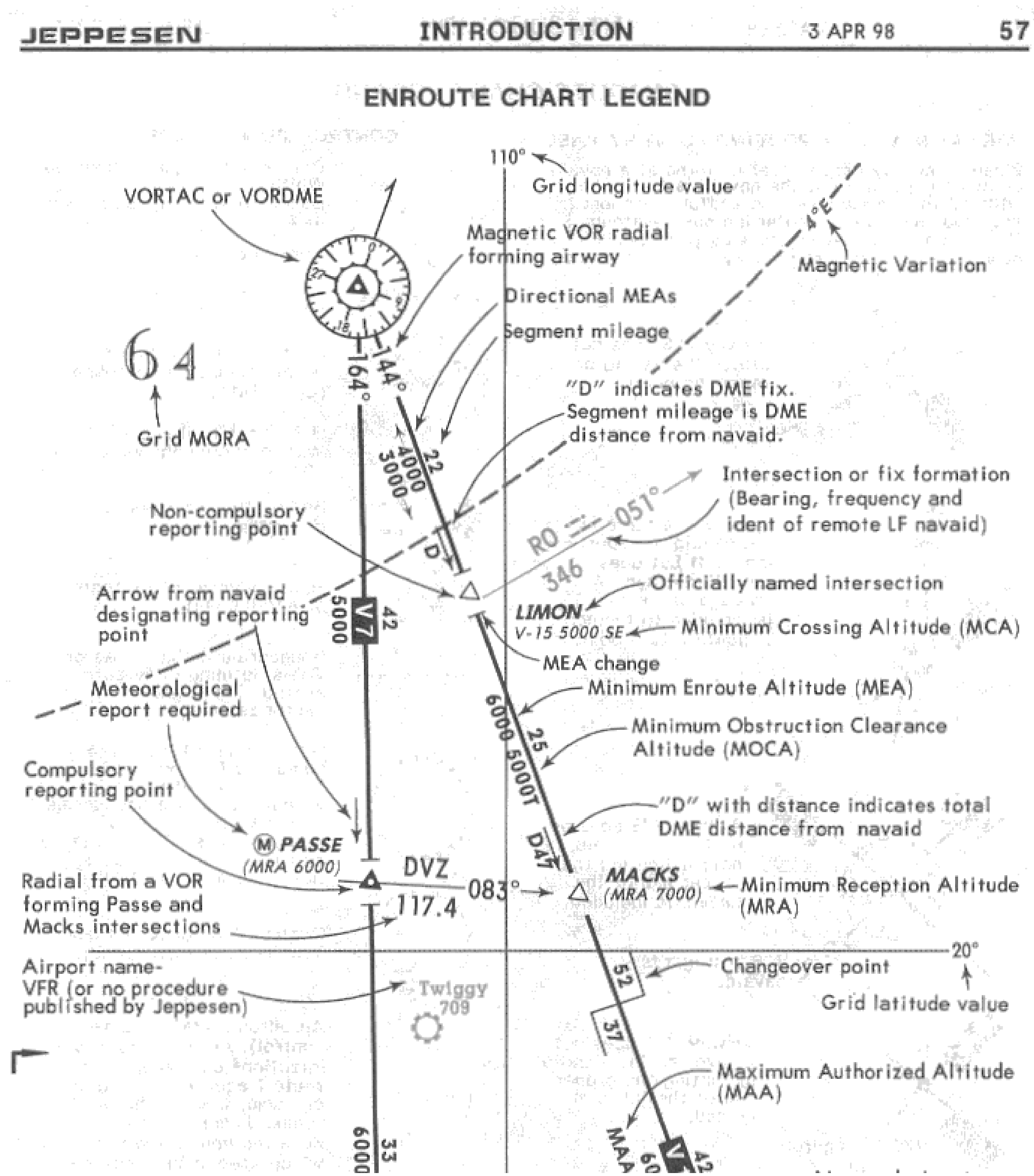
DECISION HEIGHT (DH) (USA) — With respect to

Introduction: Abbreviations

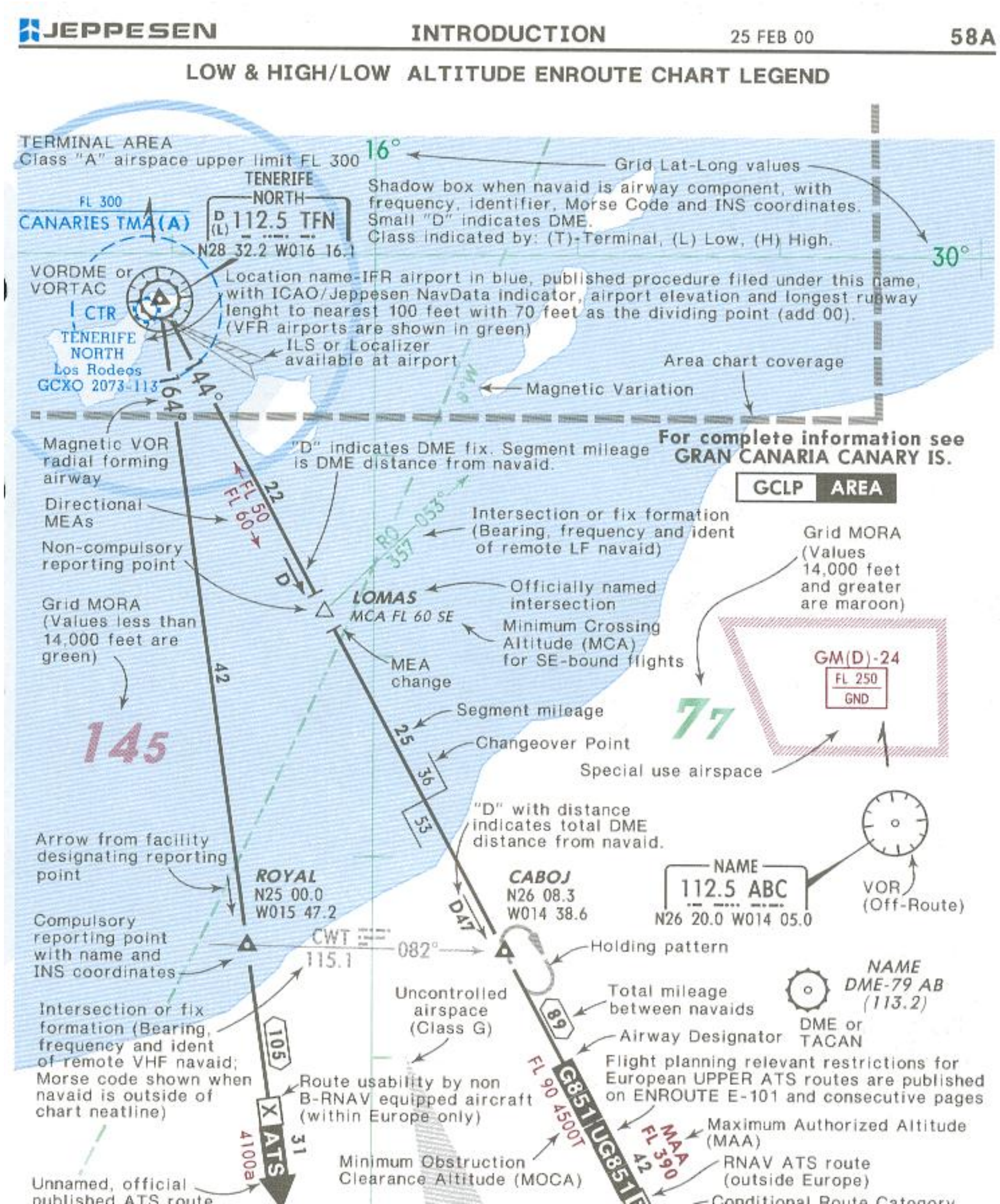
ABBREVIATIONS

A/A	Air to Air
AAF	Army Air Field
AAL	Above Aerodrome Level
AAS	Airport Advisory Service
AB	Air Base
ABM	Abeam
ABN	Aerodrome Beacon
AC	Air Carrier
ACA	Arctic Control Area
ACA	Approach Control Area
ACAS	Airborne Collision Avoidance System
ACARS	Airborne Communications Addressing and Reporting System
ACC	Area Control Center
ACFT	Aircraft
ACN	Aircraft Classification Number
AD	Aerodrome
ADA	Advisory Area
ADF	Automatic Direction Finding
ADIZ	Air Defense Identification Zone
ADR	Advisory Route
ADV	Advisory Area
AFIS	Aeronautical Enroute Information Service
AEH	Approach End Runway
AERADIO	Air Radio
AERO	Aerodrome
AF Aux	Air Force Auxiliary Field
AFB	Air Force Base
AFIS	Aerodrome Flight Information Service
AFN	American Forces Network
AFRS	Armed Forces Radio Stations
AFS	Air Force Station
AOE	Airport/Aerodrome of Entry
AOR	Area of Responsibility
APAPI	Abbreviated Precision Approach Path Indicator
APC	Area Positive Control
APCH	Approach
APP	Approach Control
APT	Airport
ARB	Air Reserve Base
ARINC	Aeronautical Radio, Inc.
ARO	Aerodrome Reporting Officer
ARR	Arrival
ARTCC	Air Route Traffic Control Center
ASDA	Accelerate Stop Distance Available
ASOS	Automated Surface Observation System
ASR	Airport Surveillance Radar
ATA	Actual Time of Arrival
ATCAA	Air Traffic Control Assigned Airspace
ATCC	Air Traffic Control Center
ATCT	Air Traffic Control Tower
ATD	Actual Time of Departure
ATF	Aerodrome Traffic Frequency
ATFM	Air Traffic Flow Management
ATIS	Automatic Terminal Information Service
ATS	Air Traffic Service
ATZ	Aerodrome Traffic Zone
AUTH	Authorized
AUW	All-up Weight
AUX	Auxiliary
AVBL	Available
AWIB	Aerodrome Weather Information Broadcast
AWOS	Automated Weather Observing

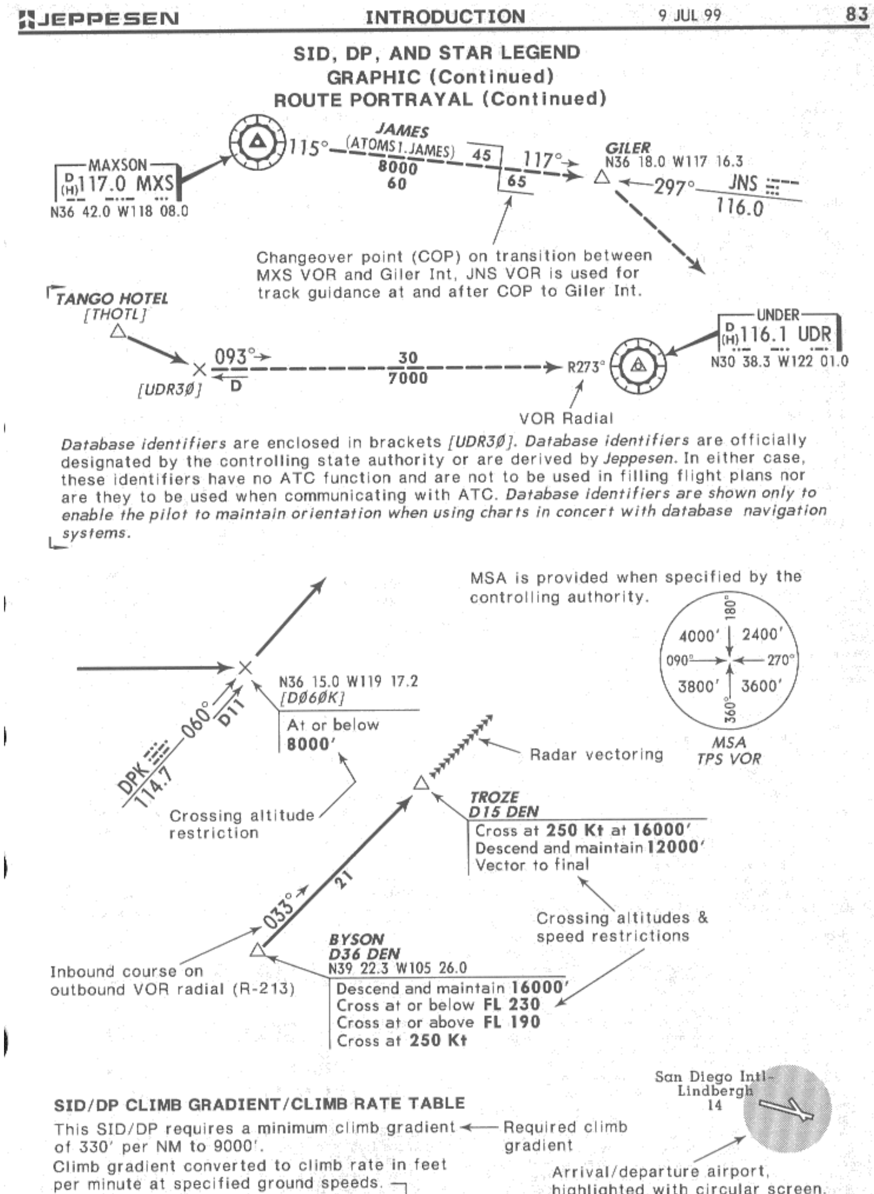
Introduction: En route chart legend



Introduction: En route legend details



Introduction: SID/STAR legend



Introduction: Approach chart legend

APPROACH CHART LEGEND

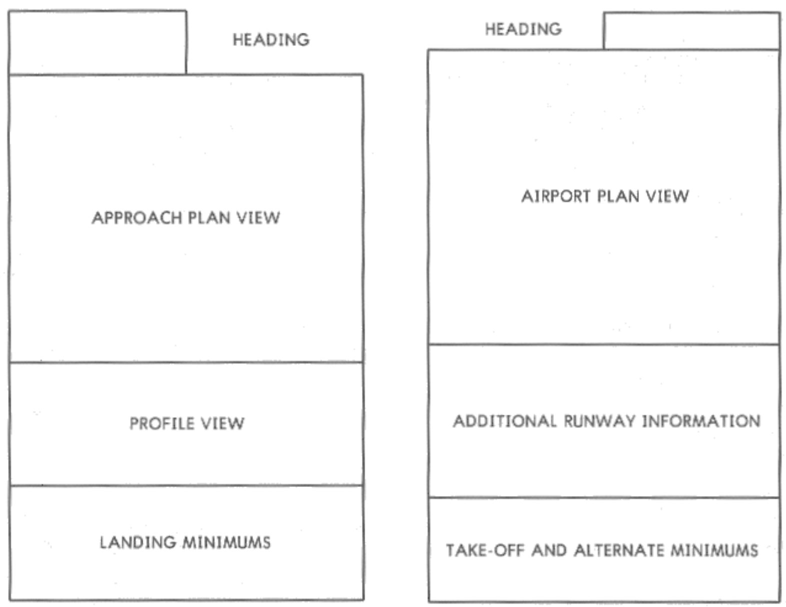
Approach charts are graphic illustrations of instrument approach procedures prescribed by the governing authority. All charts meet FAA requirements for aeronautical charts. The following legend pages briefly explain symbology used on approach charts throughout the world. *Not all items apply to all locations.* The approach chart is divided into specific areas of information as illustrated below.

FORMATS

The first approach procedure published for an airport has the procedure chart published on the front side with the airport chart on the back side. On major airports, the airport chart may proceed the first approach procedure. These locations will have expanded airport information that may occupy more than one side. When an airport has more than one published approach procedure, they are shown front and back on additional sheets. Blank pages will indicate "INTENTIONALLY LEFT BLANK".

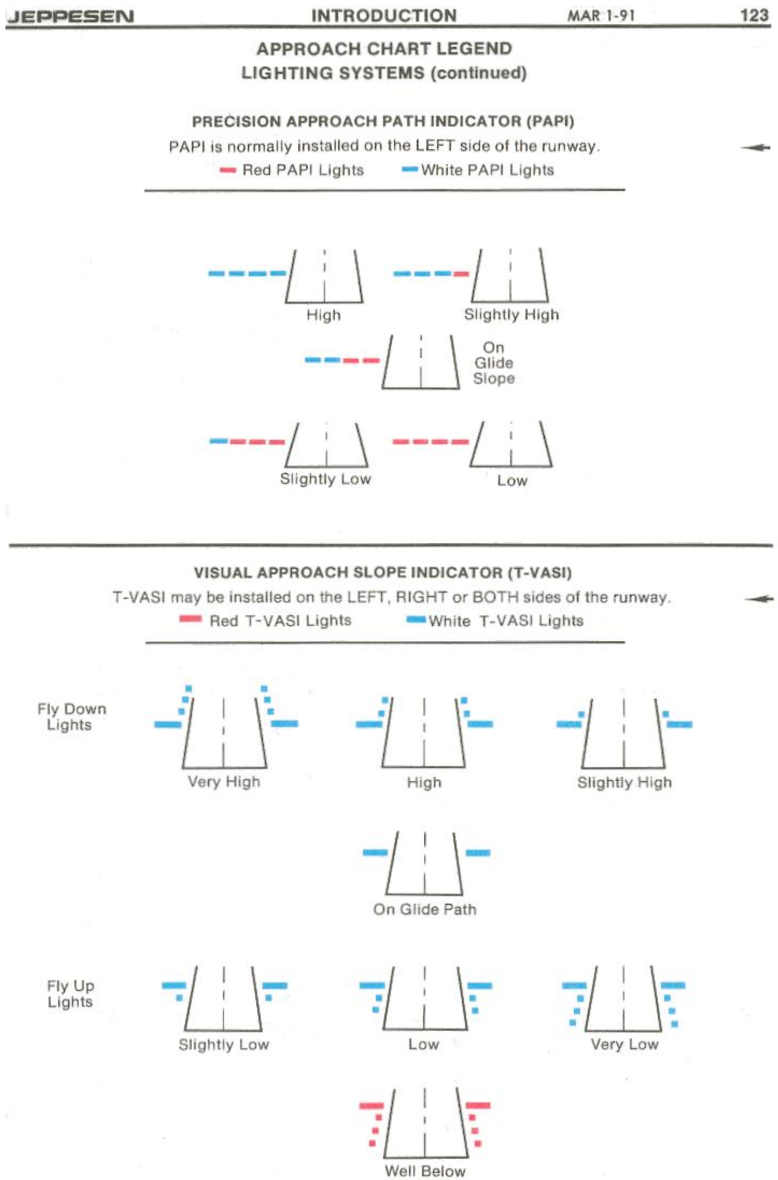
APPROACH PROCEDURE CHART FORMAT

AIRPORT CHART FORMAT



On charts dated on and after 10 MAR 95, key information is displayed in bold type. Key information includes Communication frequencies, Primary NAVAID frequency and identifier, Procedure bearings and Altitudes, Airport and runway end elevation, Decision Altitude and Minimum Descent Altitude, and Missed Approach turn limit and direction, course and altitude.

Introduction: VASI:PAPI



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APPROACH CHART LEGEND
LIGHTING SYSTEMS (continued)

PRECISION APPROACH PATH INDICATOR (PAPI)
PAPI is normally installed on the LEFT side of the runway.
Red PAPI Lights White PAPI Lights

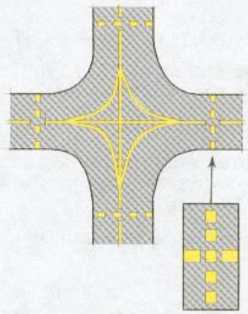
VISUAL APPROACH SLOPE INDICATOR (T-VASI)
T-VASI may be installed on the LEFT, RIGHT or BOTH sides of the runway.
Red T-VASI Lights White T-VASI Lights

Introduction: ICAO airport signs

ICAO RECOMMENDED AIRPORT SIGNS AND RUNWAY MARKINGS

TAXIWAY INTERSECTION MARKING

A taxiway intersection marking is displayed at the intersection of two paved taxiways.



RUNWAY INTERSECTION SIGNS

15-33	Taxiway intersects runway 15-33. Runway 15 threshold is to the left, runway 33 threshold is to the right. Arrangement of numbers indicates direction to the corresponding runway threshold.
33	Taxiway intersects the beginning of the take-off runway 33. (The sign at each take-off end contains only the runway number for the take-off runway while all others contain both runway numbers).
CAT II	Indicates a category II taxi-holding position.
CAT III	Indicates a category III taxi-holding position.
CAT II/III	Indicates a joint category II/III taxi-holding position.
NO ENTRY	Indicates that entrance to an area is prohibited.

TYPICAL SIGNS AT TAXI-HOLDING POSITIONS
ASSOCIATED TAXIWAY MARKING ALSO SHOWN

Chart Notams

EUROPE

Jeppesen Chart NOTAMS highlight only ***significant*** changes affecting *Jeppesen Charts*, also regularly updated at **www.jeppesen.com**
IMPORTANT: ASK FOR OTHER PERTINENT NOTAMS PRIOR TO FLIGHT

ENROUTE CHARTS

GENERAL

BELGIUM

Operations of state acft without FM immunized VHF receivers are only allowed when acft are equipped and certified according RNP/5 requirements without the use of VOR; non-FM immune state acft withoutRNP/5 requirements still get the opportunity to fly as OAT along TACAN Routes or under radar control.

Italy replaces the UK as provider of ATS, AIS and MET Service on behalf of the United Nations mission in Kosovo (UNMIK). Civil Operators should note, that the MIL procedures may not be compatible with ICAO standards. NOTAM are issued from LIIAYNYX as LYPR series K to disseminate supplementary information or changes to UNMIK AIC 2/2000. All acft opr into Kosovo are to note that only NOTAM for Kosovo issued by NOF Italy (LIIAYNYX) are valid. All other NOTAM from other agencies are to be disregarded. UNMIK AIC 2/2000 will remain avbl ufn on the UK AIS Website. ..

U.K.

STANDARD ROUTES - UNITED KINGDOM previously published on Enroute pages E-24 to

AUSTRIA

ATS ROUTES changed:

B5, Klagenfurt VORDME to Salzburg VORDME. SFOR flights exempted from restrictions ufn. **E LO-12/6ACD**.
L604, Graz VORDME to Linz VORDME, SFOR flights exempted from restrictions ufn. **E LO-12/6ABD**.
L156 GOLVA (N4642.5 E01539.1) via Gleichenberg NDB, Sollenau VORDME and Wagram VORDME to LANUX. also available for NON-RNAV equipped aircraft. **LOWW**.
L725 extended: Sollenau VORDME to ABLOM, 068°/248°, 34NM, MEA 6000'. **LOWW**.
R23, FIR [FIR18] (N4816.3 E01654.5) replaced by NCRP TOVKA at N4816.2 E01655.6. **LOWW**.
T161, Linz VORDME to SUBEN, SFOR flights exempted from restrictions ufn. **E LO-12/6A**.
UB5, Klagenfurt VORDME to Salzburg VORDME. SFOR flights exempted from restrictions ufn. **E HI-9/2&3**, **E HI-11/2&3**.
UL604, Graz VORDME to Linz VORDME, SFOR flights exempted from restrictions ufn. **E HI-9/2&3**.
UT161, Linz VORDME to SUBEN, SFOR flights exempted from restrictions ufn. **E HI-8/1**. **E HI-9/2**.

BELARUS, REP OF

ATS ROUTES changed:

→ A68, SOMIS (N5227.9 E02937.0) withdrawn;

EUROPE

UW640, OKBIS (N4947.1 E02624.3) withdrawn; Zolochiv NDB - SORON, 114NM. **E HI-15/7**.
UW689, MALBE (N4849.4 E02222.5) to BUKOV, redesignated UN190. **E HI-9/5**, **E HI-15/7&8**.

YUGOSLAVIA

ATS ROUTES changed:

B5 changed/extended: NCRP LOKSI (N4434.3 E01910.8) - Valjevo VORDME, 113°/293°, 34NM, MEA 7000'; Valjevo VORDME to VEMAK, changed to two-way. **E LO-13/4AB**.
G16 extended: Niksic NDB - NCRP PERAN, 321°/141°, 19NM, MEA 10000'. **E LO-13/4C**.
G23 extended: Valjevo VORDME - NCRP BATRU, 246°/066°, 24NM, MEA 8000'. **E LO-13/4AB**.
UG16, GAC rplcd by CRP Gacko NDB at same position. **E HI-11/5**.

TERMINAL CHARTS

GENERAL

FRANCE

For all Circle-to-land with or without prescribed flight tracks proc, CEIL values no longer required for standard operational minima.

ITALY

Coded designators introduced for all Italian STARs and SIDs which terminate at a radio navigation facility. The coded designator consists of the three-letter radio navigation facility identification followed by the validity and the route indicator. Example: BOLZANO 5R (BZO 5R). Level acceleration portion note in missed apch to be deleted from all Italian IAP, PANS OPS 4 applies.

SWEDEN

All IAPs: Acceleration portion in missed apch no longer applicable, PANS OPS 4 applies.
Omnidirectional dep procedures avbl at certain airports.
SIDs, minimum climb gradient of 400' per NM

BELGIUM

State acft not equipped with FM immune VHF receivers operating in civ TMAs are subject to PPR, those acft may expect an approval provided they operate under VMC or provided that the respective ATC unit has the possibility to assist in the approach by radar surveillance without VOR/ILS or is able to offer a NDB approach procedure; non-FM immune state acft operating within mil TMAs are not allowed to use their VOR/ILS equipment.

A Coruna, Spain, (10-9A) Low Visibility Procedures not avbl.
Aalborg, Denmark, GND svc 'AALBORG Handling' avbl on freq 131.55.
Albenga, Italy, (10-6/General Rules) Ufn ldg on rwy 09 is allowed by day, following a left-hand circuit, north of rwy at an altitude not below 2000'.
(11-1) VMC minimums chgd: Ceiling not below 2500' by day and 3000' by night.
Alderney, U.K., (16-1/16-2) IAPs no longer estbld outside controlled airspace.
(16-2) Until 30 NOV 02 IAP Cat A NDB rwy 26 suspended, except when Radar Vectors provided.
Alghero, Italy, (Fertilia) ARRs 10-2, MINKA - x-break Alghero R-067/24 DME - D17 ALG, MEA FL100/5000'.
Ufn ILS OM rwy 21 u/s.
Alicante, Spain, Ufn twy C4 clsd.
→ **Alta**, Norway, Ufn (chart 13-3) IAP VORDME 185° suspended.
Amsterdam, Netherlands, (Schiphol) After obtaining and read-back enroute clearance pilots are strongly requested to switch to 'SCHIPHOL Start-up' on freq 119.90 without ATC instructions.
Caution: Due to construction of new rwy 18/36 2100m northwest of rwy 01L/19R, do not mistake Highway for rwy 18/36 or 01L/19R ufn.
Daily 2300-0600LT rwy 09/27 clsd for T/O.
Daily 2300-0600LT straight-in approaches rwy 22 and rwy 24 not permitted.
'SCHIPHOL Delivery' 121.97(5) solely ops as clearance delivery and provides no longer start-up svc.

Radio Aids: General information

GENERAL INFORMATION

FREQUENCY ALLOCATION

Frequency allocation is established to provide a clear channeling between the various functions performed by aeronautical navaids and communications facilities. Although a general allocation plan is recognized on a world-wide basis, variations may occur within certain ranges. The listing below is intended to provide that allocation most generally used by civil operators.

NAVIGATION AIDS

200 - 415 kHz	Non-directional Beacon (low power) and Radio Range (low power).
200 - 1750 kHz	Non-directional Beacon (standard).
75 MHz	Marker Beacon.
108.0 - 118.0 MHz	VOR test facility (VOT).
108.10 - 111.95 MHz	ILS localizer (on odd-tenths decimal frequencies, 108.1, 108.3 etc.)
108.0 - 111.8 MHz	VOR (even tenths of MHz).
112.0 - 117.95 MHz	VOR (even and odd tenths of MHz).
329.15 - 335.00 MHz	ILS glide slope.
960.0 - 1215.0 MHz	DME and TACAN.

AIRBORNE STATIONS

410 kHz	International DF (outside continental USA).
475 kHz	Working frequency exclusively for aircraft on sea flights desiring an intermediate frequency
500 kHz	International frequency for aircraft and ships over the seas. Transmission on this frequenc (except for urgent and safety messages and signals) must cease twice each hour, for thre minute periods beginning at 15 and 45 minutes past each hour.
3281 kHz	Lighter-than-air craft.
121.5 MHz	Universal simplex clear channel frequency for use by aircraft in distress or condition of emergency. Emergency Locator Beacons transmit on 121.5 MHz.
122.9 & 123.1 MHz	Air-to-air or air-to-ground communications with aeronautical search and rescue stations when engaged in search and rescue operations

118.0 - 135.975 MHz AIR TRAFFIC CONTROL OPERATIONS

NOTE: The radio spectrum between 118.0 and 136.0 MHz on 50 kHz channeling (i.e., 50 kHz spacing) could contain 360 channels. 100 kHz protection is provided to 121.5 MHz (emergency), so the number is 358. 25 kHz channeling is implemented in some areas and could afford 720 channels - - with 121.5 MHz, 100 kHz protection it will afford 714 channels. Jeppesen charts only "Two" decimal places; i.e., xxx.025 is shown as xxx.02, xxx.050 as xxx.05, xxx.075 as xxx.07, etc.

121.6 - 121.925 MHz	Airport utility frequencies (ground control). In addition, may be used for control of airport lights by transmissions of brief keyed RF signals from aircraft.
121.975 - 123.075 MHz	Also available to private aircraft for air traffic control operations.
123.175 - 123.425 MHz & 3281 kHz	Available for ground and aircraft flight test

Radio data: identifier

NAVIGATION AIDS LISTED BY IDENTIFIER - EUROPE

Ident	Name
A	
A	ALFA (JYVASKYLA), FINLAND
A	TIRANA, ALBANIA
AA	BARAJAS (MADRID), SPAIN
AAL	AALBORG, DENMARK
AB	ABSAM, AUSTRIA
AB	AKRABERG, FAROE IS
AB	ALBI, FRANCE
AB	HAGFORS, SWEDEN
ABB	ABBEVILLE, FRANCE
ABD	ABYAD (DAMASCUS), SYRIA
ABN	ALBENGA, ITALY
ABU	ABU ARGUB, LIBYA, SPA JAMAHIRIYA
ABU	ALTENBURG, GERMANY
AC	GLASGOW, UNITED KINGDOM
AC	HULTSFRED, SWEDEN
ACD	ALCOBENDAS, SPAIN
AD	ANDA (SANDANE), NORWAY
ADA	ADANA, TURKEY
ADM	AL-MASSIRA (AGADIR), MOROCCO
ADN	ABERDEEN, UNITED KINGDOM
ADR	ADRAR, ALGERIA
ADV	ADVENT (SVALBARD), NORWAY
ADX	ANDRAITX, SPAIN
AFI	AFFLIGEM, BELGIUM
AG	AGEN, FRANCE
AGB	AUGSBURG, GERMANY
AGD	ALTENBURG, GERMANY
AGH	NEA ANCHIALOS, GREECE
AGN	AGEN, FRANCE
AGO	ANGOULEME, FRANCE
AH	ALGHERO, ITALY
AH	ANGELHOLM, SWEDEN
AHO	ALGHERO, ITALY
AI	ALICANTE, SPAIN
AJO	AJACCIO, FRANCE
AK	OUKACHA, MOROCCO
AKC	TRABZON, TURKEY
AKI	AKHISAR, TURKEY
AKR	AKROTIRI, CYPRUS
AKU	AKUJARVI (IVALO), FINLAND
AL	ALENCON, FRANCE
AL	LEPSOY (ALESUND), NORWAY
ALB	ALBORAN, SPAIN
ALB	ALLERSBERG, GERMANY
ALD	ALDERNEY, UNITED KINGDOM
ALE	ALEPPO, SYRIA

Ident	Name
ANE	ARLANDA (STOCKHOLM), SWEDEN
ANG	ANGERS, FRANCE
ANK	ANKARA, TURKEY
ANL	NEA ANGHIALOS, GREECE
ANS	ANSBACH, GERMANY
ANT	ANTON, FINLAND
ANT	ANTWERP (DEURNE), BELGIUM
ANW	ARLANDA (STOCKHOLM), SWEDEN
ANY	ANDOYA, NORWAY
AOG	ROTA, SPAIN
AOR	ASTOR, SWEDEN
AP	ABERPORTH, UNITED KINGDOM
AQ	ABERDEEN, UNITED KINGDOM
AQ	AQABA, JORDAN
AQA	AQABA, JORDAN
AQB	AQABA, JORDAN
AQC	AQABA, JORDAN
AR	AURILLAC, FRANCE
ARA	ARAXOS, GREECE
ARB	TORTOLI, ITALY
ARF	ARIFIYE (TOPEL), TURKEY
ARH	EL ARISH, EGYPT
ARI	AGRI, TURKEY
ARL	ARLANDA (STOCKHOLM), SWEDEN
ARN	ARBANCON, SPAIN
ARP	ARPE, GERMANY
ARS	AROS, SWEDEN
ARW	NADOR, MOROCCO
ARX	ARAXOS, GREECE
AS	ALFA-SIERRA (KEMI), FINLAND
AS	ANGERS, FRANCE
AS	ARVIDSJAUR, SWEDEN
AS	ASMARA, ERITREA
ASK	ASKOY (BERGEN), NORWAY
ASM	ASMARA, ERITREA
ASN	ASWAN, EGYPT
AST	ASYUT, EGYPT
ASW	ARLANDA (STOCKHOLM), SWEDEN
AT	ANNECY, FRANCE
ATA	ALTA, NORWAY
ATF	ABERDEEN, UNITED KINGDOM
ATH	ATHENS, GREECE
ATL	ASTIPALAIA, GREECE
ATN	AUTUN, FRANCE
ATR	ALCANTARILLA (MURCIA), SPAIN
AU	STAUNING, DENMARK
AUB	AUBENAS, FRANCE

Radio data: list by state

JEPPESEN 1 NOV 02 **RADIO AIDS** **FRANCE-1**

NAVIGATION AIDS

Name	Ident	Freq.	Class	INS Coordinates		Var.	Elev.
France							
Abbeville	ABB	108.45	V D H W	N50 08.1	E001 51.3	W003	220
Agen	AG	400.0	H L W	N44 09.0	E000 40.4	W003	204
Agen	AGN	114.8	V D H W	N43 53.3	E000 52.4	W002	896
Aix-Les-Milles	ALM	430.0	H L W	N43 30.5	E005 21.6	W001	367
Ajaccio	AJO	114.8	V D H W	N41 46.2	E008 46.5	E000	2142
Ajaccio	CT	387.0	H L W	N41 47.7	E008 43.4	E000	18
Ajaccio	IS	341.0	H W	N41 53.9	E008 36.7	E000	
Ajaccio	RB	365.0	H L W	N41 54.9	E008 48.1	E000	18
Albi	AB	323.0	H L W	N43 54.8	E002 03.9	W001	564
Alencon	AL	380.0	H L W	N48 26.1	E000 06.7	W003	479
Ales	DA	402.0	H L W	N44 03.1	E004 08.5	W002	712
Amberieu	AMU	116.3	T H	N45 59.3	E005 19.9	W001	820
Amboise	AMB	113.7	V D H W	N47 25.7	E001 03.9	W002	387
Amboise	AMB	341.0	H W	N47 25.1	E001 02.5	W002	
Amiens	GI	339.0	H L W	N49 50.5	E002 29.1	W003	197
Angers	ANG	113.0	V H W	N47 32.2	W000 51.1	W003	
Angers	AS	392.0	H L W	N47 34.6	W000 09.1	W003	194
Angouleme	AGO	404.0	H L W	N45 42.6	E000 25.6	W003	436
Annecy	AT	384.0	H L W	N45 51.7	E006 01.2	W000	1521
Aubenas	AUB	427.0	H L W	N44 26.5	E004 21.7	W001	919
Auch	LMT	420.0	H L W	N43 43.1	E000 36.2	W003	410
Aurillac	AR	343.0	H L W	N44 57.3	E002 22.1	W002	2096
Autun	ATN	114.9	V D H W	N46 48.4	E004 15.5	W001	2244
Auxerre	AX	417.0	H L W	N47 55.2	E003 30.1	W001	523
Avignon	AVN	112.3	V U W	N43 59.7	E004 44.8	W001	148
Avord	AV	306.5	H L W	N46 52.9	E002 55.7	W001	577
Avord	AVD	110.6	T H	N47 03.5	E002 37.8	W001	597
Avord	AVD	288.5	H W	N47 07.2	E002 47.1	W002	577
Basle-Mulhouse	BLM	117.45	V D U W	N47 37.1	E007 29.1	W000	886
Basle-Mulhouse	BN	353.0	H M W	N47 39.7	E007 28.7	W000	864
Basle-Mulhouse	BS	376.0	H L W	N47 34.8	E007 32.2	W000	885
Bastia	BP	369.0	H L W	N42 25.5	E009 32.2	E000	26
Bastia	BTA	114.15	V D H W	N42 34.4	E009 28.5	E000	36
Beauvais	BV	391.0	H L W	N49 29.5	E002 01.8	W002	359
Beauvais	BVS	115.9	V H W	N49 26.2	E002 09.2	W002	
Bergerac	BGC	374.0	H L W	N44 49.2	E000 37.2	W002	171
Besancon	BSV	370.0	H M W	N47 16.0	E006 12.2	W000	1271
Beziers	ZR	397.0	H L W	N43 19.6	E003 16.1	W002	55
Biarritz	BTZ	114.15	V D H W	N43 27.1	W001 30.6	W002	259
Biarritz	BZ	341.0	H M W	N43 28.2	W001 24.2	W002	245
Biscarrosse	BRS	358.0	H L W	N44 21.1	W001 07.7	W003	98
Blois	BLB	397.0	H L W	N47 40.4	E001 12.1	W003	398

Meteorology: General Informations

METEOROLOGICAL SERVICE FOR INTERNATIONAL AIR NAVIGATION – ANNEX 3

NOTE: In this part of the METEOROLOGY section, selected Chapters and paragraphs have been extracted from ICAO Annex 3 - Meteorological Service for International Air Navigation. Chapter and paragraph numbers reflect those contained in the Annex.

CHAPTER 1 – DEFINITIONS

1.1 DEFINITIONS

When the following terms are used in the Standards and Recommended Practices for Meteorological Service for International Air Navigation, they have the following meanings:

AERODROME CLIMATOLOGICAL SUMMARY – Concise summary of specified meteorological elements at an aerodrome, based on statistical data.

AERODROME CLIMATOLOGICAL TABLE – Table providing statistical data on the observed occurrence of one or more meteorological elements at an aerodrome.

AERODROME METEOROLOGICAL OFFICE – An office, located at an aerodrome, designated to provide meteorological service for international air navigation.

AERONAUTICAL FIXED TELECOMMUNICATION NETWORK (AFTN) – A world-wide system of aeronautical fixed circuits provided, as part of the aeronautical fixed service, for the exchange of messages and/or digital data between aeronautical fixed stations having the same or compatible communications characteristics.

AERONAUTICAL METEOROLOGICAL STATION – A station designated to make observations and meteorological reports for use in international air navigation.

AIRCRAFT OBSERVATION – The evaluation of one or more meteorological elements made from an aircraft in flight.

AIRMET INFORMATION – Information issued by a meteorological watch office concerning the occurrence or expected occurrence of specified en-route weather phenomena which may affect the safety of low-level aircraft operations and which was not

AREA OF RESPONSIBILITY (World Area Forecast System) – A geographical area for which a regional area forecast centre prepares significant weather forecasts.

AUTOMATIC DEPENDENT SURVEILLANCE (ADS) – A surveillance technique in which aircraft automatically provide, via a data link, data derived from on-board navigation and position-fixing systems, including aircraft identification, four-dimensional position and additional data as appropriate.

BRIEFING – Oral commentary on existing and/or expected meteorological conditions.

CONSULTATION – Discussion with a meteorologist or another qualified person of existing and/or expected meteorological conditions relating to flight operations; a discussion includes answers to questions.

EXTENDED RANGE OPERATION – Any flight by an aeroplane with two turbine power-units where the flight time at the one power-unit inoperative cruise speed (in ISA and still air conditions), from a point on the route to an adequate alternate aerodrome, is greater than the threshold time approved by the State of the Operator.

FLIGHT DOCUMENTATION – Written or printed documents, including charts or forms, containing meteorological information for a flight.

FORECAST – A statement of expected meteorological conditions for a specified time or period, and for a specified area or portion of airspace.

GAMET AREA FORECAST – An area forecast in abbreviated plain language for low-level flights for a flight information region or sub-area thereof, prepared by the meteorological office designated by the meteorological authority concerned and exchanged with meteorological offices in adjacent flight informa-

TAF decode

APPENDIX 4. TECHNICAL SPECIFICATIONS FOR AERODROME FORECASTS IN THE TAF CODE FORM

Table A4-1. Template for Aerodrome Forecasts in the TAF Code Form

Key:
M = inclusion mandatory, part of every message
C = inclusion conditional, dependent on meteorological conditions or method of observation
O = inclusion optional

Element as specified in Chapter 6	Detailed content	Template(s)	Examples
Identification of the type of report (M)	Type of report (M)	TAF or TAF AMD	TAF TAF AMD
Location indicator (M)	ICAO location indicator (M)	nnnn	YUDO[1]
Date and time of origin of forecast (M)	Date and time of the origin of the forecast in UTC (M)	nnnnnnZ	160000Z
Date and period of validity of forecast (M)	Date and period of the validity of the forecast in UTC (M)	nnnnnn	160624 080918
Surface wind (M)	Wind direction (M)	nnn or VRB[3]	24015KMH; VRB06KMH (24008KT); (VRB03KT) 19022KMH (19011KT)
	Wind speed (M)	[P]nn[n]	00000KMH (00000KT) 140P199KMH (140P99KT)
	Significant speed variations (C)[2]	G[P]nn[n]	12012G35KMH (1206G18KT) 24032G54KMH (24016G27KT)
	Units of measurement (M)	KMH or KT	
Visibility (M)	Minimum visibility (M)	nnnn C A V O K	0350 7000 9000 9999 CAVOK

Met Broadcast

AVAILABILITY OF MET BROADCASTS - EUROPE
Radiotelephony

Identify location for which weather is desired and find station(s) disseminating broadcast. Contents of broadcast of each station starts on page E-6.

WEATHER FOR:	AVAILABLE FROM STATIONS:
Aalborg	Copenhagen
Aberdeen	Scottish
Adana	Royal Air Force
Agadir	Casablanca, Las Palmas
Aigen/Ennstal	Gerlitzen
Ajaccio	Marseille
Akrotiri	Royal Air Force
Alexandria	Cairo
Algiers	Algiers, Alicante
Alicante	Alicante, Madrid
Allentsteig	Rauchenwarth
Alpe Rauz	Zugspitze
Alta	Bodo
Altenrhein	Zugspitze
Amman	Beirut, Tel Aviv
Amsterdam	Amsterdam, Brussels, Frankfurt, Hannover, London (Main), Paris, Shannon
Ancona	Royal Air Force
Andoya	Bodo
Andravida	Athens, Brindisi
Angelholm	Jonkoping
Ankara (Esenboga)	Ankara, Beirut, Istanbul
Annaba	Algiers
Antalya	Istanbul
Asturias	Santiago
Aswan	Cairo
Athens (Eleftherios Venizelos Intl)	Athens, Nicosia
Athens (Hellinikon)	Athens, Brindisi, Cairo, Istanbul
Aviano	Royal Air Force
Barcelona	Algiers, Barcelona, Bordeaux, Madrid, Marseille, Santiago, Shannon
Bardufoss	Bodo, Royal Air Force
Basle-Mulhouse	Frankfurt, Geneva, Paris, Pisa, Zurich

WEATHER FOR:	AVAILABLE FROM STATIONS:
Bolzano	Zugspitze
Bournemouth	London (South)
Bremen	Hannover
Brest	Santiago
Brindisi	Brindisi, Royal Air Force
Bristol	London (South)
Brize Norton	Royal Air Force
Bruggen	Royal Air Force
Brussels	Amsterdam, Brussels, Frankfurt, London (Main), Paris, Shannon
Bucharest (Otopeni)	Belgrade, Istanbul
Cairo	Athens, Beirut, Cairo
Cardiff	London (South)
Casablanca (Anfa)	Las Palmas, Seville (San Pablo)
Casablanca (Mohammed V)	Casablanca
Catania (Fontanarossa)	Malta, Rome
Cologne-Bonn	Brussels, Frankfurt, Hannover, Shannon
Coltishall	Royal Air Force
Constantine	Algiers
Copenhagen	Amsterdam, Berlin (Schonefeld), Copenhagen, Hannover, Oslo, Shannon, Stockholm, Warsaw
Cork	Dublin
Cranwell	Royal Air Force
Damascus	Beirut, Cairo, Nicosia
Djerba	Tunis
Dresden	Berlin (Schonefeld)
Dublin	Dublin, London (Main), Shannon
Dubrovnik	Zagreb
Dusseldorf	Amsterdam, Brussels, Frankfurt, Shannon
East Midlands	London (North)
Edinburgh	Scottish
Evenes	see Harstadt-Narvik
Faro	Lisbon, Santiago, Seville (San

Met broadcast

MET BROADCASTS IN PLAIN LANGUAGE - EUROPE
Radiotelephony

NOTE: ¢ as available

STATION	IDENT	FREQS.	BROADCAST TIMES PERIOD	BROADCAST TIMES H+	FORM	CONTENTS & SEQUENCE
Brindisi	Volmet	127.60	H24	cont.	MET Report TREND (T)	Brindisi (T), Pisa (T), Rome (Fiumicino) (T), Rome (Ciampino), Naples (Capodichino), Athens (Hellinikon) (T), Thessaloniki (Makedonia)(T), Kerkira (Ioannis Kapodistrias) (T), Andravida (T)
Brussels	Volmet	127.80	H24	cont.	MET Report QNH TREND	Brussels, Ostend, London (Heathrow), Luxembourg, Amsterdam, Paris (Orly), Frankfurt, Cologne-Bonn, Dusseldorf
		132.47				Brussels
	BUB	114.6				
	BUN	110.6				
Cairo	Volmet	126.2	H24	cont.	MET Report TREND	Cairo, Alexandria, Aswan, Luxor, Hurghada (Intl), Beirut, Damascus, Larnaca, Athens (Hellinikon), Benghazi, Khartoum, Jeddah
Casablanca	Casa-blanca Met	127.60	H24	cont.	MET Report QNH (Q) TREND (T)	Casablanca (Mohamed V) (Q) (T), Rabat(T), Marrakech (T), Tanger (T), Agadir(T), Oujda, Gran Canaria, Malaga, Seville (San Pablo)
Copenhagen (Kastrup)	Volmet	127.00	H24	cont.	MET Report QNH (Q) TREND (T) ¢	Copenhagen (Kastrup) (Q) (T), Billund, Aalborg (T), Hamburg (T), Malmo (T), Goteborg (T), Stockholm (Arlanda) (T), Oslo (Gardermoen) (T), Stavanger (T)
Dublin	Volmet	127.00	H24	cont.	MET Report TREND	Dublin, Shannon, Cork, Belfast, Glasgow, Prestwick, Manchester, London (Heathrow), London (Gatwick)
Fischamend	FMD	110.4	H24	cont.	SIGMET	Vienna FIR
Frankfurt	Volmet	127.60	H24	cont.	MET Report QNH TREND	Frankfurt, Brussels, Amsterdam, Zurich, Geneva, Basle-Mulhouse, Vienna, Prague, Paris

ATIS

AUTOMATIC TERMINAL INFORMATION SERVICE (ATIS) - EUROPE

The listing below includes airports served by an ATIS within Europe Charts coverage. Airports are listed alphabetically, under the name shown on the Jeppesen Instrument Approach Charts, with civil airports by city name, followed by airport name when different, or with military airport/facility name. ATIS information for frequencies lower than 137.00 MHz is also provided on Jeppesen Instrument Approach Charts and Jeppesen Enroute Charts communications tabulations.

AIRPORT LOCATION	VOICE FACILITY	ATIS FREQ (MHz)	INFORMATION BROADCAST	HOURS UTC
Aalborg	VHF	120.47	Arr & Dep	H24
Aberdeen (Dyce)	ADN VOR VHF	114.30 121.85	Arr & Dep Arr & Dep	Mon-Fri: 0645-2150 ⊗ Sat/Sun: 0700-2150 ⊗
Agen (La Garenne)	VHF	129.60	Arr & Dep	H24
Ajaccio (Campo dell'Oro)	VHF	126.92	Arr & Dep	H24
Alesund (Vigra)	VHF	128.65	Arr & Dep	HR of ATS OPS
Alta	VHF	118.17	Arr & Dep	Mon-Fri: 0600-2200 ⊗ Sat: 0600-1300 ⊗ Sun: 1300-2200 ⊗
Amman (Queen Alia Intl)	VHF	127.60	Arr & Dep	H24
Amsterdam	SPL VOR VHF VHF VHF	108.40 132.97 122.20 130.15	Arr Arr Dep Arr & Dep	H24 H24 H24 Peak hrs.
Ankara (Esenboga)	VHF	123.60	Arr & Dep	H24
Antalya	VHF	118.27	Arr & Dep	H24
Athens (Eleftherios Venizelos)	VHF	136.12	Arr & Dep	H24
Athens (Hellinikon)	VHF	123.40	Arr & Dep	H24
Augsburg	VHF	124.57	Arr & Dep	Mon-Sat: 0530-2000 ⊗ Sun: 0800-2000 ⊗
Avignon	VHF	120.82	Arr & Dep	H24
Barcelona	VHF	118.65	Arr & Dep	H24
Bardufoss	VHF	129.72	Arr & Dep	Mon-Fri: 0550-1850 ⊗
Basle-Mulhouse	VHF	127.87	Arr & Dep	H24
Bastia (Poretta)	VHF Tel: +33 (0)4 95591940	125.92	Arr & Dep	H24
Beauvais(Tille)	VHF	118.37	Arr & Dep	0600 1800
Beirut	KAD VOR	112.60	Arr & Dep	H24
Belfast (Aldergrove)	VHF	128.20	Arr & Dep	H24
Belfast (City)	VHF	136.62	Arr	HR of TWR OPS
Bergen (Flesland)	VHF	125.25	Arr & Dep	H24
Berlin (Schonefeld)	VHF	126.12	Arr & Dep	0500-2300 ⊗
Berlin (Tegel)	TGL VOR VHF	112.30 125.90	Arr & Dep Arr & Dep	0500-2300 ⊗ 0500-2300 ⊗
Bern (Belp)	VHF Tel: +42 (0)31 9615015/16	125.12	Arr & Dep	0450-1820 ⊗
Beziers (Vias)	VHF	127.52	Arr & Dep	PTO

Operation of Met stations

TELEPHONE/FAX NUMBERS AND HOURS OF OPERATIONS OF MET STATIONS - EUROPE

Station	Telephone/ Fax Number	Hours
CYPRUS		
Larnaca (Intl)	(04) 630394	H24
Paphos (Intl)	(06) 236816	H24
DENMARK		
Aalborg	98171117	Mon-Thu: 0600-1400 ⊗, Fri: 0600-1300 ⊗ exc Hol;
	39157272	O/T
Aarhus	39157272	H24
Billund	39157272	H24
Bornholm (Ronne)	39157272	H24
Copenhagen (Kastrup)	39157272, 32516611	H24
Copenhagen (Roskilde)	39157272	H24
Esbjerg	39157272	H24
Herning	39157272	H24
Karup	97101550	Mon-Thu: 0500-1430 ⊗ Fri (Exc Hol): 0500-1300 ⊗
	97101795	O/T
Kolding (Vamdrup)	39157272	H24
Lolland Falster (Maribo)	39157272	H24
Odense	39157272	H24
Sindal	39157272	H24
Skive	39157272	H24
Sonderborg	39157272	H24
Stauning	39157272	H24
Thisted	39157272	H24
Vojens/ Skrydstrup	74590900 Ext. 3451	Mon-Thu: 0500-1430 ⊗, Fri: 0500-1300 ⊗ exc Hol;
	97101795	O/T
EGYPT		
Abu Simbel	(097) 440528	SR/SS & O/R
Alexandria (Borg El Arab Intl)	(03) 4278774	H24
Alexandria (Intl)	(03) 4278774	H24
Aswan (Intl)	(097) 480320 Ext. 421	H24
Asyut (Intl)	(088) 325648	H24
Cairo (Intl)	(02) 4157348	H24
Dakhla	(092) 820635	H24
El Arish (Intl)	(068) 320856	H24
El Kharga	(092) 920485	H24
El Tor	(062) 770252	H24
Giza (Embaba)	(02) 7100441	0600-SS

Station	Telephone/ Fax Number	Hours
FINLAND		
Enontekio	016-5548420	H24
Halli	03-18166435	H24
Helsinki (Vantaa)	09-82773804, 82773331 0600-9-3808 (forecaster)	H24
Helsinki (Malmi)	09-82774031,	H24
Ivalo	016-6758620 Fax: 6758692	H24
Joensuu	013-2727024	H24
Jyvaskyla	014-4455820,	H24
Kajaani	08-6897520	H24
Kauhava	06-1813309	H24
Kemi (Tornio)	016-2118720	H24
Kittila	016-3668920	H24
Kokkola-Pietarsaari (Kruunupyy)	06-8696820	H24
Kuopio	017-477211, 0600-9-3805 (forecaster)	HO
Kuusamo	08-8518820	H24
Lappeenranta	05-4156428	H24
Mariehamn	018-634420	H24
Mikkeli	015-366427	HO
Oulu	08-5207802 Fax: 5207798	H24
Pori	02-61006055	H24
Rovaniemi	016-3636608, 0600-9-3807 (forecaster/H24)	H24
Savonlinna	015-523376	H24
Seinajoki	06-4219620 Fax: 4219699	HO
Sodankyla	016-610109	H24
Tampere-Pirkkala	03-2835521, 0600-9-3803 (forecaster/H24)	HO
Turku	02-2714803	H24
Utti	05-18125354	H24
Vaasa	06-2126130	H24
Varkaus	017-571278	H24
Ylivieska	08-421866	H24
FRANCE (also see Meteorology E-41 for locations not listed)		
International airports		
Agen (La Garenne)	(0)553771206	0445-1645 ⊗
Ajaccio (Campo dell'Oro)	(0)495237675	0500-1730 ⊗

Phonetic alphabet and morse code

PHONETIC ALPHABET AND MORSE CODE

LETTER	CODE	WORD	LATIN ALPHABET REPRESENTATION
A	• —	Alfa	AL FAH
B	— • • •	Bravo	BRAH VOH
C	— • — •	Charlie	CHAR LEE or SHAR LEE
D	— • •	Delta	DELL TAH
E	•	Echo	ECK OH
F	• • — •	Foxtrot	FOKS TROT
G	— — •	Golf	GOLF
H	• • • •	Hotel	HOH TELL
I	• •	India	IN DEE AH
J	• — — —	Juliett	JEW LEE ETT
K	— • —	Kilo	KEY LOH
L	• — • •	Lima	LEE MAH
M	— —	Mike	MIKE
N	— •	November	NO VEM BER
O	— — —	Oscar	OSS CAH
P	• — — •	Papa	PAH PAH
Q	— — • —	Quebec	KEH BECK
R	• — •	Romeo	ROW ME OH
S	• • •	Sierra	SEE AIR RAH
T	—	Tango	TANG GO
U	• • —	Uniform	YOU NEE FORM or OO NEE FORM
V	• • • —	Victor	VIK TAH
W	• — —	Whiskey	WISS KEY
X	— • • —	X-ray	ECKS RAY
Y	— • — —	Yankee	YANG KEY
Z	— — • •	Zulu	ZOO LOO

NUMERAL OR NUMERAL ELEMENT	CODE	PRONUNCIATION
1	• — — — —	WUN
2	• • — — —	TOO
3	• • • — —	TREE
4	• • • • —	FOW-er
5	• • • • •	FIFE
6	— • • • •	SIX
7	— — • • •	SEV-en
8	— — — • •	AIT
9	— — — — •	NIN-er
0	— — — — —	ZE-RO

Conversions

CONVERSIONS

METERS PER SECOND to FEET PER MINUTE (mps = 196.85 fpm)

MPS	FPM	MPS	FPM	MPS	FPM	MPS	FPM
1	197	6	1181	11	2165	16	3150
1.5	295	6.5	1279	11.5	2263	16.5	3248
2	394	7	1378	12	2362	17	3346
2.5	492	7.5	1476	12.5	2460	17.5	3444
3	591	8	1575	13	2559	18	3543
3.5	689	8.5	1673	13.5	2657	18.5	3641
4	787	9	1772	14	2756	19	3740
4.5	885	9.5	1870	14.5	2854	19.5	3838
5	984	10	1969	15	2953	20	3937
5.5	1082	10.5	2067	15.5	3051		

METERS PER SECOND to KNOTS (1 mps = 1.9438 knots)

Meters p/sec.	0	1	2	3	4	5	6	7	8	9
	Knots									
0	-	1.9	3.9	5.8	7.8	9.7	11.7	13.6	15.6	17.5
10	19.4	21.4	23.3	25.3	27.2	29.2	31.1	33.0	35.0	36.9
20	38.9	40.8	42.8	44.7	46.6	48.6	50.5	52.5	54.4	56.4
30	58.3	60.3	62.2	64.1	66.1	68.0	70.0	71.9	73.9	75.8
40	77.8	79.7	81.6	83.6	85.5	87.5	89.4	91.4	93.3	95.2
50	97.2	99.1	101.1	103.0	105.0	106.9	108.8	110.8	112.7	114.7
60	116.6	118.6	120.5	122.5	124.4	126.3	128.3	130.2	132.2	134.1
70	136.1	138.0	140.0	141.9	143.8	145.8	147.7	149.7	151.6	153.6

TEMPERATURES (CELSIUS/FAHRENHEIT)

°C	°F	°C	°F	°C	°F
-40	-40.0	-8	17.6	24	75.2
-39	-38.2	-7	19.4	25	77.0
-38	-36.4	-6	21.2	26	78.8
-37	-34.6	-5	23.0	27	80.6
-36	-32.8	-4	24.8	28	82.4
-35	-31.0	-3	26.6	29	84.2
-34	-29.2	-2	28.4	30	86.0
-33	-27.4	-1	30.2	31	87.8
-32	-25.6	0	32.0	32	89.6
-31	-23.8	1	33.8	33	91.4

WEIGHT

Lbs	Kgs / Lbs	Kgs
2.2046	1	.45359
4	2	1
7	3	1
9	4	2
11	5	2
13	6	3
15	7	3
18	8	4
20	9	4

Units of measurements

COUNTRY	TABLE
Niger	ICAO (56) (60)
Nigeria	Blue (63)
Niue Is	Blue
Norfolk Is	Blue
Okinawa	. . . . (31)
Oman	Blue (60)
Pago Pago	Blue
Panama	. . . . (31) (62)
Papua New Guinea	Blue
Paraguay	ICAO (30) (37)
Peru	ICAO/Blue (3) (37)
Philippines	Blue (6) (36)
Poland	SI (64) (73)
Portugal	SI (59) (63)
Qatar	Blue (63)
Reunion	ICAO (60)
Romania	ICAO (44) (64)
Rwanda	ICAO (8) (60)
St. Helena I	.
Sao Tome & Principe	. . . .
Senegal	ICAO (56) (60)
Seychelles	Blue (60)
Sierra Leone	ICAO (4) (60)
Singapore	Blue (27)
Solomon Is	Blue (51) (60) (70)
Somalia	Blue
Spain	ICAO (63) (68)
Sri Lanka	ICAO (51) (60)

COUNTRY	TABLE
Sudan	Blue
Suriname	. . . . (36)
Swaziland	Blue (60)
Switzerland	SI/Non-SI
Syria	Blue
Taiwan	Blue
Tanzania	Blue (60)
Thailand	SI/Non-SI
Togo	ICAO (56) (60)
Tonga	Blue
Transkei	Blue
Tunisia	ICAO (60)
Turkey	Blue
Tuvalu	Blue
Uganda	Blue
United Arab Emirates	Blue
United Kingdom	Blue (63)
United States	Blue (33)
Uruguay	ICAO (37)
USSR	ICAO (23) (34)
Vanuatu	Blue (60)
Venezuela	Blue
Vietnam	ICAO (60)
Western Samoa	Blue
Yemen	Blue (60)
Yugoslavia	SI/Non-SI (60)
Zaire	ICAO (57) (60)
Zambia	Blue
Zimbabwe	Blue (60)

Notes:

(1) ICAO table on request

(3) ICAO for MET: Blue for ATC.

NOTAMS

NOTAMS

NOTAMs promulgating significant information changes are disseminated from locations all over the world. NOTAMs are intended to supplement Aeronautical Information Publications and provide a fast medium for disseminating information at a short notice. The following format and codes are used in disseminating NOTAMs.

TYPICAL NOTAM — in the SYSTEM NOTAM Format

```
GG EDZZNAEG
020610 EGGNYNYX

(A0623/91 NOTAMN

Q)EGXX/QRDCA/IV/NBO/W/000/400/5510N00520W050

A) EGTT/EGPX
B) 9104030730
C) 9104281500
D) APR 03 07 12 21 24 AND 28 0730 TO 1500
E) DANGER AREA DXX IS ACTIVE
F) GND
G) 12200M (40000 FT.) MSL.)
```

Priority, address, and date-time group for use by NOTAM office.

NOTAM series A number 0623 of 1991. New NOTAM.

Q) More than one FIR is affected; Danger Area activated; IFR/VFR; NOTAM is for the immediate attention of aircraft operators, for inclusion in pre-flight bulletins and operationally significant for IFR flights; navigational warning; ground; 40,000 ft; centered at N 5510 W 00520 for a radius of 50 NM.

A) ICAO four letter indicator for Scottish and London FIR/UIR.
B) Beginning 1991, 4th month, 3rd day, 0730 UTC.
C) Ending 1991, 4th month, 28th day, 1500 UTC.
D) Danger area is only active 4th month, day 03, 07, 12, 21, 24 and 28, between 0730 to 1500 hr.
E) Danger Area DXX is activated.
F) Lower limits beginning at ground level.
G) Upper limits to 12,200m (40,000 ft) MSL.

FORMAT EXPLANATION of SYSTEM NOTAM

NOTAMN — New NOTAM
NOTAMR — Replaces a previous NOTAM
NOTAMC — Cancels a NOTAM
NOTAMS — SNOWTAM (see page 41)

NOTAM format item Q is divided into eight separate qualifier fields.

a. FIR — ICAO location indicator plus "XX" if applicable to more than one FIR.

b. NOTAM CODE — For NOTAM code decode see page 33. If the subject of the NOTAM (second and

SNOWTAM

SNOWTAM

Extracted from ICAO Annex 15 — AERONAUTICAL INFORMATION SERVICES

ORIGINATION AND DISTRIBUTION

Notification of the presence or removal or significant changes in hazardous conditions due to snow, slush, ice or water on the movement area is to be made preferably by use of the SNOWTAM format, or the NOTAM Code and plain language.

Information concerning snow, ice and standing water on aerodrome pavements shall, when reported by means of a SNOWTAM, contain information in the order shown in the following SNOWTAM Format.

SNOWTAM FORMAT

(COM heading)	(PRIORITY INDICATOR)	(ADDRESSES)		≪≡
	(DATE AND TIME OF FILING)	(ORIGINATOR'S INDICATOR)		≪≡
(Abbreviated heading)	(SWAA* SERIAL NUMBER) S W * *	(LOCATION INDICATOR)	DATE/TIME OF OBSERVATION	(OPTIONAL GROUP) ≪≡(

SNOWTAM	(Serial number) →

(AERODROME LOCATION INDICATOR)	A)	→
(DATE/TIME OF OBSERVATION *(Time of completion of measurement in UTC)*)	B)	→
(RUNWAY DESIGNATORS)	C)	→
(CLEARED RUNWAY LENGTH, IF LESS THAN PUBLISHED LENGTH *(m)*)	D)	→
(CLEARED RUNWAY WIDTH, IF LESS THAN PUBLISHED WIDTH *(m; if offset left or right of centre line add "L" or "R")*)	E)	→
(DEPOSITS OVER TOTAL RUNWAY LENGTH *(Observed on each third of the runway, starting from threshold having the lower runway designation number)* NIL — CLEAR AND DRY 1 — DAMP 2 — WET or water patches 3 — RIME OR FROST COVERED *(depth normally less than 1 mm)* 4 — DRY SNOW 5 — WET SNOW 6 — SLUSH 7 — ICE 8 — COMPACTED OR ROLLED SNOW 9 — FROZEN RUTS OR RIDGES)	F)	→
(MEAN DEPTH *(mm)* FOR EACH THIRD OF TOTAL RUNWAY LENGTH)	G)	→
(FRICTION MEASUREMENTS ON EACH THIRD OF RUNWAY AND FRICTION MEASURING DEVICE MEASURED OR CALCULATED COEFFICIENT *or* ESTIMATED SURFACE FRICTION 0.40 and above — GOOD — 5 0.39 to 0.36 — MEDIUM/GOOD — 4 0.35 to 0.30 — MEDIUM — 3 0.29 to 0.26 — MEDIUM/POOR — 2 0.25 and below — POOR — 1 9 - unreliable — UNRELIABLE — 9 *(When quoting a measured coefficient use the observed two figures, followed by the abbreviation of the friction measuring device used. When quoting an estimate use single digit)*)	H)	→
(CRITICAL SNOWBANKS *(If present, insert height (cm)/distance from the edge of runway (m) followed by "L", "R" or "LR" if applicable)*)	J)	→

Sunset/Sunrise tables

SUNRISE											SUNSET									
January											January									
2	5	8	11	14	17	20	23	26	29	LAT	2	5	8	11	14	17	20	23	26	29
h m	h m	h m	h m	h m	h m	h m	h m	h m	h m	°	h m	h m	h m	h m	h m	h m	h m	h m	h m	h m
■	■	■	■	■	■	■	■	12 04	11 03	N 72	■	■	■	■	■	■	■	■	12 25	13 24
■	■	■	■	■	12 05	11 11	10 46	10 25	10 06	70	■	■	■	■	■	12 21	13 11	13 39	14 01	14 21
■	11 34	11 13	10 56	10 41	10 27	10 13	09 59	09 46	09 33	68	■	12 36	13 00	13 20	13 37	13 54	14 09	14 25	14 39	14 54
10 27	10 21	10 14	10 06	09 58	09 49	09 39	29	09 19	09 09	66	13 41	13 49	13 59	14 09	14 20	14 32	14 43	14 55	15 06	15 18
09 49	09 45	09 41	09 35	29	22	09 15	09 07	08 59	08 50	64	14 19	14 25	14 32	14 40	14 49	14 58	15 07	15 17	27	36
22	20	09 16	09 12	09 07	09 02	08 56	08 49	43	35	62	14 45	14 50	14 57	15 03	15 10	15 18	26	35	43	15 52
09 02	09 00	08 58	08 54	08 50	08 46	08 40	08 35	08 29	08 22	N 60	15 06	15 10	15 16	15 22	15 28	15 35	15 42	15 49	15 57	16 05
08 45	08 44	42	39	36	32	27	22	17	11	58	22	27	32	37	43	15 49	15 55	16 02	16 09	15
31	30	28	26	23	20	16	12	08 07	08 02	56	37	41	45	15 50	15 55	16 01	16 06	12	19	25
19	18	17	15	12	09	08 06	08 02	07 58	07 53	54	15 49	15 53	15 57	16 01	16 06	11	16	22	28	34
08 08	08 07	08 06	08 05	08 02	08 00	07 57	07 54	50	46	52	16 00	16 03	16 07	11	16	20	25	30	36	41
07 58	07 58	07 57	07 56	07 54	07 52	07 49	07 46	07 42	07 39	N 50	16 09	16 13	16 16	16 20	16 24	16 29	16 33	16 38	16 43	16 48
38	38	38	37	36	34	32	30	27	24	45	29	32	35	39	42	16 46	16 50	16 54	16 58	17 03
22	22	22	22	21	20	18	16	14	12	40	16 46	16 48	16 51	16 54	16 57	17 01	17 04	17 08	17 11	15
07 08	07 08	07 08	07 08	07 08	07 07	07 06	07 05	07 03	07 01	35	17 00	17 02	17 04	17 07	17 10	13	16	19	22	25
06 56	06 57	06 57	06 57	06 57	06 56	06 56	06 55	06 54	06 52	30	12	14	16	19	21	24	26	29	32	34
06 35	06 36	06 37	06 37	06 38	06 38	06 38	06 38	06 37	06 37	N 20	17 32	17 34	17 36	17 38	17 40	17 42	17 44	17 46	17 48	17 50
17	18	19	20	21	22	22	22	23	23	N 10	17 50	17 52	17 54	17 55	17 57	17 58	18 00	18 01	18 02	18 04
06 00	06 02	06 03	06 04	06 05	06 06	06 07	06 08	06 09	06 10	0	18 07	18 09	18 10	18 11	18 12	18 14	14	15	16	17
05 43	05 44	05 46	05 48	05 49	05 51	05 52	05 54	05 55	05 56	S 10	25	26	27	28	29	29	29	30	30	30
24	26	28	30	32	34	36	38	40	42	20	18 43	18 44	18 45	18 45	18 46	18 46	18 46	18 45	18 45	18 44
05 03	05 05	05 07	05 10	05 12	05 15	05 17	05 20	05 23	05 25	S 30	19 05	19 05	19 06	19 06	19 05	19 05	19 04	19 03	19 02	19 00

ATC définitions

TABLE OF CONTENTS

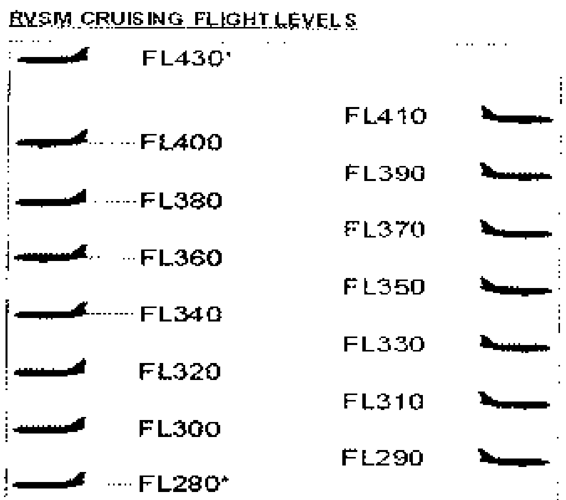
Below is a complete list of the standard contents of Airway Manual. *Limited or special coverages may not contain all items,* but that material which is included should be arranged in the order outlined.

RVSM

Reduced Vertical Separation Minimum (RVSM) EUR

1. AREA OF APPLICABILITY

RVSM shall be applicable in that volume of airspace between FL 290 and FL 410 inclusive in the flight information regions (FIR/UIR) as shown on ATC page E-7.



*non RVSM level

AIRCRAFT EQUIPMENT

The minimum equipment list (MEL) fulfilling the MASPS consists of: (see JAA TGL6)
Two independent altitude measurement systems each equipped with:
cross-coupled static/source system with

compliance with the criteria of the RVSM minimum aircraft systems performance specifications (MASPS).

2. They have instituted procedures in respect of continued airworthiness (maintenance and repair) and programs, and

3. They have instituted flight crew procedures for operations in the EUR RVSM airspace (dimensions specified on ATC page E-7).

Note1:
An RVSM approval is not restricted to a specific region instead. it is valid globally on the understanding that any operating procedures specific to a given region in this case the EUR Region, should be stated in the operations manual or appropriate crew guidance.

Note 2:
Aircraft that have received State approval for RVSM operations will be referred to as 'RVSM approved aircraft'.

Note 3:
Aircraft that have not received State approval for RVSM operations will

Flight procedures

3 ARRIVAL AND APPROACH SEGMENTS

3.2 STANDARD INSTRUMENT ARRIVALS

3.2.1 When necessary or where an operational advantage is obtained, arrival routes from the en-route phase to a fix or facility used in the procedure are published. When arrival routes are published, the width of the associated area decreases from the "en-route" value until the "initial approach" value with a convergence angle of 30° each side of the axis. This convergence begins at 46 km (25 NM) before the IAF if the length of the arrival route is greater than or equal to 46 km (25 NM). It begins at the starting point of the arrival route if the length of the arrival route is less than 46 km (25 NM). The arrival route normally ends at the initial approach fix. Omnidirectional or sector arrivals can be provided taking into account minimum sector altitudes (MSA) (see 1.2.5).

3.2.2 Terminal radar is a suitable complement to published arrival routes. When terminal radar is employed the aircraft is vectored to a fix, or onto the intermediate or final approach track, at a point where the approach may be continued by the pilot through reference to the instrument approach chart.

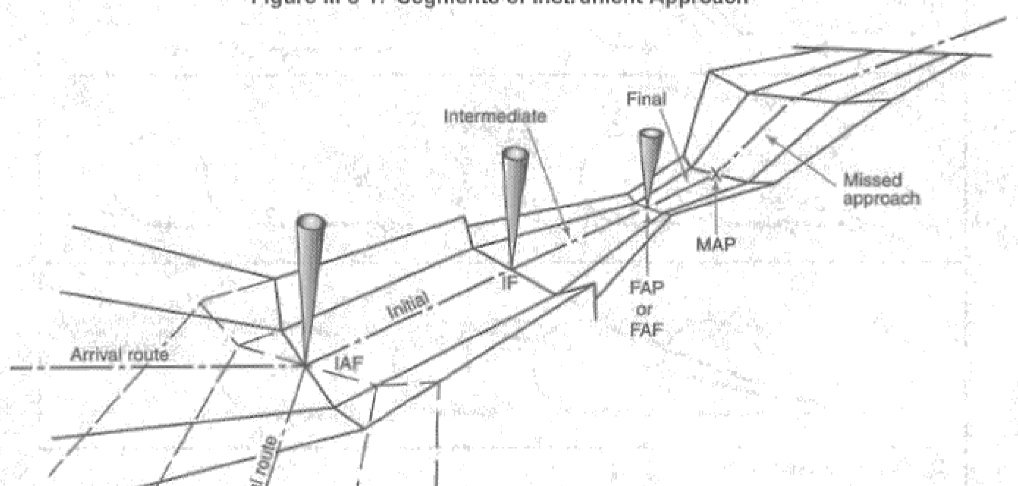
3.3 INITIAL APPROACH SEGMENT

3.3.1 General

3.3.1.1 The initial approach segment commences at the initial approach fix (IAF) and ends at the intermediate fix (IF). In the initial approach, the aircraft has departed the en-route structure and is manoeuvring to enter the intermediate approach segment. Aircraft speed and configuration will depend on the distance from the aerodrome, and descent required. The initial approach segment provides at least 300m (984 ft) of obstacle clearance in the primary area.

3.3.1.3 Where no suitable initial approach fix or intermediate fix is available to construct the instrument procedure in the form shown in Figure III-3-1, a reversal procedure, racetrack or holding pattern is required.

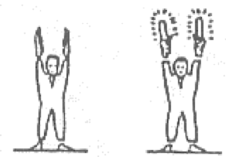
Figure III-3-1. Segments of Instrument Approach



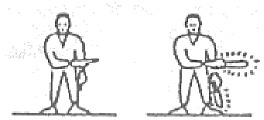
ICAO rules of the air

ICAO RULES OF THE AIR
ANNEX 2

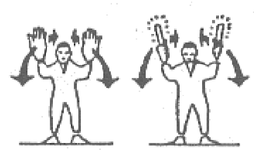
5.1.1.2 ***This Bay*** — Arms above head in vertical position with palms facing inward.



5.1.1.3 ***Proceed To Next Signalman*** — Right or left arm down, other arm moved across body and extended to indicate direction of next signalman.

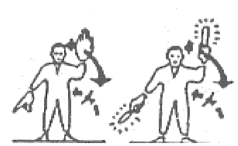


5.1.1.4 ***Move Ahead*** — Arms a little aside, palms facing backward and repeatedly moved upward-backward from shoulder height.

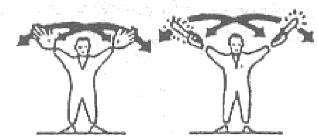


5.1.1.5 ***Turn*** —

a. *Turn to your left:* Right arm downward, left arm repeatedly moved upward-backward. Speed of arm movement indicating rate of turn.



5.1.1.6 ***Stop*** — Arms repeatedly crossed above head (the rapidity of the arm movement should be re-lated to the urgency of the stop; i.e., the faster the movement the quicker the stop).



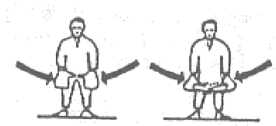
5.1.1.7 ***Brakes*** —

a. *Engage brakes:* Raise arm and hand, with fingers extended, horizontally in front of body then clench fist.

b. *Release brakes:* Raise arm, with fist clenched, horizontally in front of body, then extend fingers.

5.1.1.8 ***Chocks*** —

a. *Chocks inserted:* Arms down, palms facing inwards, move arms from extended position inwards.



b. *Chocks removed:* Arms down, palms facing

State rules

RULES AND PROCEDURES

GENERAL

In general, French air traffic rules and procedures are in conformity with ICAO Standards, Recommended Practices and Procedures.

Units of measurement used in connection with all air traffic services in France:

Distance used in navigation, position reporting, etc. - generally in excess of 2 to 3 nautical miles	NM and TENTHS
Relatively short distances such as those relating to aerodromes (e.g., runway lengths)	METERS
Altitude, elevations, and heights	METERS, FEET*
Horizontal speed including **wind speed**	KNOTS
Vertical speed	METERS per SEC
Wind direction for landing and taking off	DEG/MAG
Wind direction except for landing and taking off	DEG/TRUE
Visibility including **runway visual range**	KM or METERS
Altimeter setting, atmospheric pressure	hPa
Temperature	DEG/CELSIUS (CENTIGRADE)
Weight	mt or KGS
Time	HR & MIN, the DAY of 24 HR BEGINS AT MIDNIGHT UTC

* Unit used on IAC and TMA charts.

FLIGHT PROCEDURES

HOLDING

b) If the instrument approach procedure is in controlled airspace, the minimum altitude is determined from the limit of that airspace, or a point near the airspace limits.

When an Approach Control Service is not provided, the pilot-in-command shall:
Transmit position reports on the local frequency and indicate his intentions
- before initiating the approach procedure,
- at each characteristic position of the instrument approach procedure:

Ensure adequate separation between his aircraft and other aircraft flying IFR, after eventual radio contacts between the pilots, with taking into account the established missed approach procedure.

Entry into a racetrack procedure is normally carried out at the exit from a holding pattern. (The aircraft must mandatorily first integrate into the holding pattern respecting the outbound flight time for that holding pattern.) Racetrack entry restrictions are charted.

Reference datum for QFE Altimeter Setting:
The threshold of the runway is used as the QFE reference datum for precision and non-precision approaches when the threshold elevation is more than 5 meters (16 feet) below the aerodrome elevation, otherwise the aerodrome elevation shall be used.

When no reference is given in the missed approach to level acceleration, the level acceleration segment has not been calculated.

French regulations distinguish between two types of visual maneuvering (circling):

a) For free visual maneuvering (MVL) the ICAO principles for the construction of the visual maneuvering area are the same used by France except for the radius from threshold. These are:
   Cat A - 2.2 NM

Entry requirements

1. **PASSPORT:**
   Required.

2. **VISA:**
   Required, except of nationals of the following countries:
   Andorra, Argentina, Austria, Belgium, Brunei, Canada, Chile, Cyprus, Czech, Denmark, Finland, Germany, Great Britain, Greece, Hungary, Iceland, Ireland, Israel, Italy, Japan, Republic of Korea, Liechtenstein, Luxembourg, Malta, Monaco, Netherlands, New Zealand, Norway, Poland, Portugal, San Marin, Singapore, Slovakia, Slovenia, Spain, Sweden, Switzerland, U.S.A., and Vatican.

   Transit visas are required from nationals of the following countries:
   Afghanistan, Albania, Angola, Bangladesh, Ethiopia, Ghana, Haiti, Iran, Iraq, Liberia, Libya, Nigeria, Pakistan, Sierra Leone, Somalia, Sri Lanka and the Democratic Republic of Congo, except if they are holders of a diplomatic or service passport or are in possession of a normal entry visa for France.

3. **HEALTH:**
   Neither a certificate of vaccination against smallpox nor against cholera is required of passengers arriving on international flights.

4. **AIRCRAFT ENTRY REQUIREMENTS:**
   a) **Scheduled Air Traffic**
   Scheduled operations are governed by interstate bilateral air agreements or specific clearances granted by the French Government.
   Overflights of other airlines require a clearance from the Ministry of Foreign Affairs requested through diplomatic channels.
   b) **Non-Scheduled Air Traffic**
   For overflights and technical landings of

- 8 days in advance in all other cases.

Long series of flights (i.e. within the Europe-Mediterranean region at least 4 flights per month over a time period of two to twelve months with the same charterer; for other destinations 2 flights per month over a time period of two to twelve months with the same charterer)
- 2 months in advance.

Short series of flights (i.e. flights do not conform either with single flights or with long series mentioned above)
- 8 days in advance (Europe-Mediterranean region);
- 15 days (other destinations).

ii) Commercial landings of aircraft registered in a country which has ratified the Multilateral Agreement on Commercial Rights of Non-Scheduled Air Services in Europe are subject to the following conditions:
1. The flight is intra-European (all landings are effected in a country member of the European Civil Aviation Organization) and conforms with the following categories of flights:
   - humanitarian flights;
   - taxi flights with a seating capacity of not more than six passengers;
   - flights where the whole capacity is hired by a single person (except tourist agencies for transport of personnel and merchandise);
   - non-regular flights for

Emergency procedures

INTERNATIONAL CIVIL AVIATION ORGANIZATION (ICAO)

3.3.3 When an aircraft subjected to an act of unlawful interference must depart from its assigned track or its assigned cruising level without being able to make radiotelephony contact with ATS, the pilot-in-command should, whenever possible:

a. attempt to broadcast warnings on the VHF emergency frequency and other appropriate frequencies, unless considerations aboard the aircraft dictate otherwise. Other equipment such as on-board transponders, data links, etc., should also be used when it is advantageous to do so and circumstances permit; and

b. proceed in accordance with applicable special procedures for in-flight contingencies, where such procedures have been established and promulgated in ICAO Document 7030 — *Regional Supplementary Procedures*; or

c. if no applicable regional procedures have been established, proceed at a level which differs from the cruising levels normally used for IFR flight in the area by 300m (1,000') if above Fl 290 or by 150m (500') if below FL 290.

(Annex 2, Attachment B, 2.2)

4 EMERGENCY DESCENT

4.1 INITIAL ACTION BY THE AIR TRAFFIC CONTROL UNIT

4.1.1 Upon receipt of advice that an aircraft is making an emergency descent through other traffic, all possible action shall be taken immediately to safeguard all aircraft concerned. When deemed necessary, air traffic control units shall immediately broadcast by means of the appropriate radio aids, or if not possible, request the appropriate communications stations immediately to broadcast an emergency message. *(Doc 4444, Part III, 16.4.1)*

4.2 ACTION BY THE PILOT-IN-COMMAND

5 DISTRESS AND URGENCY RADIOTELEPHONY COMMUNICATION PROCEDURES

5.1 GENERAL

5.1.1 The radiotelephony distress signal **MAYDAY** and the radiotelephony urgency signal **PAN PAN** shall be used at the commencement of the first distress and urgency communication respectively. *(Annex 10, Vol II, 5.3.1.2)*

5.1.1.1 At the commencement of any *subsequent* communication in distress and urgency traffic, it shall be permissible to use the radiotelephony distress and urgency signals. *(Annex 10, Vol II, 5.3.1.2.1)*

5.1.2 The originator of messages addressed to an aircraft in distress or urgency condition shall restrict to the minimum the number and volume and content of such messages as required by the condition. *(Annex 10, Vol II, 5.3.1.3)*

5.1.3 If no acknowledgement of the distress or urgency message is made by the station addressed by the aircraft, other stations shall render assistance, as prescribed in 5.2.2 and 5.3.2 respectively. *(Annex 10, Vol II, 5.3.1.4)*

NOTE: "Other stations" is intended to refer to any other station which has received the distress or urgency message and has become aware that it has not been acknowledged by the station addressed.

5.1.4 Distress and urgency traffic shall normally be maintained on the frequency on which such traffic was initiated until it is considered that better assistance can be provided by transferring that traffic to another frequency. *(Annex 10, Vol II, 5.3.1.5)*

State special procedures

ICAO DIFFERENCES OR STATE SPECIAL PROCEDURES

GENERAL
In general, the Emergency, Unlawful Interference, Communications Failure, Interception and Search and Rescue procedures are in conformity with the Standards, Recommended Practices and Procedures in ICAO Annexes and Documents.

COMMUNICATIONS FAILURE
In VMC: ICAO Procedure
When the flight is executed in Class A airspace
the conditions of Class B will apply.
In IMC: ICAO Procedure, supplemented as follows:

DIFFERENCES TO THE RADIO COMMUNICATION FAILURE PROCEDURES EUROPE (EMERGENCY E-21)

para e) The point to refer is the Initial Approach Fix (IAF) instead of the designated navigational aid as this fix is the published approach procedure reference point to be notified in the flight plan.

If communication failure occurred during the arrival phase (STAR), the approach procedure phase to an airport, or the departure phase from an airport (SID), the pilot will comply with the specific procedures published for that airport, if any.

GENERAL LEAVING PROCEDURE
If unable to land at the airport of destination for any reason, the pilot shall leave TMA within 30 minutes and try to reach VMC applying appropriate procedure described under "LEAVING PROCEDURE".

Procedures for airports are listed under pertinent TMA name. (For cross-reference see below:)

Albi (Le Sequestre)	TOULOUSE TMA
Annecy (Meythet)	CHAMBERY TMA
Avignon (Caumont)	MARSEILLE/PROVENCE TMA
Beauvais (Tille)	PARIS TMA
Beziers (Vias)	MONTPELLIER TMA
Bourg (Ceyzeriat)	LYON TMA
Bretigny-Sur-Orge	PARIS TMA
Melun (Villaroche)	PARIS TMA
Merville (Calonne)	LILLE TMA
Muret (L'Herm)	TOULOUSE TMA
Pau/Pyrenees	PYRENEES TMA
Pontoise (Cormeilles-en-Vexin)	PARIS TMA
Rochefort (St. Agnant)	LA ROCHELLE TMA
St. Etienne (Boutheon)	LYON TMA
St. Nazaire (Montoir)	NANTES TMA
Tarbes (Lourdes-Pyrenees)	PYRENEES TMA
Toussus-Le Noble	PARIS TMA
Valence (Chabeuil)	LYON TMA
Villefranche (Tarare)	LYON TMA
Vichy	CLERMONT-FERRAND TMA
Villacoublay (Velizy)	PARIS TMA

COMMUNICATION FAILURE PROCEDURES

AGEN TMA

a) **Arrival Procedure**
In case of a missed approach climb along
extended runway centerline to 1000 ft, continue climb to intercept and follow 27 NM DME-arc of VORDME "AGN" until reaching
3500 ft, then turn right to NDB "AG".

b) **Leaving Procedure**
If the second attempt for landing fails, leave
the TMA by following SID SECHE 1W at the minimum safe level and try to reach VMC.

c) **After Departure**
In VMC: Return for landing on the aerodrome.
In IMC: Continue flight to the TMA boundary at the last assigned flight level, thereafter continue climb to the cruising level.

AJACCIO TMA

I. **AJACCIO (Campo Dell'Oro)**

a) **Leaving Procedure**
Rejoin and follow track 236° from locator "RB", then intercept and follow radial 244 of VORDME "AJO" and try to reach VMC.

b) **After Departure**
Continue flight to the boundary of

Legend and explanation

LEGEND AND EXPLANATION

LOCATION (AIRPORT), Apt of Entry (if applicable)

Airports are listed alphabetically by location name, with airport name in parentheses when different than location name. A cross index by airport name is provided.

ELEVATION, JEPPESEN NavData (ICAO) IDENTIFIER, IATA IDENTIFIER (if applicable), TIME ZONE (1), COORDINATES.

(1) Time Zone, in numeric format, observed by the airport as stated in the source and indicating the standard time differential of each zone from Universal Time Coordinated (UTC).

* indicates that the airport observes Daylight Savings or Summer Time.

TELEPHONE/TELEFAX NUMBERS

Telephone/Telefax numbers are provided for contact with the airport, where available.

RUNWAY DATA and runway/approach lights

All usable runways are listed indicating the following items.

1. Runway designators.
2. Total runway length, excluding stopways, overruns or clearways.
3. TORA and LDA if not identical with total runway length. TODA and ASDA when longer than take-off run (TORA) and provided by controlling authority. For explanation see reverse.

1. Type of runway surface.
2. Runway bearing strength.
   - Load classification number (LCN) supplemented (if known) by
     r (rigid pavement) - radius of relative strength in inches
     f (flexible pavement) - thickness in inches
   - Load Classification Group (LCG)
   - Wheel and/or aircraft loads in thousands of pounds

SIWL	-single isolated wheel load times number of main wheels = allowable aircraft weight.
ESWL	- Equivalent Single Wheel Load, a calculated value for multi-wheel legs. The resultant value is considered to be the same as SIWL for determining LCN as indicated on page 4.
S or SW	- (allowable aircraft weight) for single wheel per leg configuration.
T or DW	- (allowable aircraft weight) for tandem or dual wheel per leg configuration.
TT or DDW -	(allowable aircraft weight) for twin tandem or double dual wheel per leg configuration.
TDT -	Runway weight bearing capacity for aircraft with twin delta tandem landing gear.
DDT -	Runway weight bearing capacity for aircraft with double dual tandem type landing gear.

ICAO location indicators decode

JEPPESEN NavData (ICAO) LOCATION IDENTIFIERS DECODE - EUROPE

DA	**FIR/UIR**
DAAA	ALGIERS FIR

DA	**ALGERIA**
DAAD	BOU SAADA
DAAE	BEJAIA (SOUMMAM-ABANE RAMDANE)
DAAG	ALGIERS (HOUARI BOUMEDIENE)
DAAJ	DJANET (TISKA)
DAAP	ILLIZI (TAKHAMALT)
DAAT	TAMANRASSET (AGUENAR)
DAAV	JIJEL (FERHAT ABBAS)
DABB	ANNABA (RABAH BITAT)
DABC	CONSTANTINE (MOHAMED BOUDIAF)
DABS	TEBESSA (CHEIKH LARBI TEBESSI)
DAOB	TIARET (A. BOUSSOUF AIN BOUCHEKIF)
DAOI	ECH-CHELIFF
DAON	TLEMCEN (ZENATA-MESSALI EL HADJ)
DAOO	ORAN (ES SENIA)
DAOV	GHRISS
DATG	IN GUEZZAM
DATM	BORDJ MOKHTAR
DAUA	ADRAR (TOUAT-CH SIDI MOHAMED BELKEBIR)
DAUE	EL GOLEA
DAUG	GHARDAIA (NOUMERAT-MOUFDI ZAKARIA)
DAUH	HASSI MESSAOUD (OUED IRARA-KRIM BELKACEM)
DAUI	IN SALAH
DAUK	TOUGGOURT (SIDI MAHDI)
DAUO	EL OUED (GUEMAR)
DAUT	TIMIMOUN
DAUU	OUARGLA
DAUZ	ZARZAITINE (IN AMENAS)

DT	**FIR/UIR**
DTTC	TUNIS FIR

DT	**TUNISIA**
DTKA	TABARKA (7 NOVEMBER)
DTMB	MONASTIR (HABIB BOURGUIBA)
DTTA	TUNIS (CARTHAGE)
DTTF	GAFSA (KSAR)
DTTI	BORJ EL AMRI
DTTJ	DJERBA (ZARZIS)

EBTY	TOURNAI (MAUBRAY)
EBUL	URSEL (URSEL AB)
EBWE	WEELDE (WEELDE AB)
EBZH	HASSELT
EBZR	ZOERSEL (OOSTMALLE AB)
EBZW	GENK (ZWARTBERG)

ED	**FIR/UIR**
EDBB	BERLIN FIR
EDFF	FRANKFURT FIR
EDLL	DUSSELDORF FIR
EDMM	MUNICH FIR
EDUU	RHEIN UIR
EDVV	HANNOVER UIR
EDWW	BREMEN FIR

ED	**GERMANY**
EDAB	BAUTZEN
EDAC	ALTENBURG (NOBITZ)
EDAD	DESSAU
EDAE	EISENHUTTENSTADT
EDAG	GROSSRUCKERSWALDE
EDAH	HERINGSDORF
EDAI	SEGELETZ
EDAJ	GERA (LEUMNITZ)
EDAK	GROSSENHAIN
EDAL	FURSTENWALDE
EDAM	MERSEBURG
EDAN	NEUSTADT-GLEWE
EDAO	NORDHAUSEN
EDAP	NEUHAUSEN
EDAQ	HALLE (OPPIN)
EDAR	PIRNA (PRATZSCHWITZ)
EDAT	NARDT
EDAU	RIESA (GOHLIS)
EDAV	FINOW
EDAW	ROITZSCHJORA
EDAX	RECHLIN-LARZ
EDAY	STRAUSBERG
EDAZ	SCHONHAGEN
EDBA	ALKERSLEBEN (WULFERSHAUSEN)
EDBC	COCHSTEDT/SCHNEIDLINGEN
EDBD	DEDELOW
EDBF	FEHRBELLIN
EDBG	BURG
EDBH	BARTH
EDBI	ZWICKAU
EDBJ	JENA (SCHONGLEINA)
EDBK	KYRITZ
EDBL	LAUCHA
EDBM	MAGDEBURG
EDDL	DUSSELDORF
EDDM	MUNICH
EDDN	NURNBERG
EDDP	LEIPZIG-HALLE
EDDR	SAARBRUCKEN
EDDS	STUTTGART
EDDT	BERLIN (TEGEL)
EDDV	HANNOVER
EDDW	BREMEN
EDEB	BAD LANGENSALZA
EDEH	HERRENTEICH
EDEL	LANGENLONSHEIM
EDEM	MOSENBERG
EDEQ	MUHLHAUSEN
EDEW	WALLDURN
EDFA	ANSPACH/TAUNUS
EDFB	REICHELSHEIM
EDFC	ASCHAFFENBURG
EDFD	BAD NEUSTADT/SAALE (GRASBERG)
EDFE	EGELSBACH
EDFG	GELNHAUSEN
EDFH	HAHN
EDFI	HIRZENHAIN
EDFJ	LAGER HAMMELBURG
EDFK	BAD KISSINGEN
EDFL	GIESSEN (LUTZELLINDEN)
EDFM	MANNHEIM (MANNHEIM CITY)
EDFN	MARBURG-SCHONSTADT
EDFO	MICHELSTADT
EDFP	OBER-MORLEN
EDFQ	ALLENDORF/EDER
EDFR	ROTHENBURG O.D. TAUBER
EDFS	SCHWEINFURT (SCHWEINFURT SUD)
EDFT	LAUTERBACH
EDFU	MAINBULLAU
EDFV	WORMS
EDFW	WURZBURG (SCHENKENTURM)
EDFX	HOCKENHEIM
EDFY	ELZ
EDFZ	MAINZ (FINTHEN)
EDGA	AILERTCHEN
EDGB	BREITSCHEID
EDGE	EISENACH-KINDEL
EDGF	FULDA-JOSSA
EDGH	HETTSTADT
EDGI	INGELFINGEN (BUHLHOF)
EDGJ	OCHSENFURT
EDGK	KORBACH
EDGL	LUDWIGSHAFEN (LUDWIGSHAFEN UNFALLKLINIK)

Airport data

Airport Availability Hours in France are generally H24. The hours published in this Directory are the operating hours of the Air Traffic Service Unit at the airport. Pilots intending to use the apt outside these hours are required to follow the "auto-information-service" procedures valid within France (see France, ATC Section). Remarks are included covering the restricted use of any apt.

France

Abbeville Apt of Entry
220' LFOI +01:00* N50 08.6 E001 49.9
Apt Administration 0322242471, 0322240521; Fax 0322312438. Aeroclub 0322240848.
02/20 4101' CONCRETE. S/L 11, T/L 18, TT/L 42.
Rwy 20 (CONCRETE) Right-Hand Circuit.
02/20 2953' GRASS.
Rwy 20 (GRASS) Right-Hand Circuit.
13/31 2083' GRASS. LDA 13 2001'.
Rwy 13 Right-Hand Circuit.
Winter: Mon, Thu-Fri 1000-1315LT, 1445-1930LT. Sat Sun Hol 1000-1330LT, 1430-1930LT. Summer (15 APR-30 SEP): Mon-Fri 1100-1415LT, 1600-2015LT. Sat Sun Hol 1100-1415LT, 1545-2030LT. Customs O/R: Days, 24hr PN.
F-3. Fire 1.

Agen (La Garenne) Apt of Entry
204' LFBA AGF +01:00* N44 10.5 E000 35.4
TWR 0553770057. Apt Manager 0553770051, Fax 0553770052. Apt Operator 0553770088, Fax 0553960349. Aeroclub 0553963667.
11/29 7103' BITUMEN. PCN 23/F/C/W/T. LDA 11 6562'. LDA 29 5791'. HIRL. HIALS 29.
Rwy 11 Right-Hand Circuit.
Mon-Fri 0600-2230LT, Sat 0630-1200LT, 1400-1930LT (Winter until 1830LT), Sun/Hol 0900-1200LT, 1400-1930LT (Winter until 1830LT), O/T PPR 24hr. Customs O/R: 24hr PPR to Apt Manager.
F-3. Jet. Fire 3.

Aire-Sur-L'Adour
259' LFDA +01:00* N43 42.6 W000 14.7
Airport Operator 0558716466 (except Mon and Tue).
12/30 3281' MACADAM. S/L 11. LDA 12 2740'. LDA 30 2182'.
Rwy 30 (MACADAM) Right-Hand Circuit.
12/30 3248' GRASS. LDA 12 2838'. LDA 30 2247'.
Rwy 30 (GRASS) Right-Hand Circuit.
O/R except Mon and Tue.

Albi (Le Sequestre) Apt of Entry
564' LFCI LBI +01:00* N43 54.8 E002 07.0
Airport Operator 0563494847, Fax 0563494840. AFIS 0563543849, Fax 0563542400. Aeroclub 0563542768.
09/27 5118' MACADAM. S/L 22, T/L 33. LDA 09 4872'. LDA 27 4560'. HIRL.
Mon-Fri 0730-1900LT, Sat, Sun & Hol not provided; O/T PPR24 hr for Mon-Fri and PPR 48hr for Sat, Sun & Hol. Customs O/R: O/R to AFIS; for Mon-Fri PPR 24hr, for Sat, Sun & Hol PPR on last preceding workday.
F-3. Jet. Fire U.

Alencon (Valframbert)
479' LFOF +01:00* N48 26.8 E000 06.6
Apt Operator 0233292586, 0233276183, Fax 0233319317.
07/25 2543' PAVED. AUW-13. LDA 07 1854'. LDA 25 1952'.
Rwy 25 (PAVED) Right-Hand Circuit.
07/25 2395' GRASS. LDA 07 1804'. LDA 25 1804'.
Rwy 25 (GRASS) Right-Hand Circuit.
Days O/R.
F-3. Fire 1.

Ales (Deaux)
669' LFMS +01:00* N44 04.4 E004 08.6
Apt Administration 0466784949. ATS 0466835222; Fax 0466836491. Aeroclub 0466835618.
01/19 4593' MACADAM. S/L 20, T/L 29. LDA 01 3937'. TODA 01 4806'. RL.
Rwy 19 Right-Hand Circuit.
Mon-Thu 0830-1200LT & 1400-1800LT, Fri 0830-1200LT & 1400-1700LT. O/T for commercial, rescue and offical flights only, Mon-Fri 24hr PPR, Sat Sun 48hr PPR. Pr iority for fire fighting acft.
F-3. Fire 1.

Amberieu (Amberieu AB)
823' LFXA +01:00* N45 58.8 E005 20.3
Airport 0474382152 ext 81229, Aeroclub ext 81388.
01/19 6562' ASPHALT. S/L 44. LDA 19 5899'. TODA 01 6890'. TODA 19 6890'. ASDA 01 6890'. ASDA 19 6890'.
Rwy 19 Right-Hand Circuit. Rwy 01/19 closed every Thu 0800-1200LT.
02/20 2625' GRASS. S/L 11. LDA 20 2231'.
Rwy 02 Right-Hand Circuit.
By NOTAM. Open to public air traffic.
F-3 O/R. Jet A-1. Oxygen . Fire 6.

Ambert (Le Poyet)
1847' LFHT +01:00* N45 31.0 E003 44.8

How to serve Jeppesen

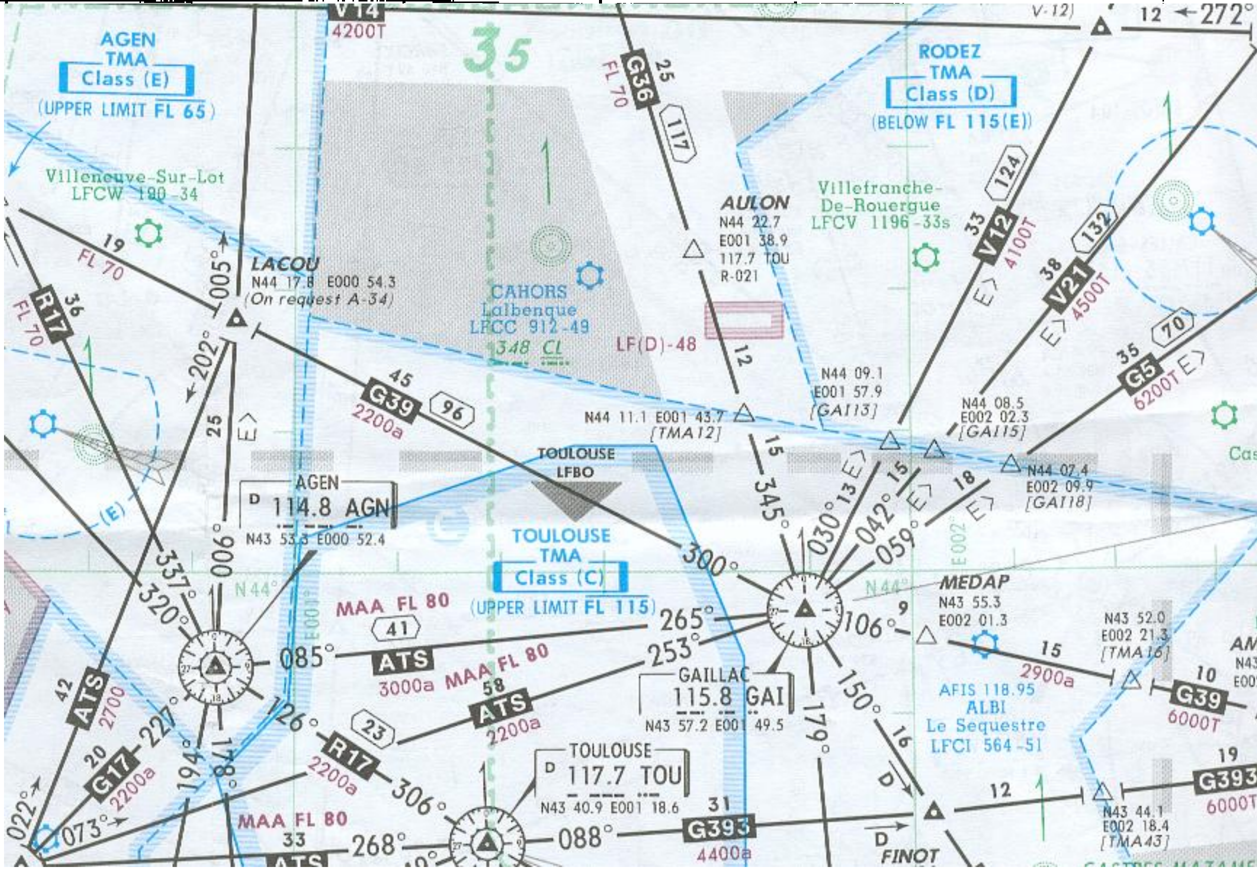
JAR-OPS 1.1045 Appendix I	Jeppesen Airway Manual
a) Minimum flight level Minimum flight altitude	1. Enroute-, Area-, SID/STAR charts. 2. INTRODUCTION section, page 57, Enroute Chart Legend.
b) Operating minima for departure-, destination- and alternate aero-dromes	1. Approach-, Airport charts. 2. 10-9X pages(JAA-Minima). 3. ATC section, JAA AOM pages 601-611.
c) Communication facilities and navigation aids	1. RADIO AIDS section. 2. Approach-, Airport charts. 3. SID/STAR charts. 4. Enroute charts.
d) Runway data and aerodrome facilities	1. AIRPORT DIRECTORY. 2. Airport charts.
e) Approach, missed approach and departure procedures including noise abatement procedures	1. Approach charts. 2. SIDs & STARs. 3. Noise pages. 4. ATC section, chapter 8. a) Noise Abatement Procedures. b) State pages.

How to serve Jeppesen

f) COM-failure procedures	1. EMERGENCY section. 2. ATC section, Emergency and Communications Failure Procedures.
g) Search and rescue facilities in the area over which the aeroplane is to be flown	1. EMERGENCY section, Search and Rescue. 2. ATC section. 3. AIRPORT DIRECTORY.
h) A description of the aeronautical charts that must be carried on board in relation to the type of flight and the route to be flown, including the method to check their validity.	1. INTRODUCTION section, Chart Legend.
i) Availability of aeronautical info and MET services	1. METEOROLOGY section. 2. ENROUTE charts. 3. AIRPORT DIRECTORY
j) Enroute COM/NAV procedures	1. ATC section, State pages 2. ENROUTE section. 3. Enroute charts

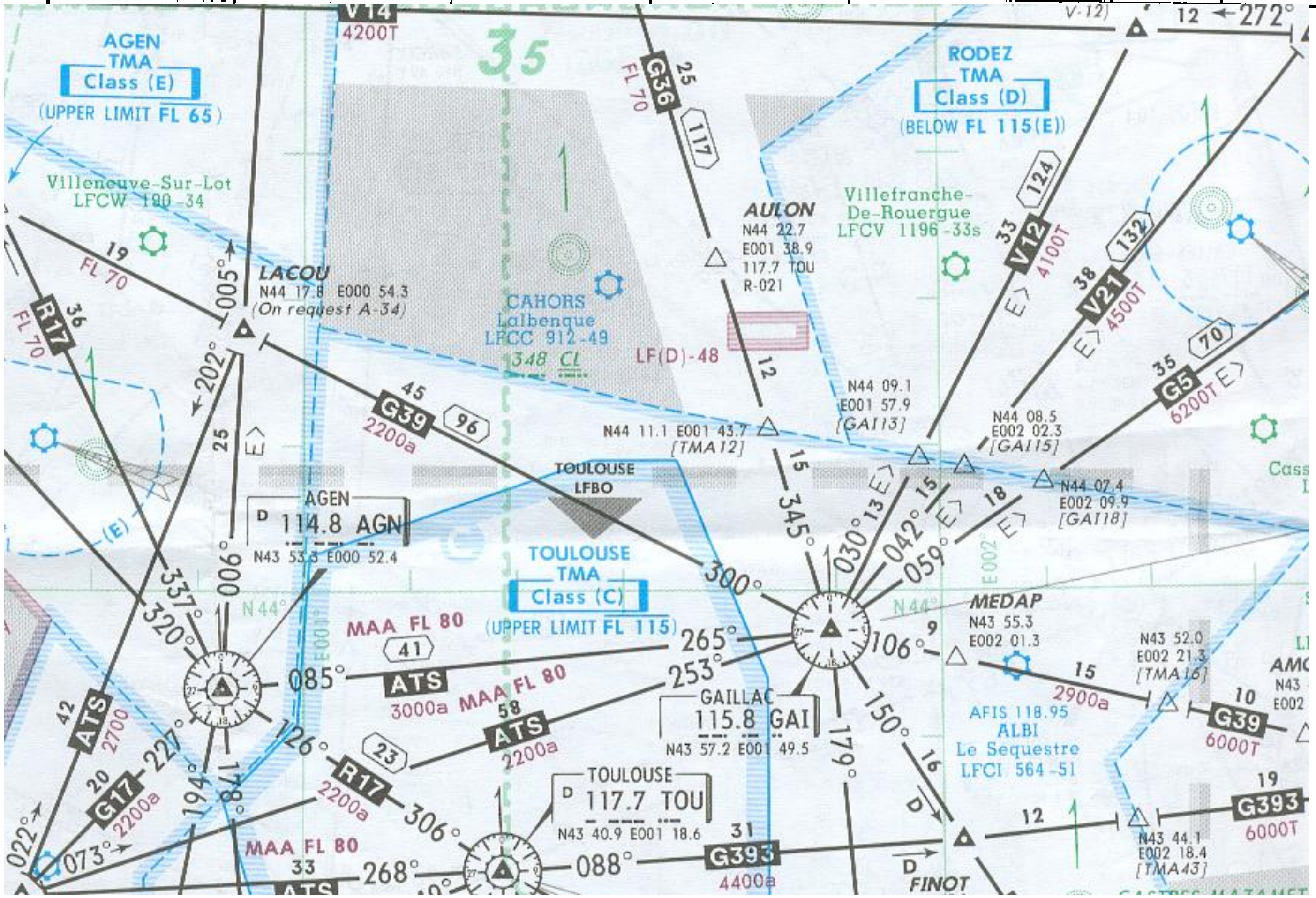
MEA

MEA	MINIMUM EN ROUTE IFR ALTITUDE	FL 40	AIRWAY (E.A.C.) (2)	Demi largeur de l'airway + 5 NM de chaque coté du Centerligne	EN FRANCE	• +1000ft • +2000ft	-Valeur CFMU/EMU Altitude ou Niveau le plus bas utilisable garantissant : -Les réceptions radio COM et radio NAV -Passage des zones P,R,D -La sécurité par rapport au relief



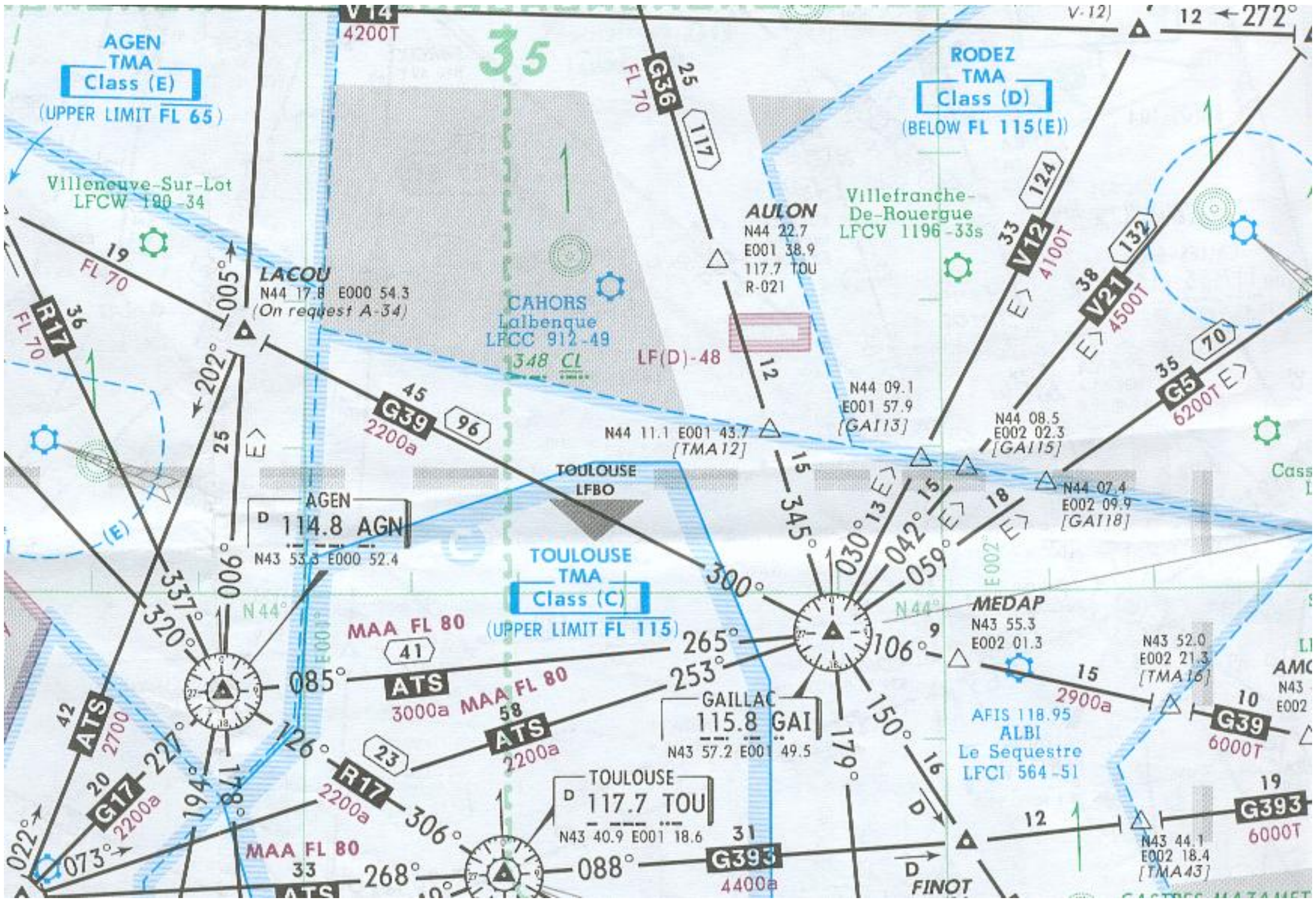
MOCA

MOCA	MINIMUM Obstruction Clearance Altitude	3300T	Sous AIRWAY (EANC) (2)	Demi largeur de l'airway + 5 NM de chaque coté du Centerligne	EN FRANCE	+1500 ft	-Valeur OFFICIELLE -Assure la sécurité par rapport au relief sur un couloir élargi.



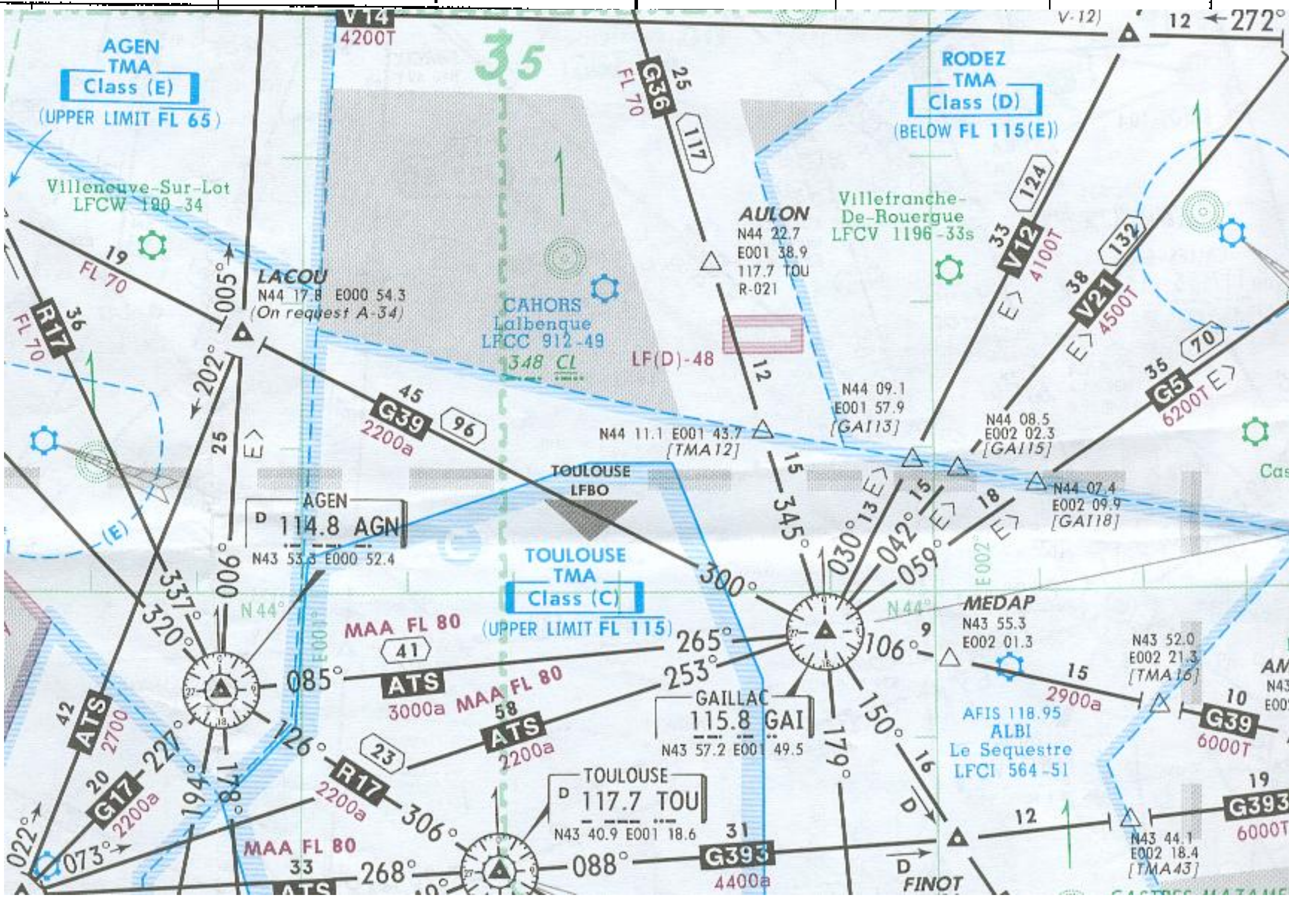
MORA

MORA (1) (Jeppesen)	MINIMUM OFF ROUTE ALTITUDE	3300a	SOUS AIRWAY (EANC) (2)	+/-10 NM DU CENTERLINE DE L'AWY ASSOCIEE	De 0 à 5000 ft ≥ 5001 ft	+1000 ft +2000 ft	- *Calcul JEPPESEN (3)* *Assure uniquement la protection sol.*



GRID MORA

GRID MORA (1) *(Jeppesen)*	MINIMUM OFF ROUTE ALTITUDE Dans Grille de 1°de longitude sur 1° de latitude		TOUT ESPACE (EAC–EANC)	1° de Longitude 1° de Latitude	de 0 à 5000 ft ≥ 5001 ft	+1000 ft +2000 ft	*–Calcul JEPPESEN (3) Permet le calcul des protections donc des niveaux de vol hors airways*



OFFICIELLE

AMA

AMA (1) *(Jeppesen)*	AREA MINIMUM ALTITUDE		TMA +CTR (EAC)	Courbes de niveaux => variables	de 0 à 5000 ft ≥ 5001 ft	+1000 ft +2000 ft	*-Calcul JEPPESEN (3)* *Sur cartes AREA, altitudes minimales exprimées sous forme de courbes de niveau (couleur verte)*